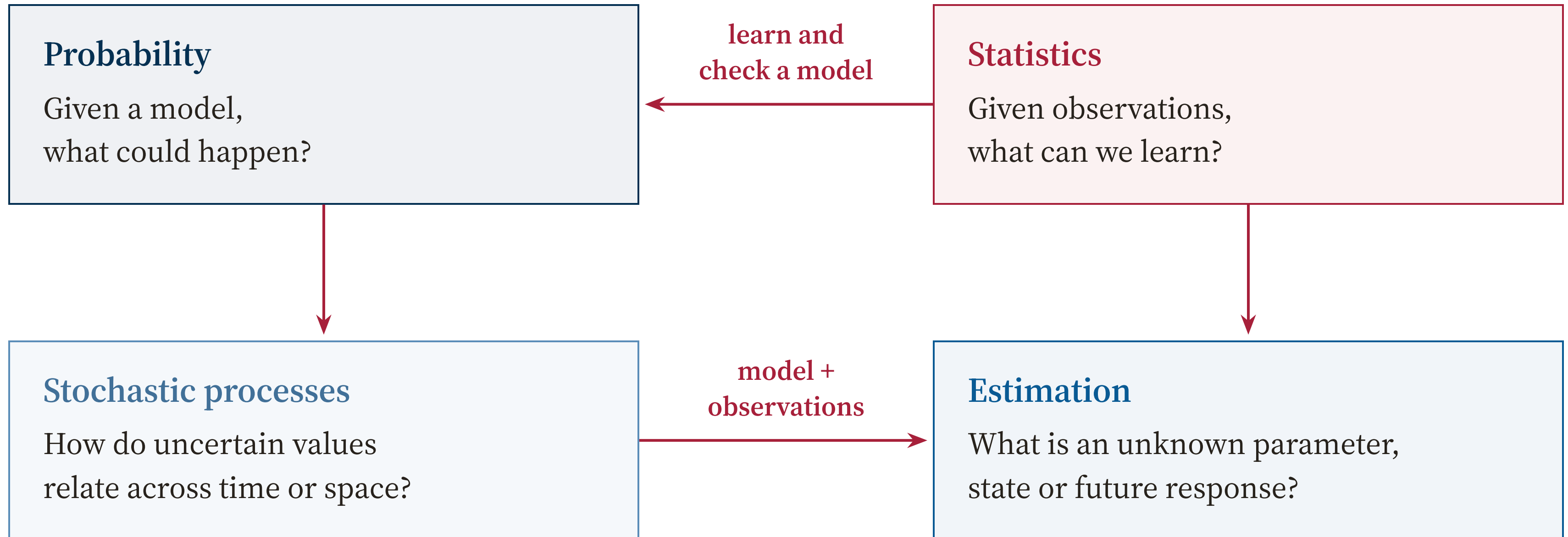


Probability, statistics and estimation

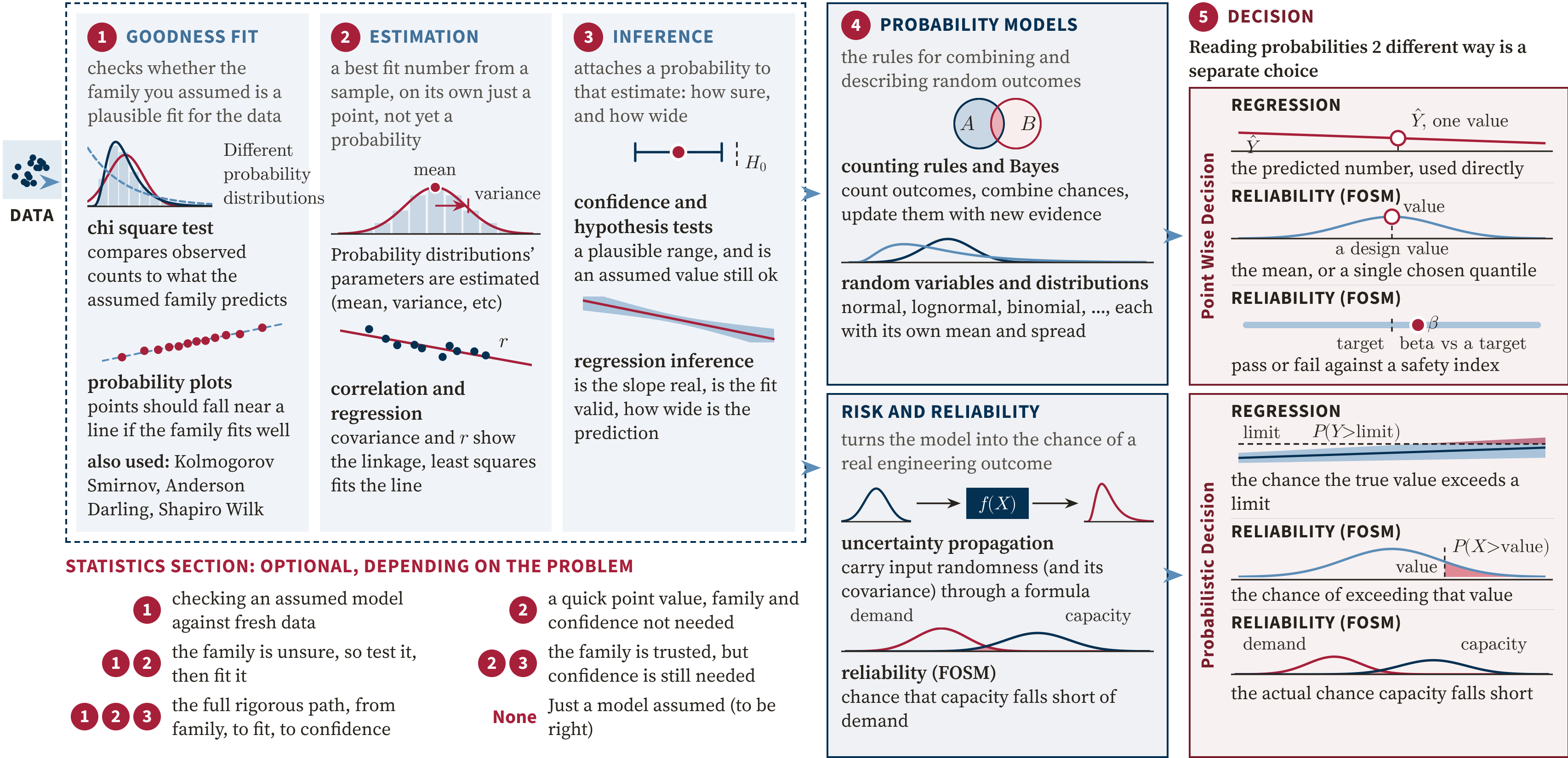
Four connected questions organize the whole subject. Decisions also require consequences, constraints and judgment.



Probability supplies the language. Statistics uses it to learn from evidence.

THE FULL WORKFLOW OF A PROBABILISTIC DECISION (FROM STATISTICS TO PROBABILISTIC DECISION – ALL STEPS INVOLVE PROBABILITY)

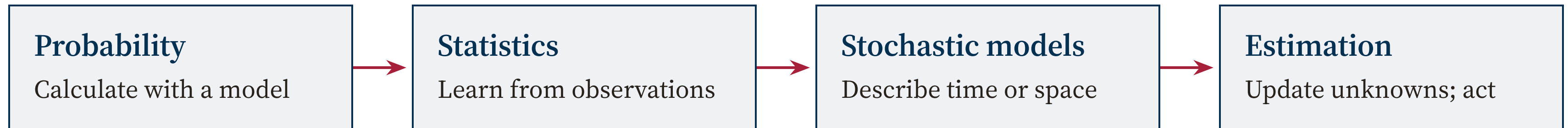
Five stages carry data to a decision. The first 3 are optional, sometimes the model is already known, and you can skip straight to using it. Probabilistic models (that we will cover in this class) are distribution and regression based.



How to read the big-picture map

Keep slide 2 as your reference. We will build its tools in a learning order, then connect them in applications.

Learning route



Fitting and checking form a loop

An existing model may save new fitting.
Its assumptions still need justification.

Point values can encode probability

A quantile or reliability index is one number
with a probabilistic meaning.

Estimation is part of inference

Inference includes estimating unknowns,
quantifying uncertainty and testing claims.

A model gives conditional conclusions

A probability is only as credible as the model.
FOSM generally approximates the response.

Use the map to locate a question. Return to the data and revise the model when needed.

Probability makes sampling risk visible

A lot has 25 items and exactly one defect. Inspect 3 distinct items selected uniformly at random.

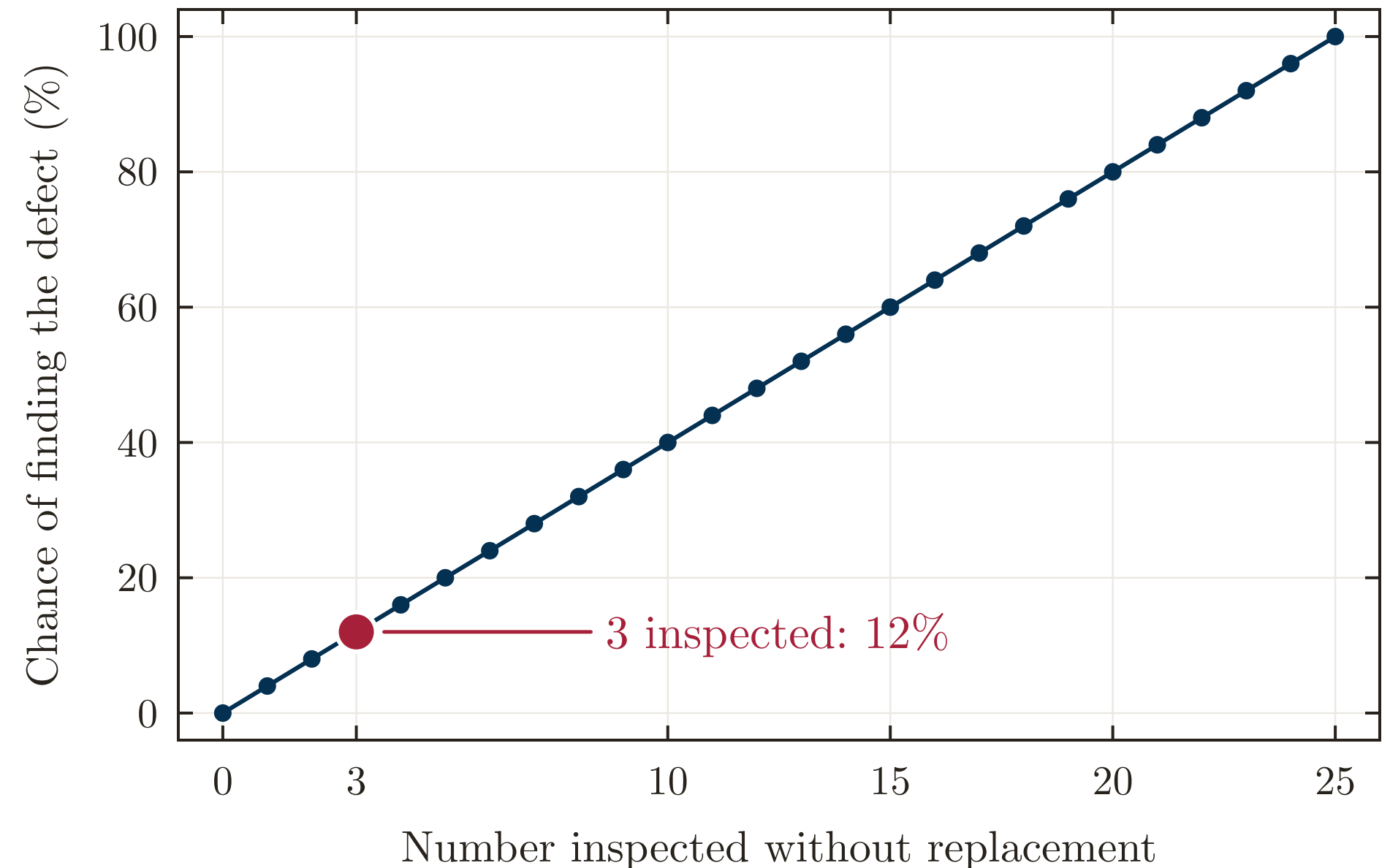
The model is specified

We know the lot size, defect count and sampling rule.

The probability is calculable

$$P(\text{find the defect}) = 3/25 = 12\%$$

$$P(\text{miss the defect}) = 88\%$$



A sample can miss a defect even when the sampling procedure is correct.

Queues: variability becomes waiting time

Hospital arrivals, toll plazas and inspection stations share the problem of variable demand meeting limited capacity.

Input uncertainty

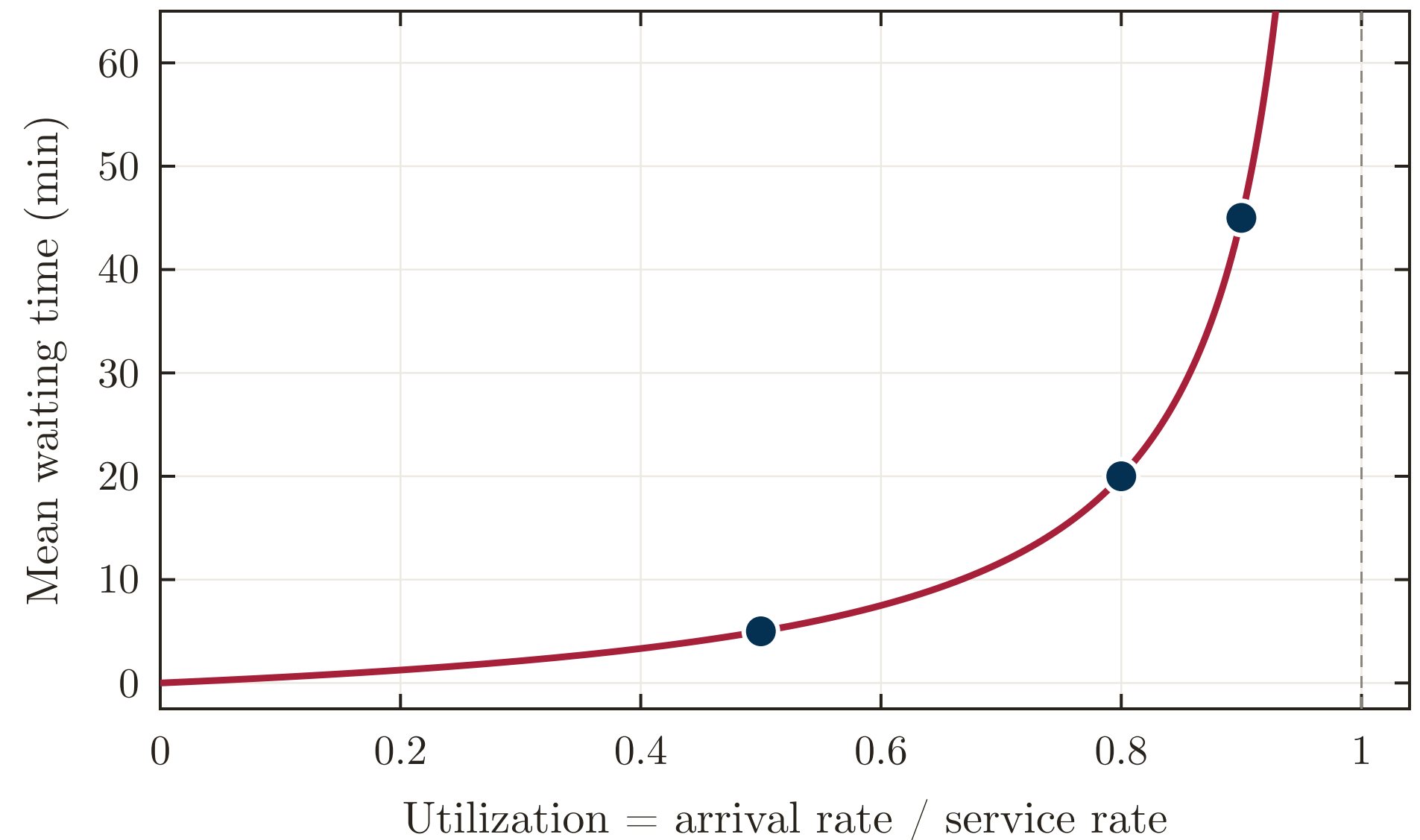
Arrival times and service durations vary from one person to the next.

Model output

A distribution of waiting times and the chance of a long delay.

Illustrative one-server model.

Mean service time: 5 min. No abandonment.



An average service rate above the arrival rate does not eliminate queues.

The probability toolkit

Each tool answers a different question. Together they support later inference, reliability and process models.

Tool	Question	Engineering example
Events and counting	Which outcomes count as success?	Find a defect in a sample
Conditional probability / Bayes	What changes when evidence arrives?	Damage probability after an alarm
Random variables and joint models	What numbers vary together?	Load, strength and their dependence
Distributions	How is probability spread over values?	Counts, waiting times and positive strengths

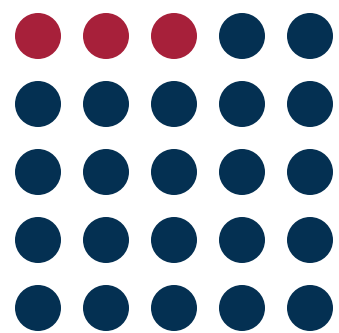
Choose a model for the mechanism, then use probability to calculate with it.

The probability toolkit in action

Events become numbers, evidence updates chances, and joint models describe quantities that vary together.

Events and counting

$$P(A) = |A| / |\Omega| = 3/25 = 12\%$$



Count favorable outcomes
among equally likely outcomes.

Conditional probability and Bayes

$$P(D|A) = P(A|D)P(D) / P(A)$$

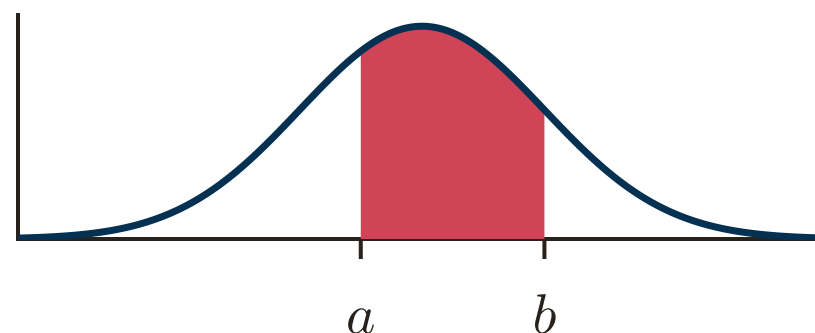
Damage D : 1% prior probability

Alarm A : 90% sensitivity, 5% false alarms

After an alarm: $P(D|A) = 15.4\%$

Random variables and distributions

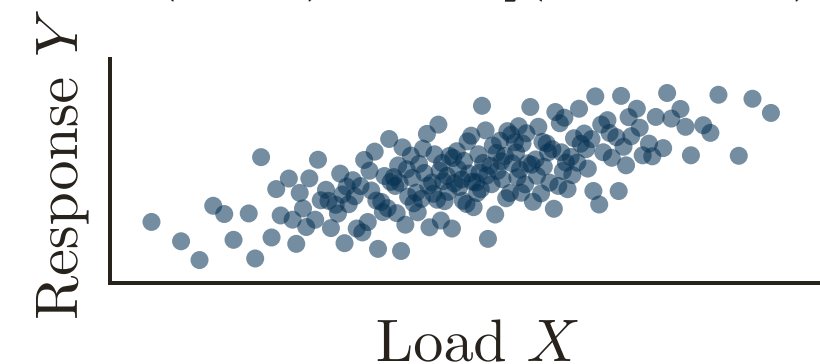
$$P(a < X < b) = \int_a^b f_x(x) \, dx$$



The shaded area is
a probability.
 X assigns a number
to each outcome.

Joint models and dependence

$$\text{Cov}(X,Y) = E[(X - \mu_x)(Y - \mu_y)]$$



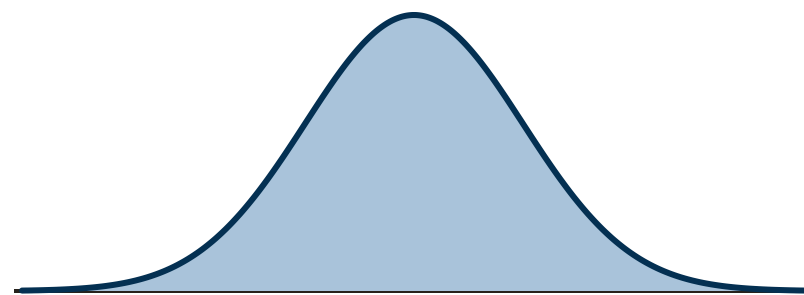
A joint model keeps
the relationship
between uncertain
quantities.

Probability starts with a model and computes the implications of that model.

Distribution models: the shape follows the mechanism

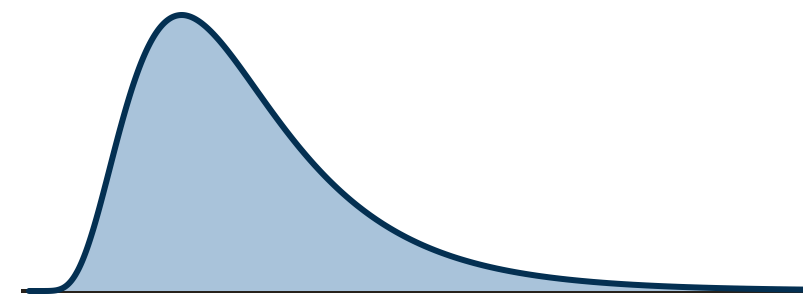
A distribution is the simplest probabilistic model: it spreads probability over the possible values of one quantity. How the quantity is produced suggests the family.

Normal



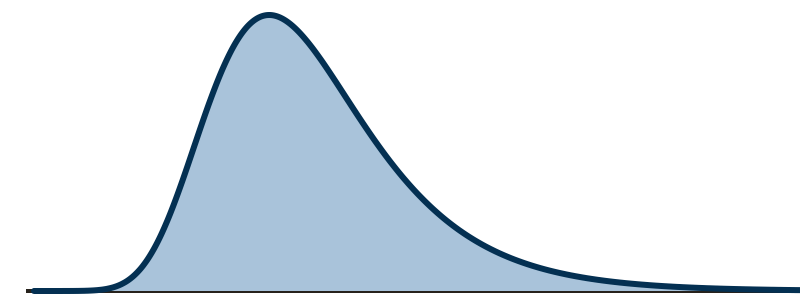
sum of many small effects
measurement error, dead load

Lognormal



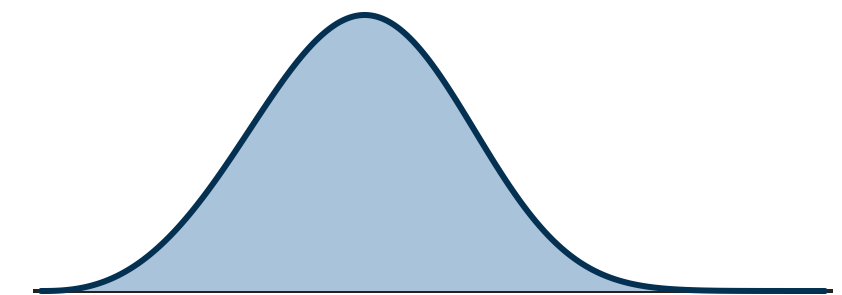
positive; product of factors
strength, stiffness, fatigue life

Gumbel (extreme value)



largest value per period
annual peak flood, wind

Weibull



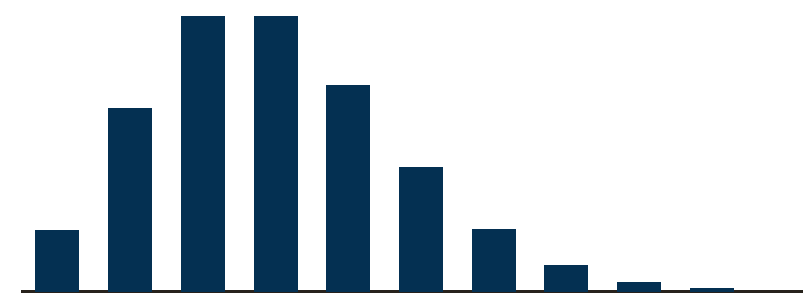
weakest link fails first
brittle strength, component life

Exponential



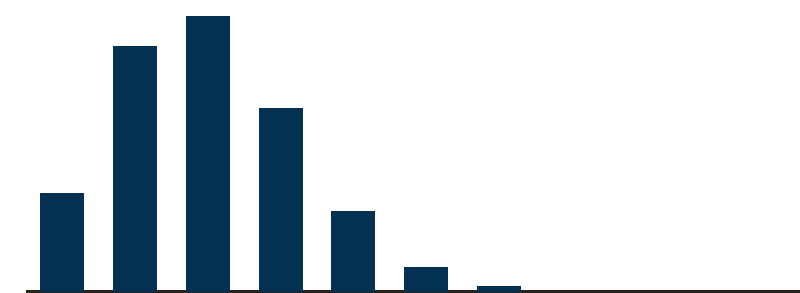
waiting time, no memory
time to next vehicle or event

Poisson



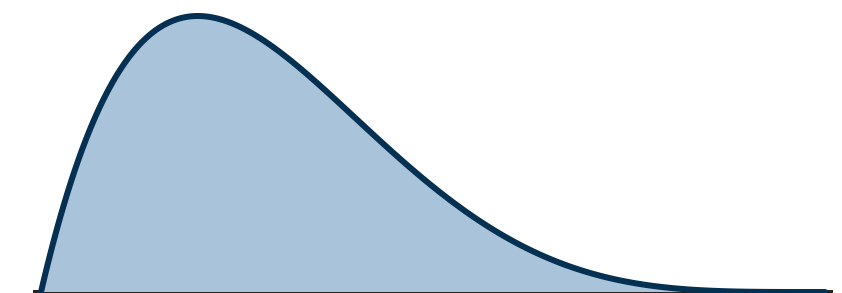
count of rare events
trucks per hour, quakes per year

Binomial



successes in n trials
defective bolts in a sample of 10

Beta



a fraction between 0 and 1
percent damaged, a probability

Ask how the value is produced; the mechanism suggests the distribution family.

Probabilistic models: distributions, regression and more

Every probabilistic model states how probability is spread. The families differ in what else they describe: predictors, dependence, time or hidden states.

Model family	What it describes	Civil engineering example	In this guide
Distribution	One uncertain quantity	Concrete strength, annual peak flow	Slides 8, 13, 25
Joint distribution	Several quantities that vary together	Load and capacity rising together	Slides 7, 37
Regression	A distribution whose mean depends on predictors	Strength vs water-to-cement ratio	Slides 22–24
Logistic regression, fragility curve	Probability of a yes/no outcome vs a predictor	$P(\text{damage})$ vs earthquake shaking	This slide
Bayesian model	Uncertainty about the parameters themselves	Updating mean strength with tests	Slides 14–20
Stochastic process	Uncertainty over time or space	Traffic, vibration, deterioration	Slides 40–60
State-space model	A hidden state seen through noisy sensors	Tracking bridge displacement	Slides 65–67

Distributions and regression are the workhorses; next slide shows they share one core.

One core idea: every model is a probability distribution

Yes: at the core there is one model, the probability distribution. Regression, Bayesian models, processes and state-space models are distributions of different things.

$$p(x)$$

One quantity

Distribution: Concrete strength

$$p(x, y)$$

Several together

Joint distribution: Load and capacity

$$p(y \mid x)$$

y given predictors x

Regression: Strength given w/c ratio

$$p(\theta \mid \text{data})$$

Unknown parameters

Bayesian model: Mean strength after 25 tests

$$p(x_1, x_2, \dots, x_t)$$

A whole history

Stochastic process: Traffic over one day

$$p(x_t \mid z_1 \dots z_t)$$

Hidden state from sensors

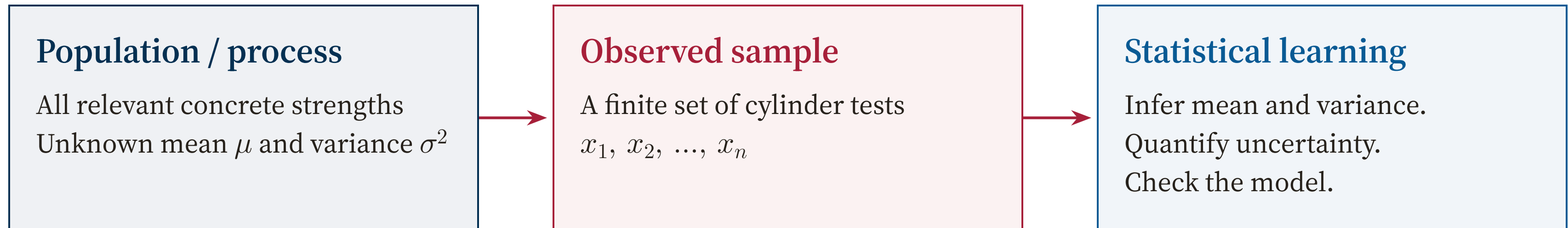
Kalman filter: Displacement now

So distributions and regression are not two separate kinds of model: regression is a distribution whose mean moves with x . The rest add more variables, uncertain parameters, time or hidden states.

Every probabilistic model answers one question: how is probability spread, and over what?

Statistics: learning about a population from a sample

Probability asks what data a model could produce. Statistics uses the observed data to learn about the model.



Describe the sample

What did we observe?
A histogram, sample mean
and sample spread.

Estimate the population

What is the unknown value?
Use $\bar{x}$ to estimate μ .
More data can improve precision.

Assess the uncertainty

How reliable is the estimate?
Use intervals, tests and
model diagnostics.

First identify the unknown. Then choose a method and explain how uncertain the answer is.

Fitting and checking form a loop

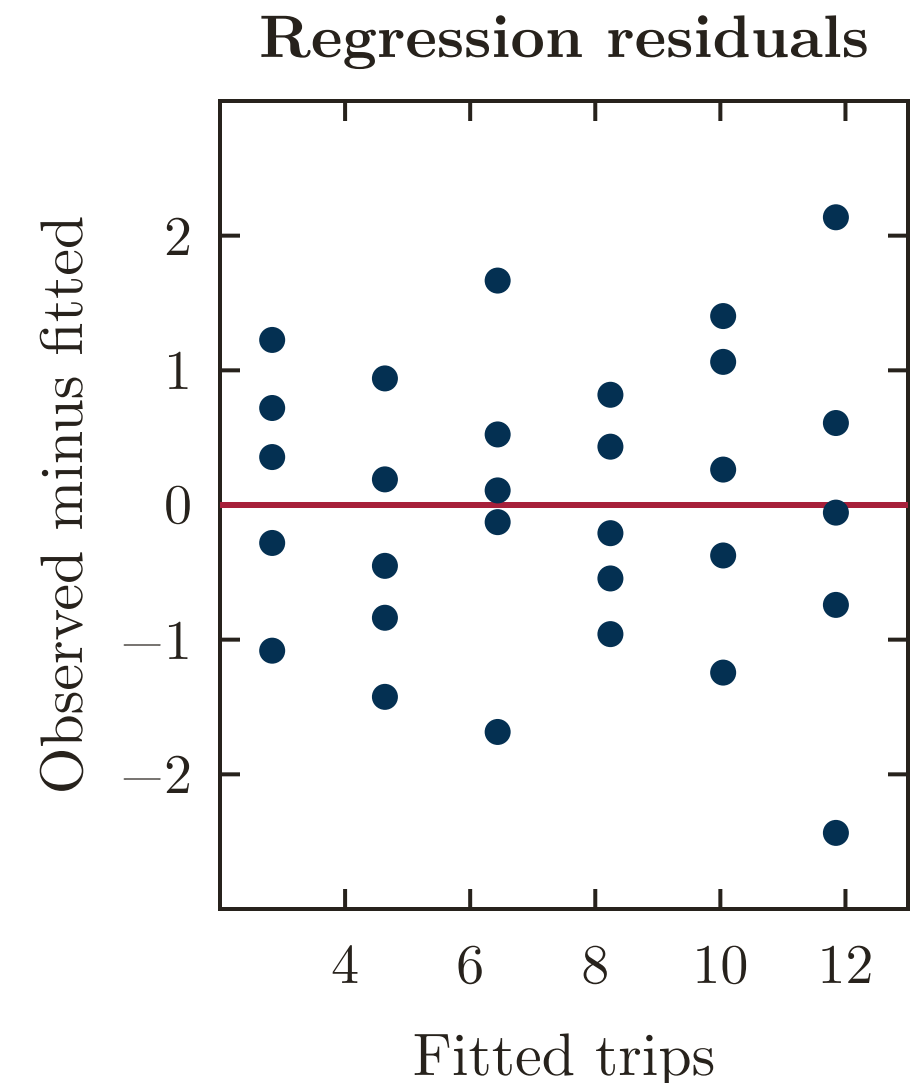
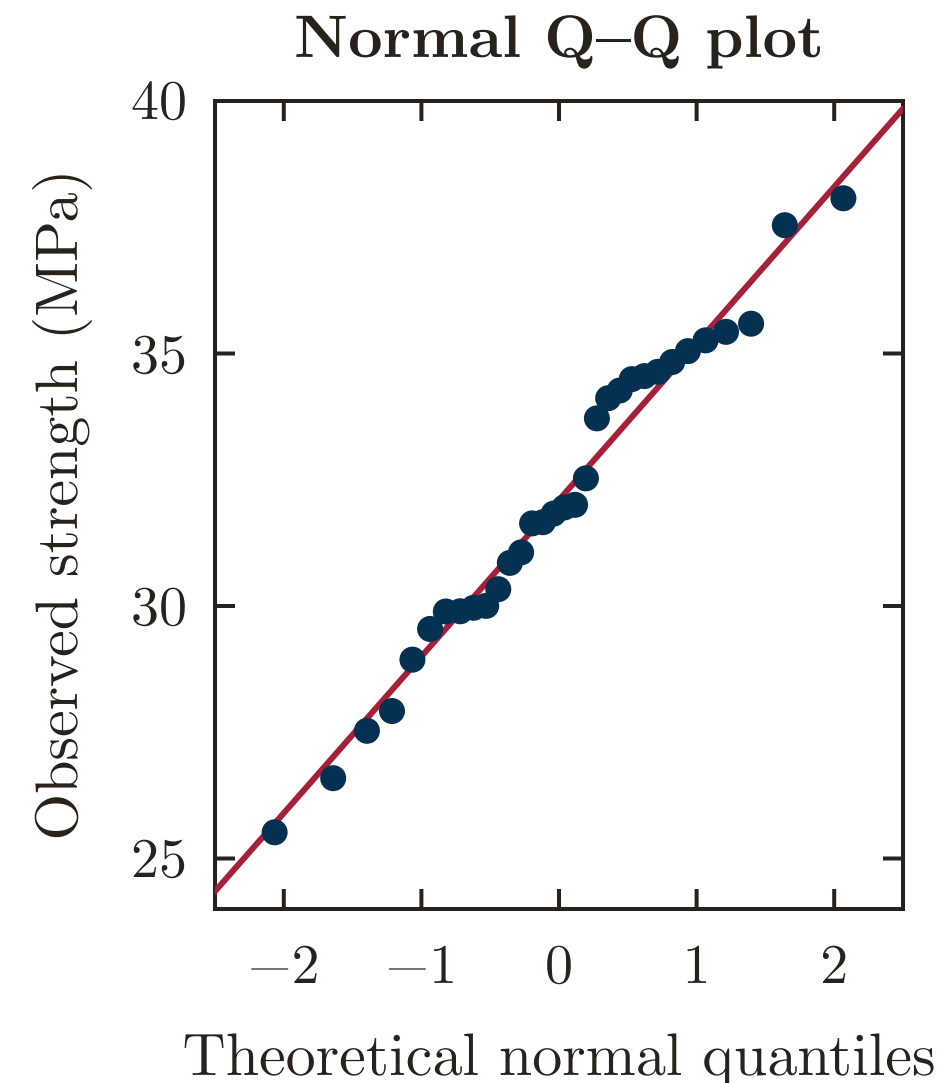
Fit and diagnose iteratively. With a defensible existing model, new fitting may be unnecessary.

Fit

Estimate a distribution or regression using the observed sample.

Check

Compare model implications with data.
Look at tails, residuals and dependence.



Methods: Q–Q plots, χ^2 , K–S, Anderson–Darling, Shapiro–Wilk, residual checks.

Check data quality and model assumptions. A goodness-of-fit test cannot prove that a model is true.

Concrete strength: data become a distribution

One measured strength is a value. Repeated tests reveal variability, and a fitted distribution summarizes it.

Estimate the model

Mean = 32.1 MPa

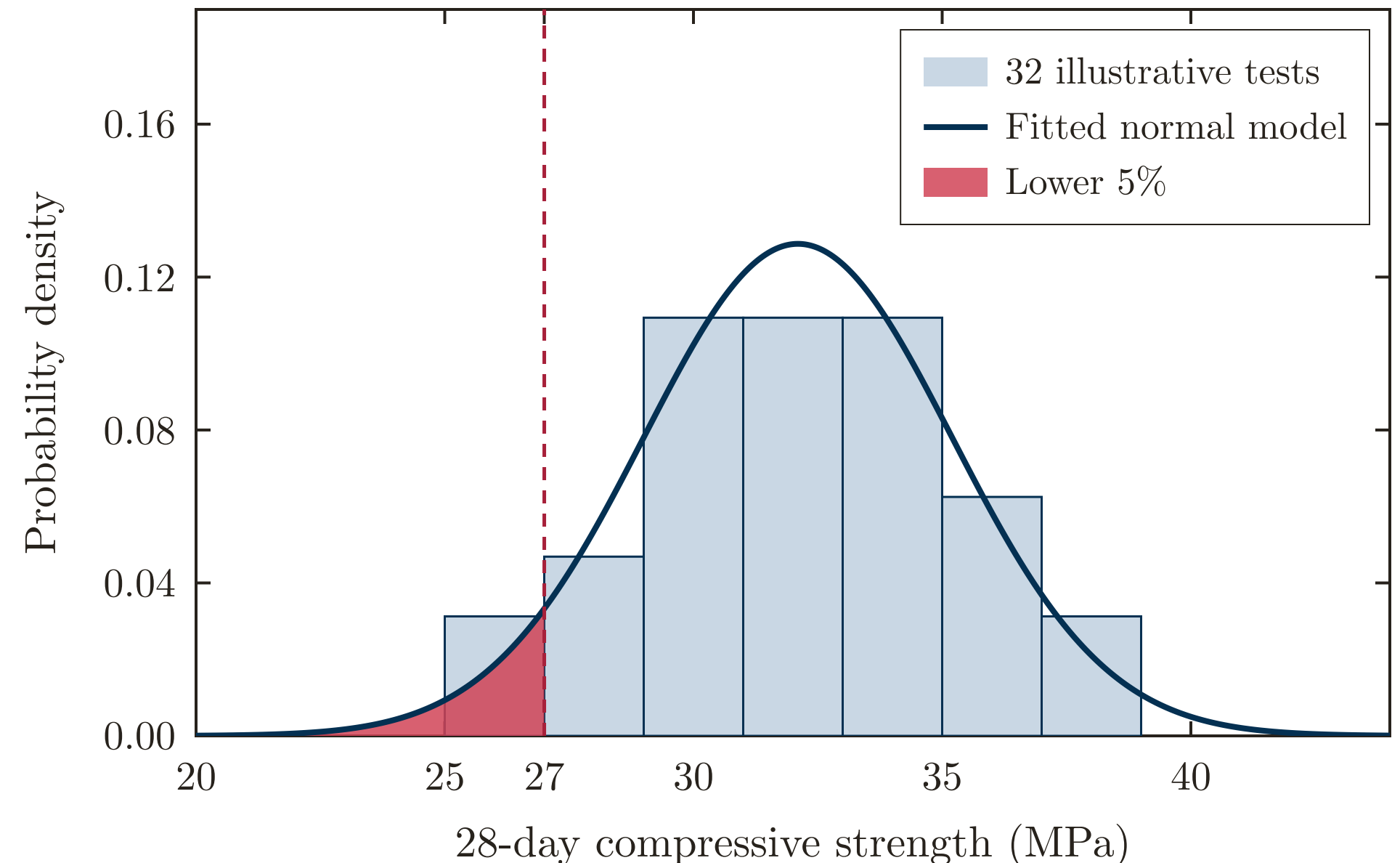
Standard deviation = 3.1 MPa

32 illustrative cylinder tests

Ask a probability question

Which strength has only 5%
of the fitted distribution below it?

Estimated answer: 27.0 MPa.



A lower percentile is a model-based quantity. Its estimate also has uncertainty.

Frequentist and Bayesian: what are we asking?

Suppose we tested 25 concrete cylinders. We want to learn the unknown population mean strength μ .

FREQUENTIST: evaluate a procedure

“If we repeated this sampling process,
how often would our method succeed?”

Imagine many possible samples of 25 tests.
Each gives a different mean and interval.
Assess the method across those samples.

The population mean stays fixed.
The data and our estimates vary.

BAYESIAN: update uncertainty

“Given these measurements, which
values of μ are now more plausible?”

Start with a prior over plausible μ values.
Use the data likelihood to reweight them.
The posterior describes updated uncertainty.

μ can be physically fixed in both frameworks.
The Bayesian distribution represents what
we do not yet know about it.

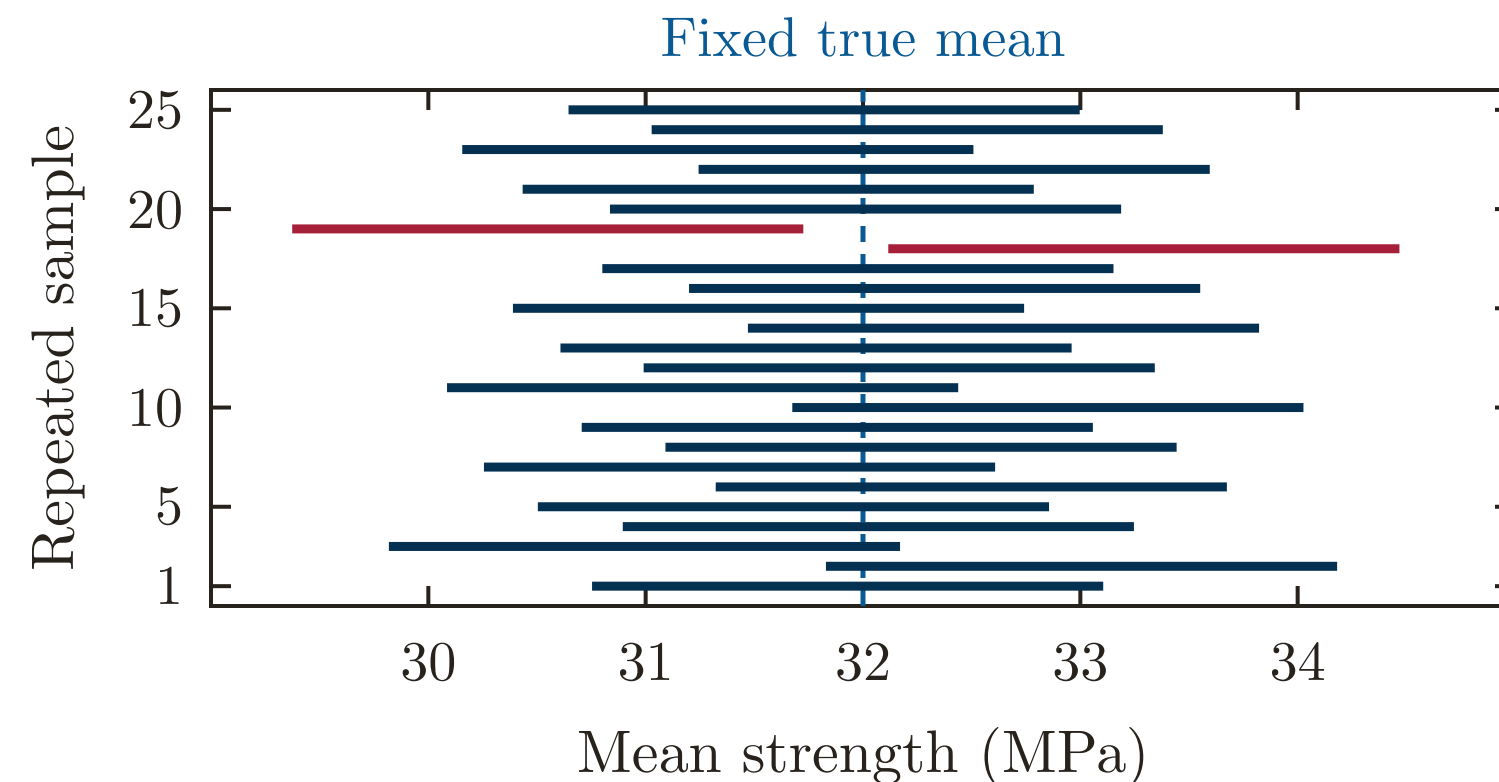
Both use a data model. They differ in how they describe uncertainty about the unknown.

Frequentist and Bayesian inference: two frameworks

D = observed data; θ = unknown parameter. Likelihood $p(D|\theta)$ measures how well θ explains the observed data.

FREQUENTIST

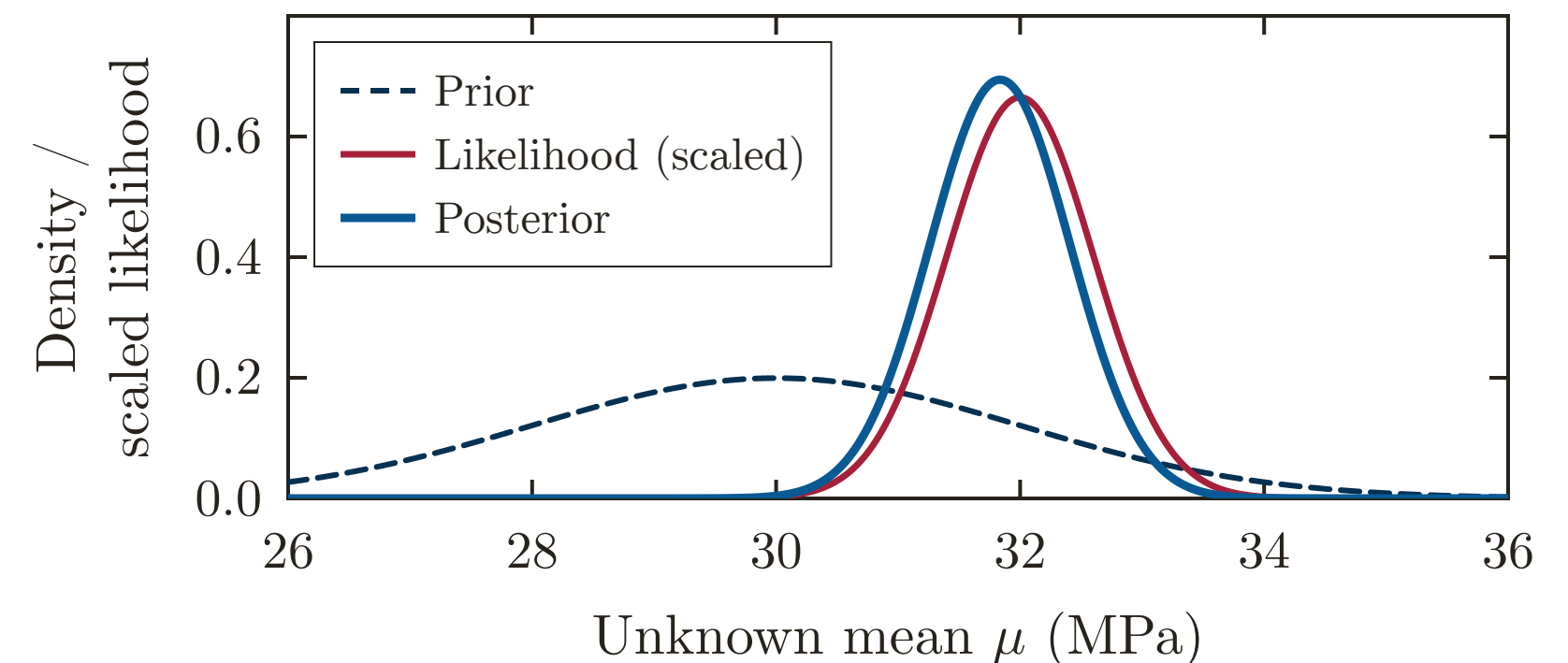
θ is fixed and unknown. Samples vary.
Build rules that work over repeated samples.



95% confidence: the procedure covers the fixed true value in 95% of repeated samples.

BAYESIAN

A distribution describes uncertainty about θ .
Update it after observing the data D .



$$p(\theta|D) \propto p(D|\theta) p(\theta)$$

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

The distinction concerns what probability describes and how we assess uncertainty.

Two kinds of spread: one test vs. the average of 25

Single cylinders scatter because the material varies. The average of 25 cylinders scatters much less, because high and low results cancel out.

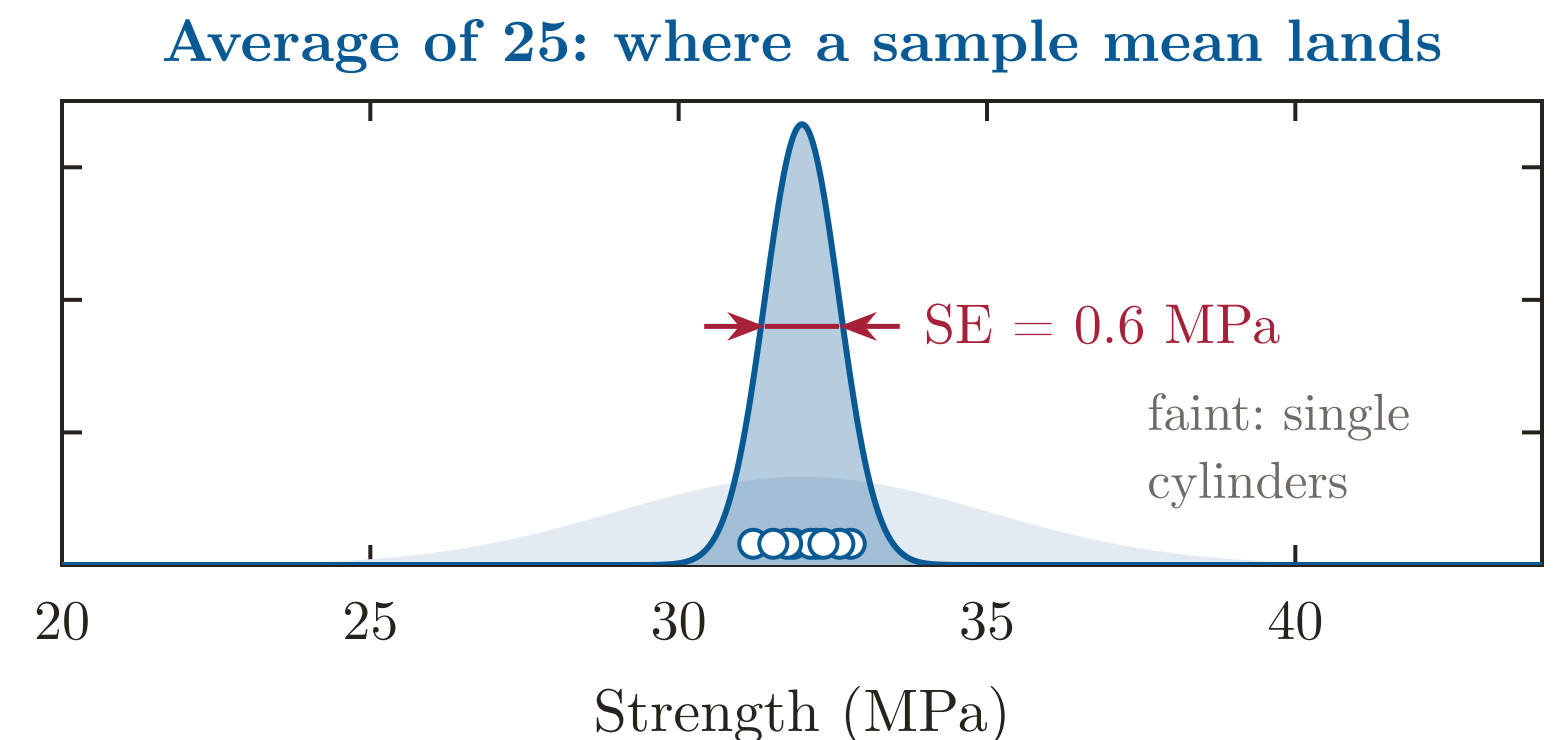
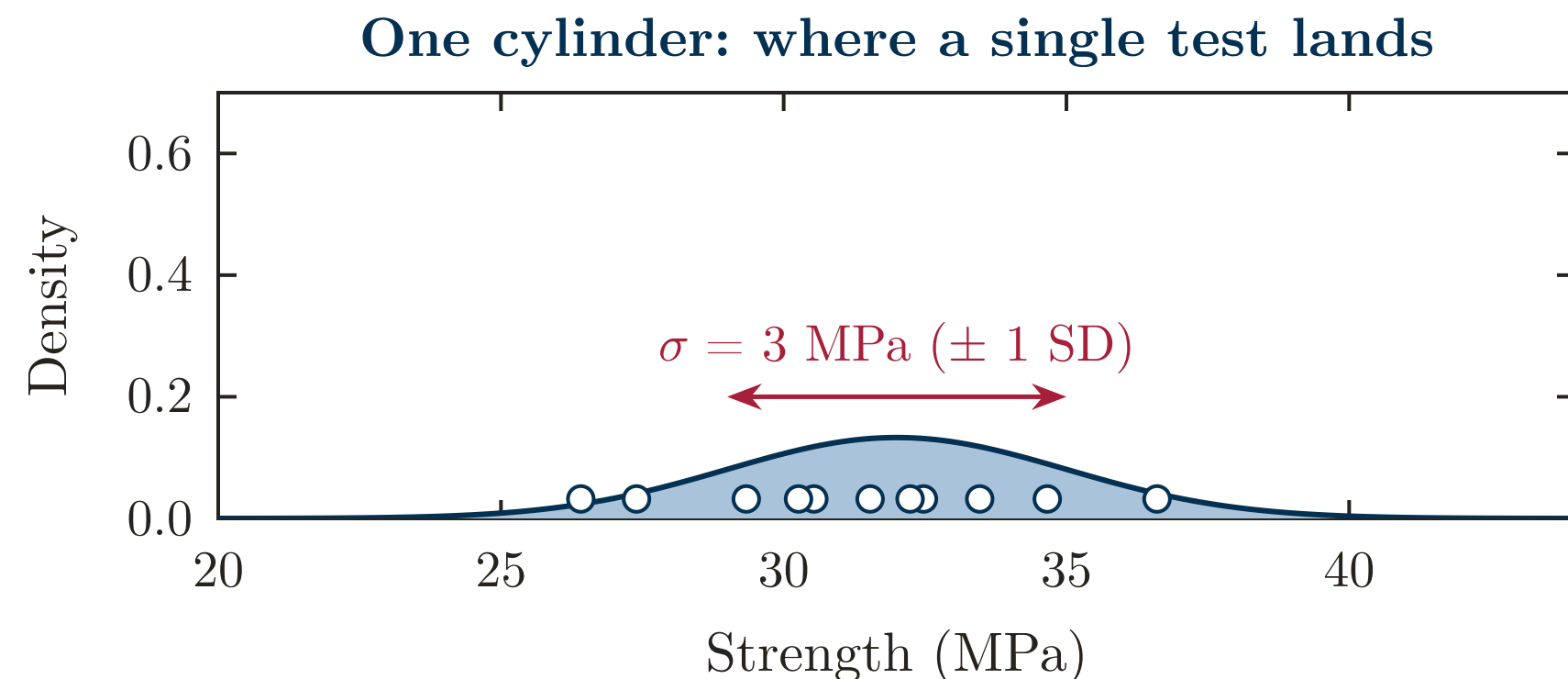
One cylinder: material variability

Lands anywhere around μ ; SD $\sigma = 3$ MPa.
More testing never shrinks σ .

More
data

Average of 25: estimation uncertainty

Averages cluster near μ ; SE = 0.6 MPa.
More tests shrink SE = $\sigma/\sqrt{n}$.



$$\text{Var}(\bar{X}) = (1/n^2) \times n\sigma^2 = \sigma^2/n$$

$$\text{SE}(\bar{X}) = \sigma/\sqrt{n} = 3/\sqrt{25} = 0.6 \text{ MPa}$$

★ More tests sharpen our estimate of μ ; they never make individual cylinders more alike.

Frequentist: why 1.96 in a 95% confidence interval?

Concrete strength: 25 independent normal tests, known population SD $\sigma = 3$ MPa, observed mean $\bar{x} = 32$ MPa.

1 Standardize the sample mean

μ = unknown population mean strength

$$SE = \sigma / \sqrt{n} = 3 / \sqrt{25} = 0.6 \text{ MPa}$$

$$Z = (\bar{X} - \mu) / SE \sim N(0, 1)$$

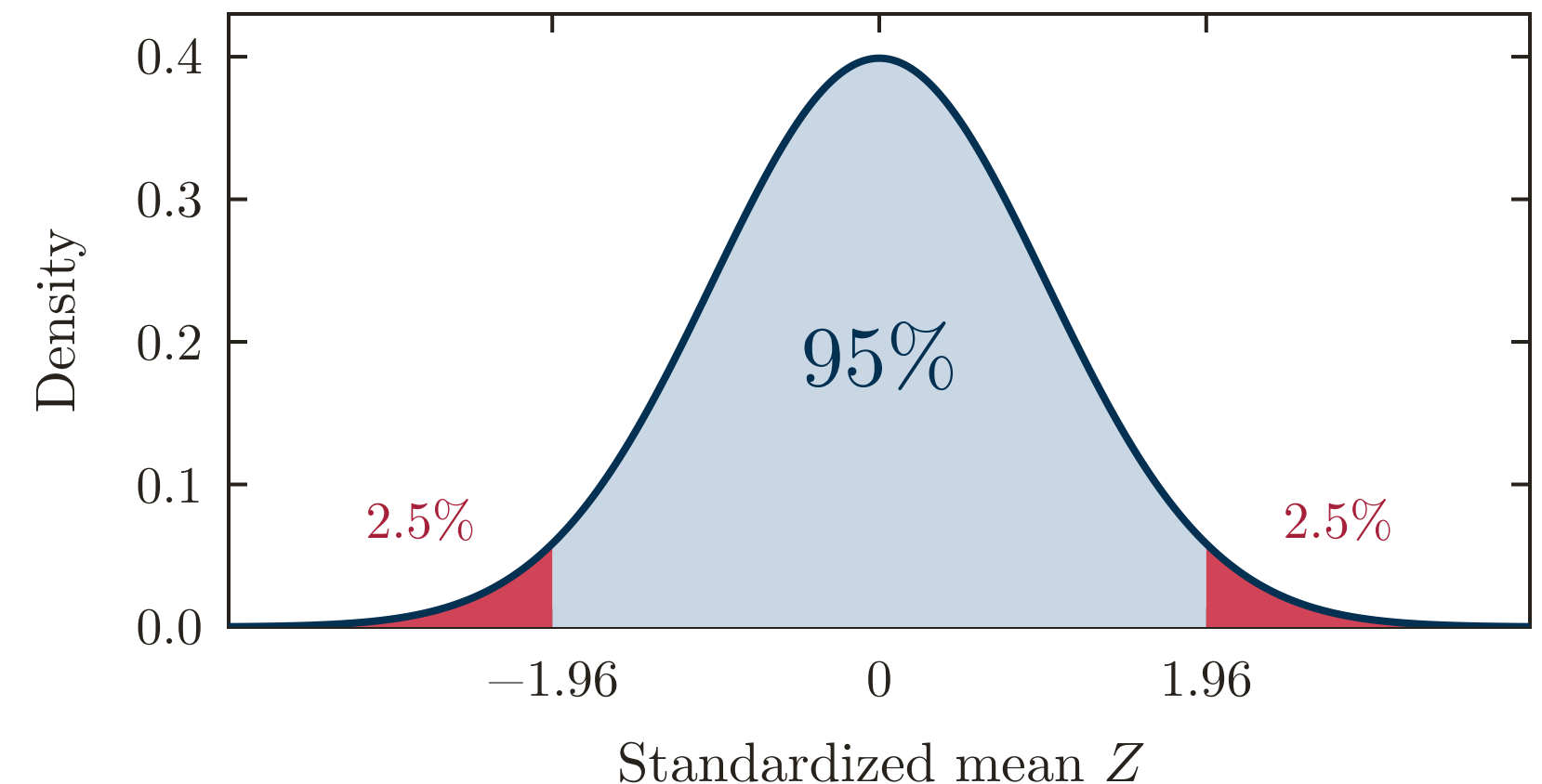
2 Keep the central 95%

$$P(-1.96 \leq Z \leq 1.96) = 0.95$$

Each tail contains 2.5%.

1.96 is the 97.5th percentile of $N(0, 1)$.

Here $N(\text{mean}, \text{variance})$ denotes a normal distribution.



3 Return to MPa

$$\bar{x} \pm 1.96 SE = 32 \pm 1.176$$

$$95\% \text{ CI} = [30.82, 33.18] \text{ MPa}$$

1.96 comes from the standard normal distribution and the chosen two-sided 95% coverage.

Bayesian updating: deriving the posterior

We describe uncertainty about the unknown mean μ with a prior, then update it using the observed tests.

Ingredients and notation

D : the observed strength measurements

Prior: $\mu \sim N(m_0, v_0)$

m_0 : prior mean; v_0 : prior variance

$v_D = \sigma^2/n$: variance of the sample mean

Bayes' rule

$$p(\mu \mid D) \propto p(D \mid \mu) \times p(\mu)$$

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

$\propto$ means “proportional to.” A normalizing constant makes the total probability equal to one.

Combine the two squared distances

$$p(\mu \mid D) \propto \exp\left\{-\frac{1}{2}\left[(\mu - m_0)^2/v_0 + (\mu - \bar{x})^2/v_D\right]\right\}$$

Expand the squares and collect terms in μ :

$$\mu^2(1/v_0 + 1/v_D) - 2\mu(m_0/v_0 + \bar{x}/v_D)$$

Then complete the square: $(\mu - m_p)^2/v_p$.

$$v_p = 1 / (1/v_0 + 1/v_D)$$

$$m_p = v_p(m_0/v_0 + \bar{x}/v_D)$$

m_p = posterior mean; v_p = posterior variance.

Multiplying two normal-shaped functions adds their precisions: precision = 1 / variance.

Bayesian strength example: the actual calculation

Same strength data: $n = 25$, $\bar{x} = 32$ MPa, known $\sigma = 3$ MPa. Prior mean 30 MPa and prior SD 2 MPa.

Prior variance $v_0 = 2^2 = 4$ MPa²

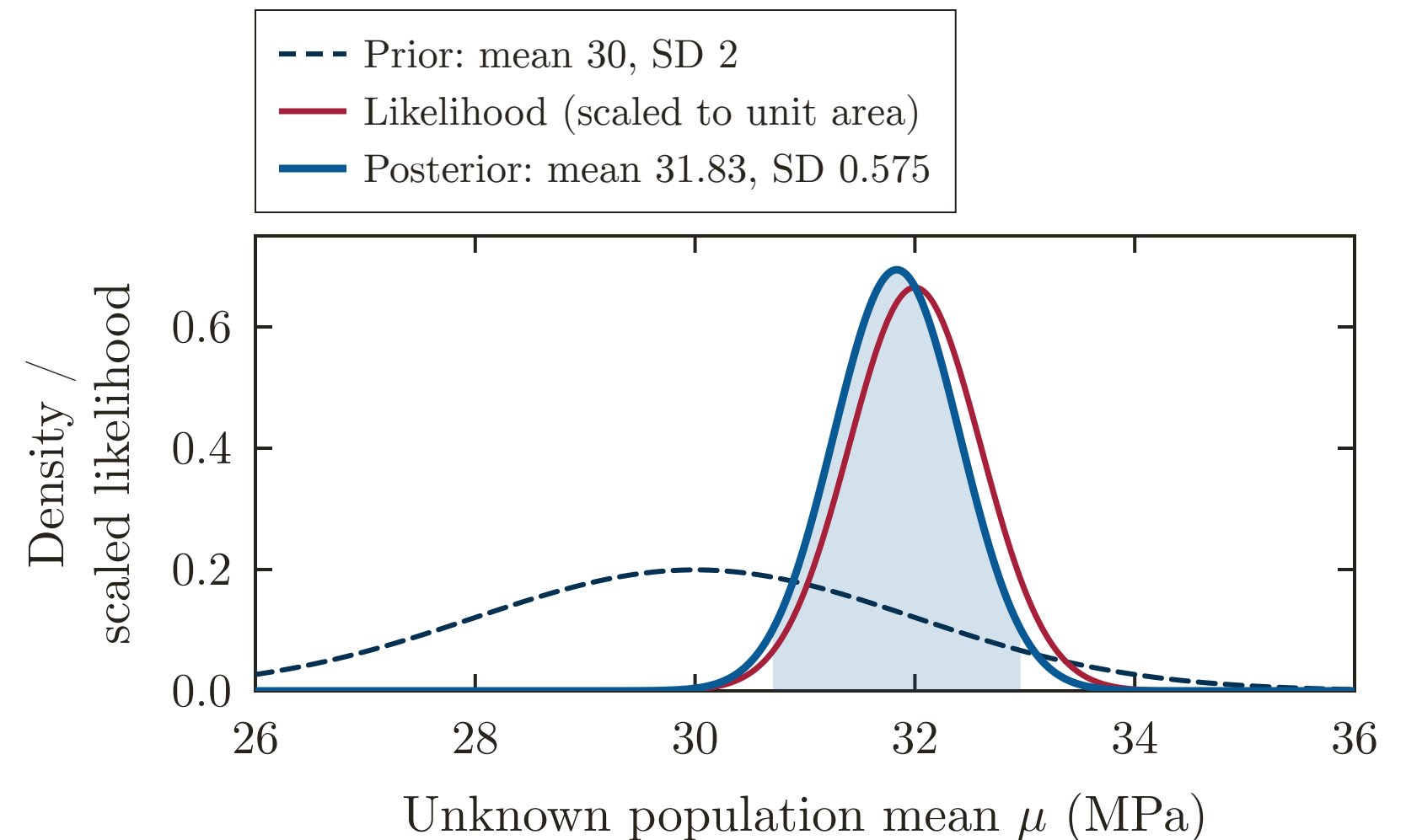
Data variance $v_D = 3^2/25 = 0.36$ MPa²

$v_p = 1/(1/4 + 1/0.36) = 0.3303$ MPa²

Posterior SD = $\sqrt{v_p} = 0.575$ MPa

$$\begin{aligned} m_p &= (30/4 + 32/0.36)/(1/4 + 1/0.36) \\ &= 0.0826 \times 30 + 0.9174 \times 32 \\ &= 31.83 \text{ MPa} \end{aligned}$$

About 8% prior weight and 92% data weight.



95% credible interval

$31.835 \pm 1.96(0.575) = [30.71, 32.96]$ MPa

95% posterior probability, given this model, prior and data.

The more precise source receives more weight. More tests increase the data's weight.

Similar intervals, different meanings

Same example: 25 tests, sample mean 32 MPa, known population SD 3 MPa.
Bayesian prior: mean 30 MPa, SD 2 MPa.

Question	Frequentist confidence interval	Bayesian credible interval
Numerical answer	[30.82, 33.18] MPa	[30.71, 32.96] MPa
What does 95% mean?	Across repeated samples, 95% of intervals built this way contain the fixed true mean.	Given these data, model and prior, 95% posterior probability lies within this interval.
Where is uncertainty represented?	In the sampling distribution of the estimator and the behavior of the procedure.	In the posterior distribution over the unknown mean.
Why are the numbers close?	The data locate the mean fairly precisely.	The data receive about 92% weight; the prior supplies about 8%.

Different target: one future cylinder

Individual strengths still have SD 3 MPa. A prediction interval must include that variability.

Neither interval describes the range containing 95% of individual cylinder strengths.

Hypothesis test: is an assumed value still plausible?

Same 25 tests: $\bar{x} = 32$ MPa, $\sigma = 3$ MPa. The supplier claims $\mu = 33$ MPa.

Would data like ours be unusual if the claim were true?

Four steps

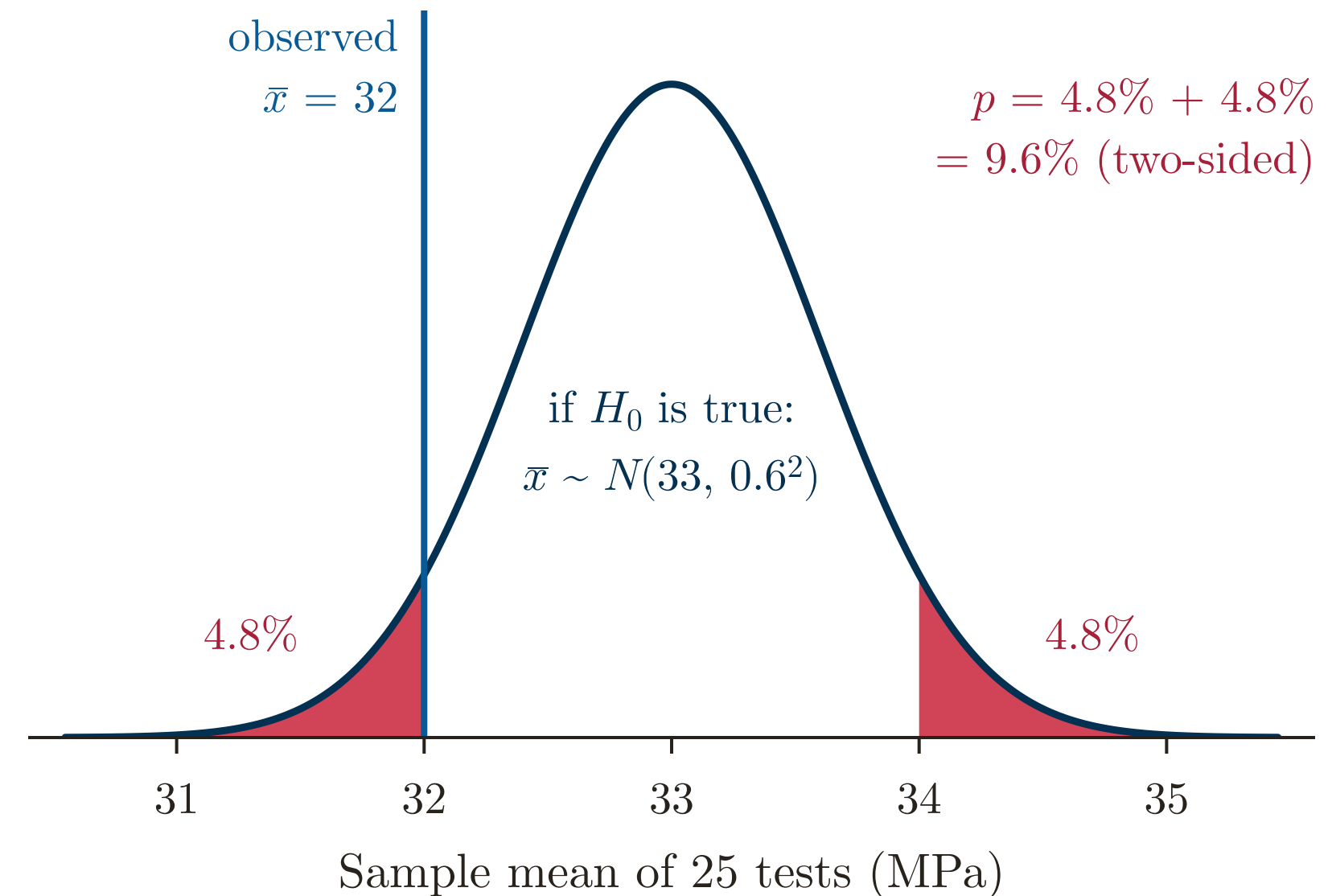
- 1 Claim: $H_0: \mu = 33$ MPa; $H_1: \mu \neq 33$ MPa
- 2 Test statistic: $z = (32 - 33)/0.6 = -1.67$
- 3 p-value: $P(|Z| \geq 1.67) = 9.6\%$
- 4 At $\alpha = 5\%$: $9.6\% > 5\%$, do not reject H_0

Read it carefully

Consistent with slide 17: the 95% CI $[30.82, 33.18]$ contains 33.

Not rejecting does not prove $\mu = 33$.

A p-value is not the probability that H_0 is true.



A test checks one assumed value; a confidence interval shows all values the data allow.

Regression: a response depends on predictors

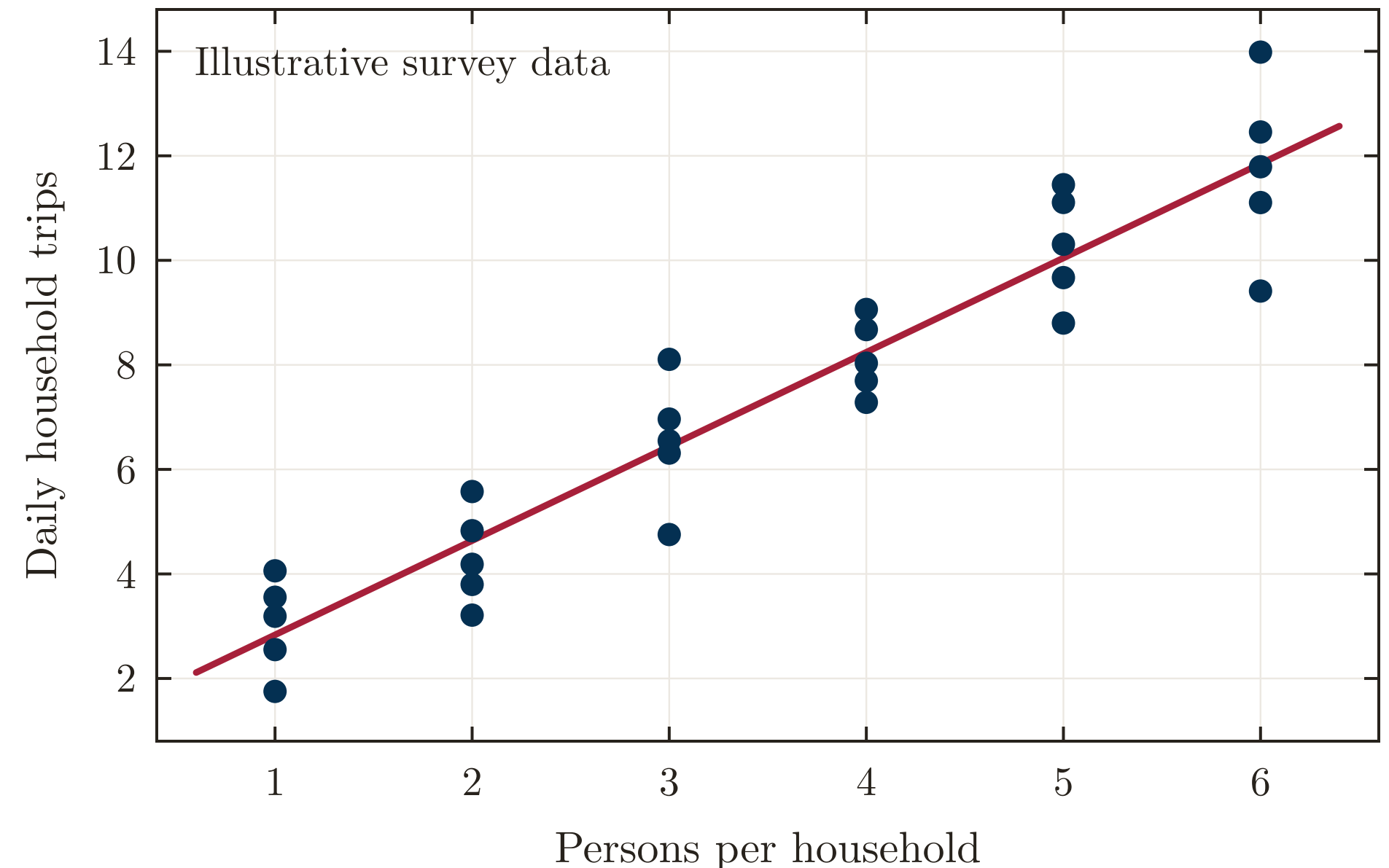
Transportation example: how do daily trips vary with household size and vehicle ownership?

Fit a relationship

Expected trips = intercept
+ effect of household size
+ effect of vehicle ownership

Describe the remaining scatter

Households with the same predictors
can still produce different trip counts.
That residual variation matters.



Regression models a conditional response. Correlation alone describes association.

Why regression is a probabilistic model

Regression does not just draw a line. It says that at each x , the response has a distribution, centered on the line.

The model

$$Y \mid x \sim N(\beta_0 + \beta_1 x, \sigma^2)$$

The line is the mean at each x ; σ is the scatter.

What that gives us

Least squares = maximum likelihood here

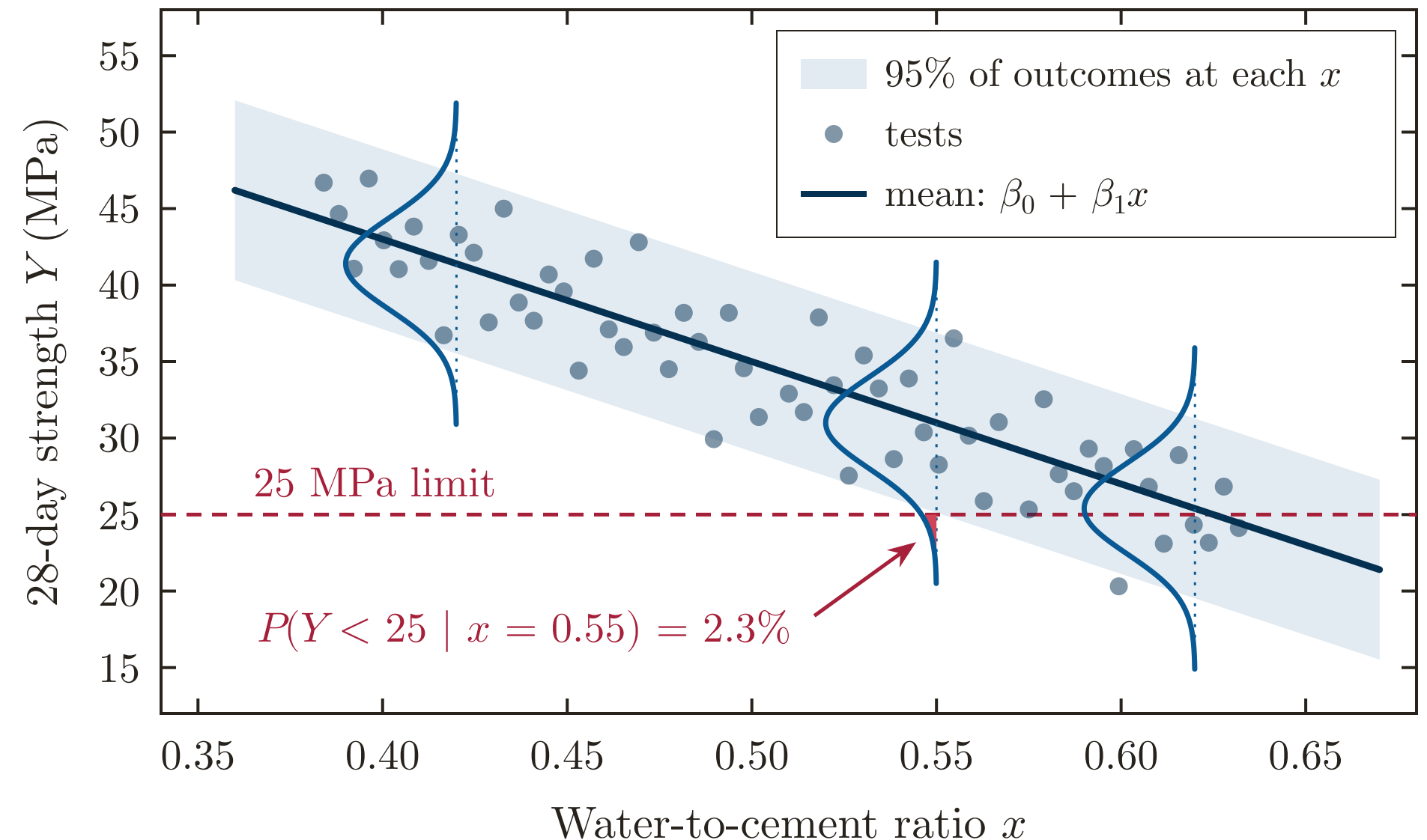
Prediction intervals for a new test (slide 24)

Probabilities: $P(Y < 25 \text{ MPa} \mid x = 0.55) = 2.3\%$

Same idea, other outcomes

Counts: Poisson regression.

Yes/no: logistic regression, fragility curves.



Illustrative data: $\beta_0 = 75 \text{ MPa}$, $\beta_1 = -80 \text{ MPa}$, $\sigma = 3 \text{ MPa}$

The line is the mean of a distribution; the scatter around it is part of the model.

Uncertainty about an average and a new outcome

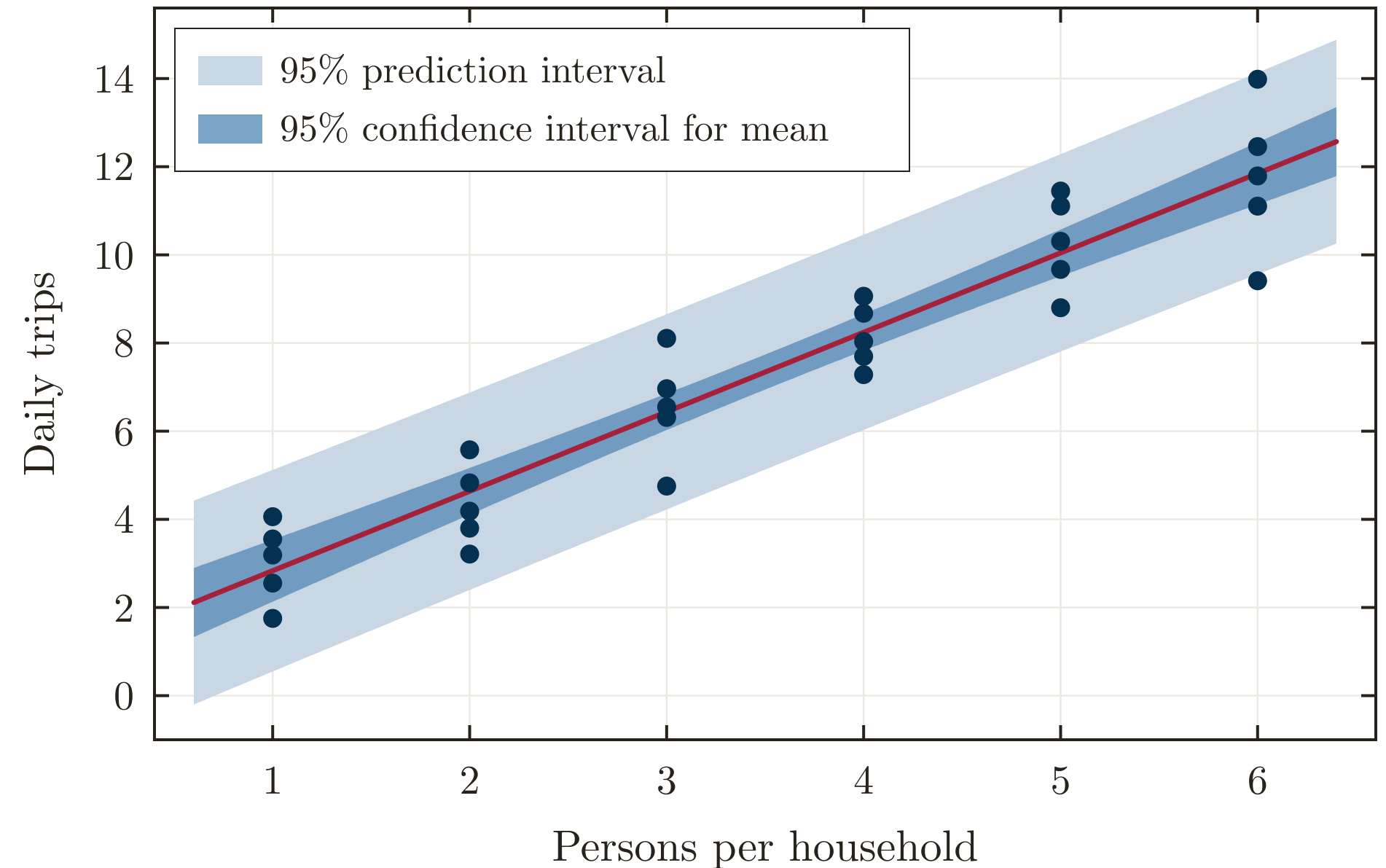
Both bands come from the same fitted regression, but they describe different unknowns.

Mean response

How many trips do households with these predictors make on average?
Use a confidence interval.

Next response

How many trips might one new household make?
Use a prediction interval.

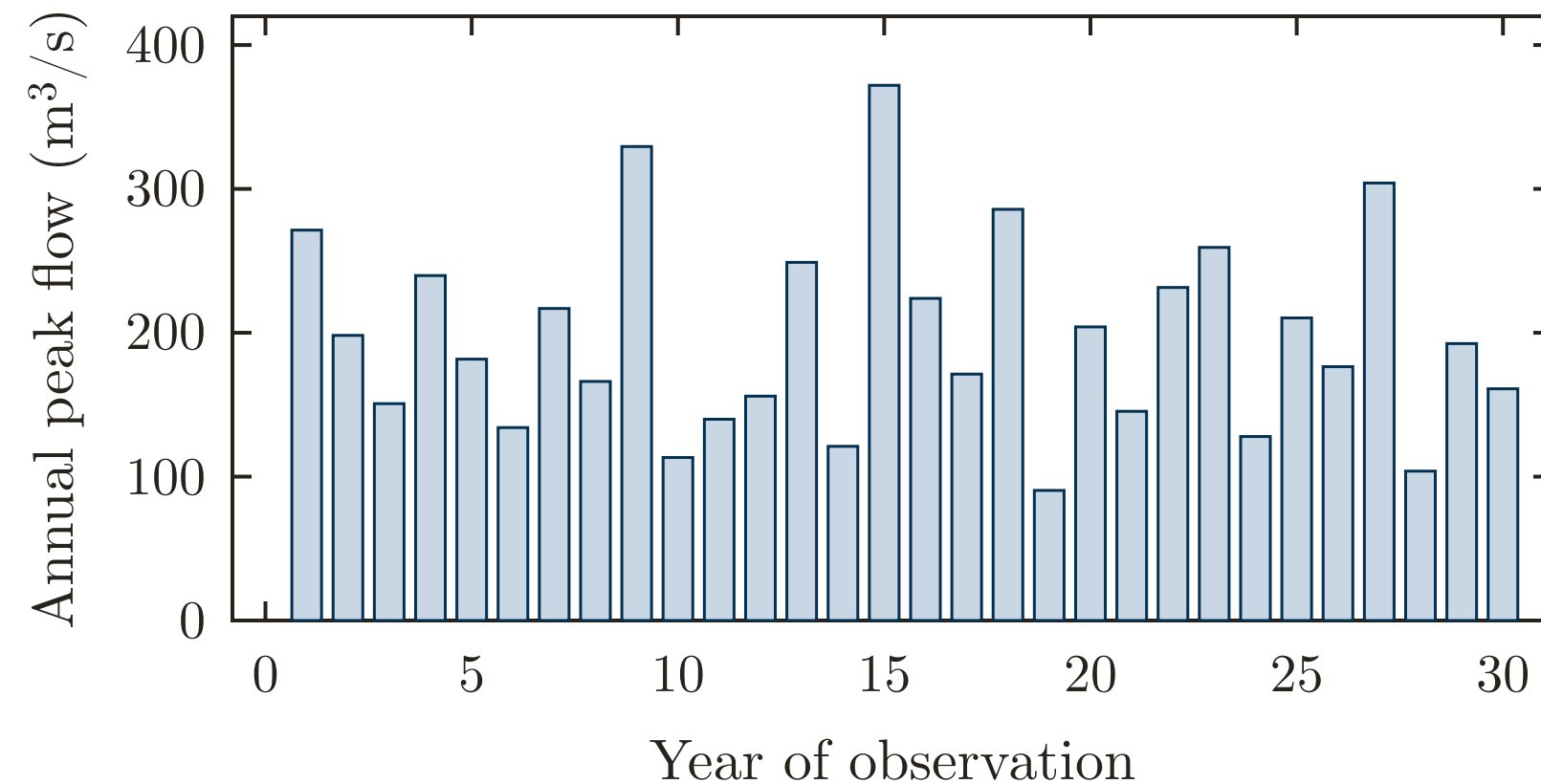


More data can narrow uncertainty about the mean while individual outcomes remain variable.

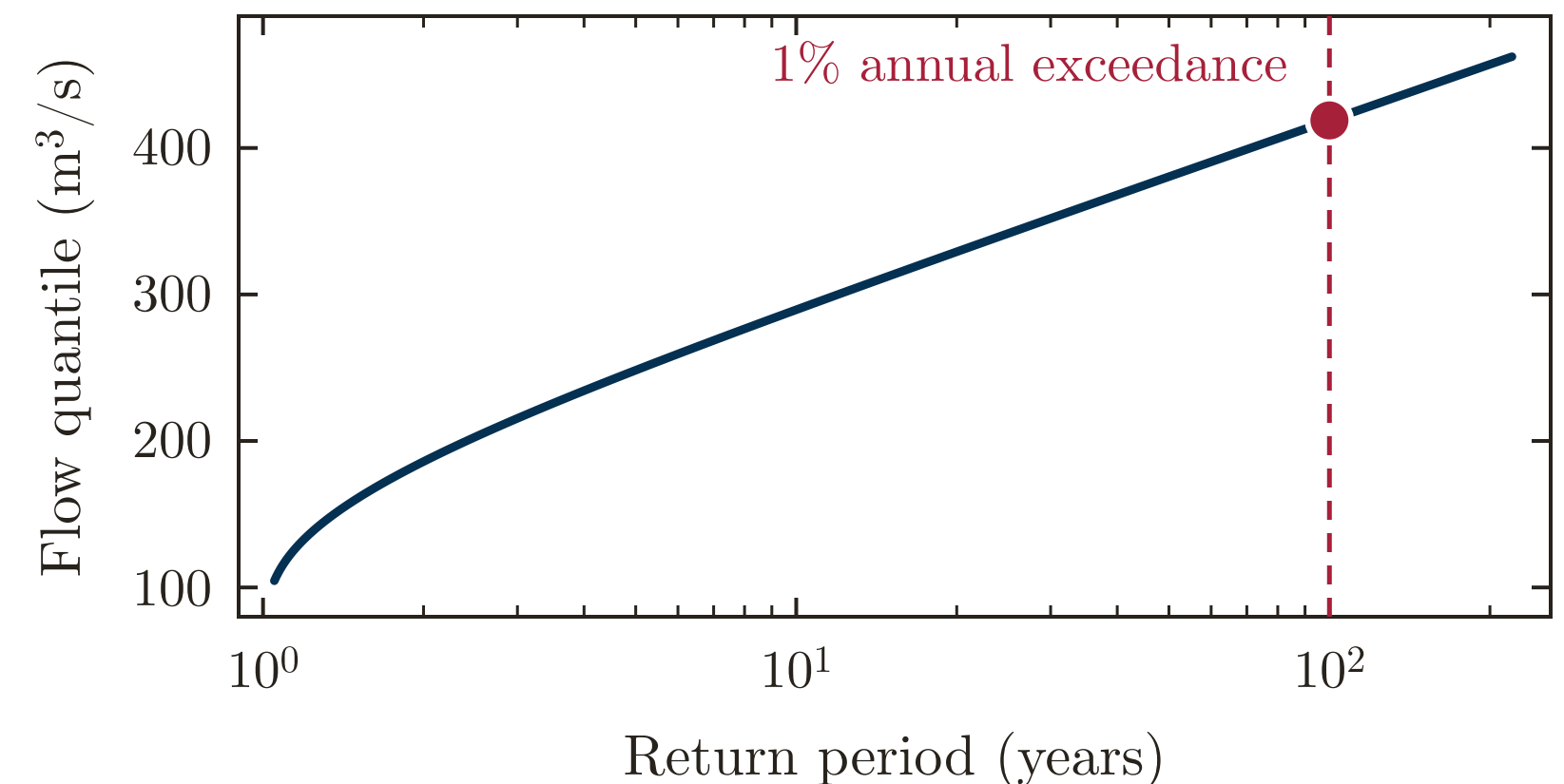
Flood frequency: a fitted model supports design

Statistics estimates a flow model. Probability turns that model into exceedance probabilities and quantiles.

Observed variability



Model-based extrapolation



DATA

Annual peak flows in m³/s

MODEL

Fit a tail and assess its uncertainty

DECISION

Choose a capacity and acceptable risk

A “100-year flood” has 1% annual exceedance probability under the assumed stationary model.

Uncertainty propagation: what goes in, what comes out

A deterministic equation gives one output per input combination. Uncertain inputs create a distribution of outputs.

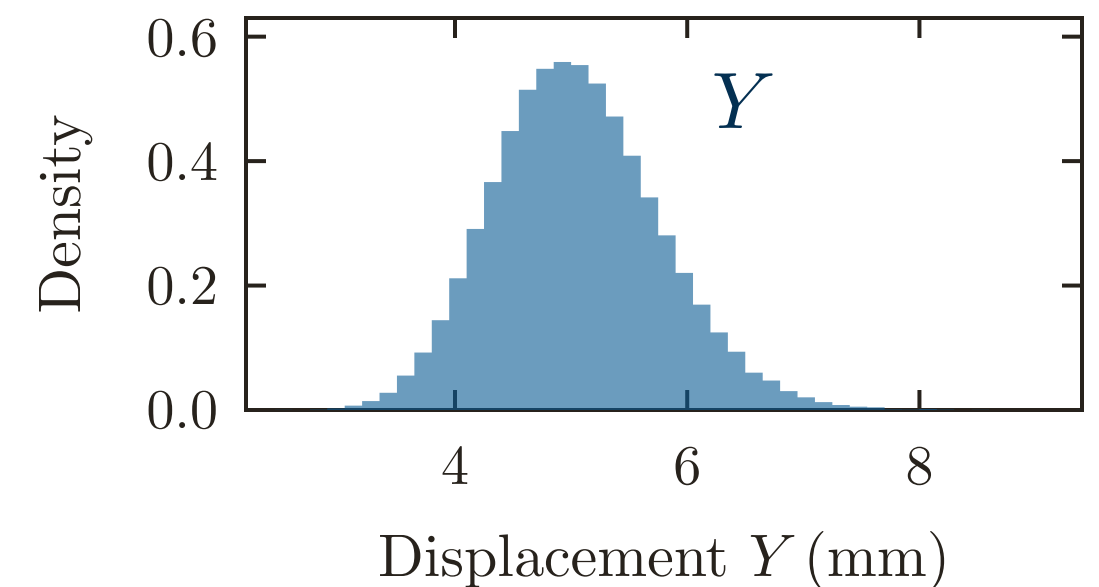
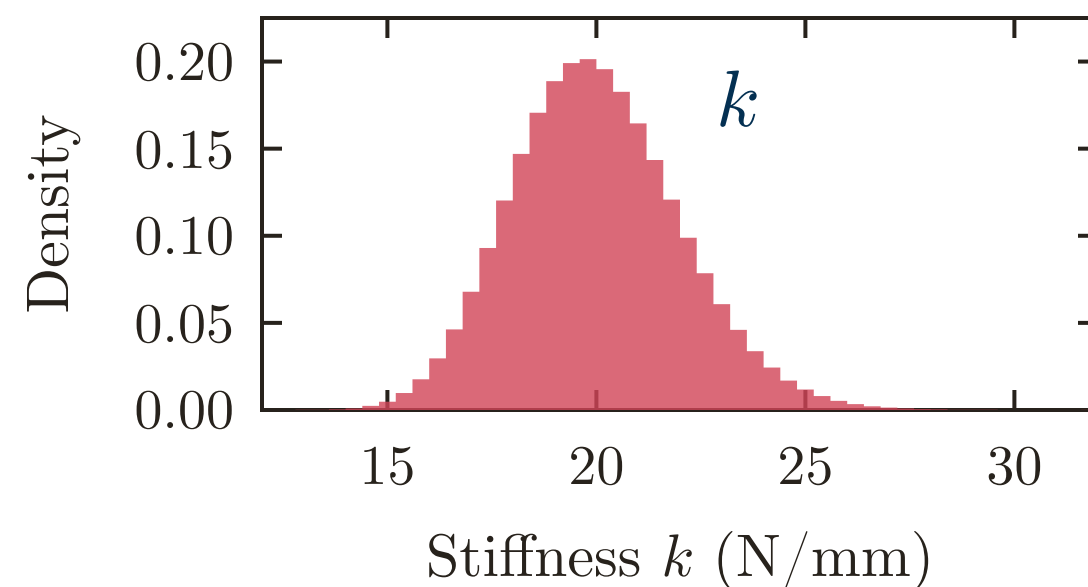
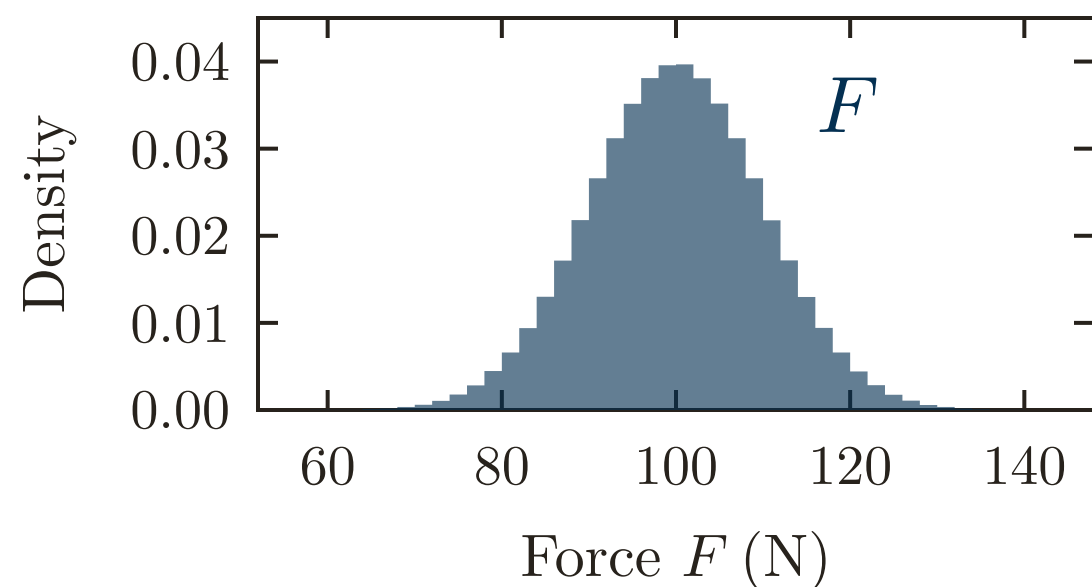
X_i = uncertain input i $\mu_i = E[X_i]$, its mean σ_i = its standard deviation

$Y = f(X_1, \dots, X_n)$: uncertain output f : the known calculation or physical model

A simple example with a simple equation: spring displacement $Y = F/k$

X_1 = force F : mean 100 N, SD 10 N X_2 = stiffness k : mean 20 N/mm, SD: 2 N/mm

Assume normal F , positive lognormal k , and independence. At the mean inputs, $Y = 5$ mm.



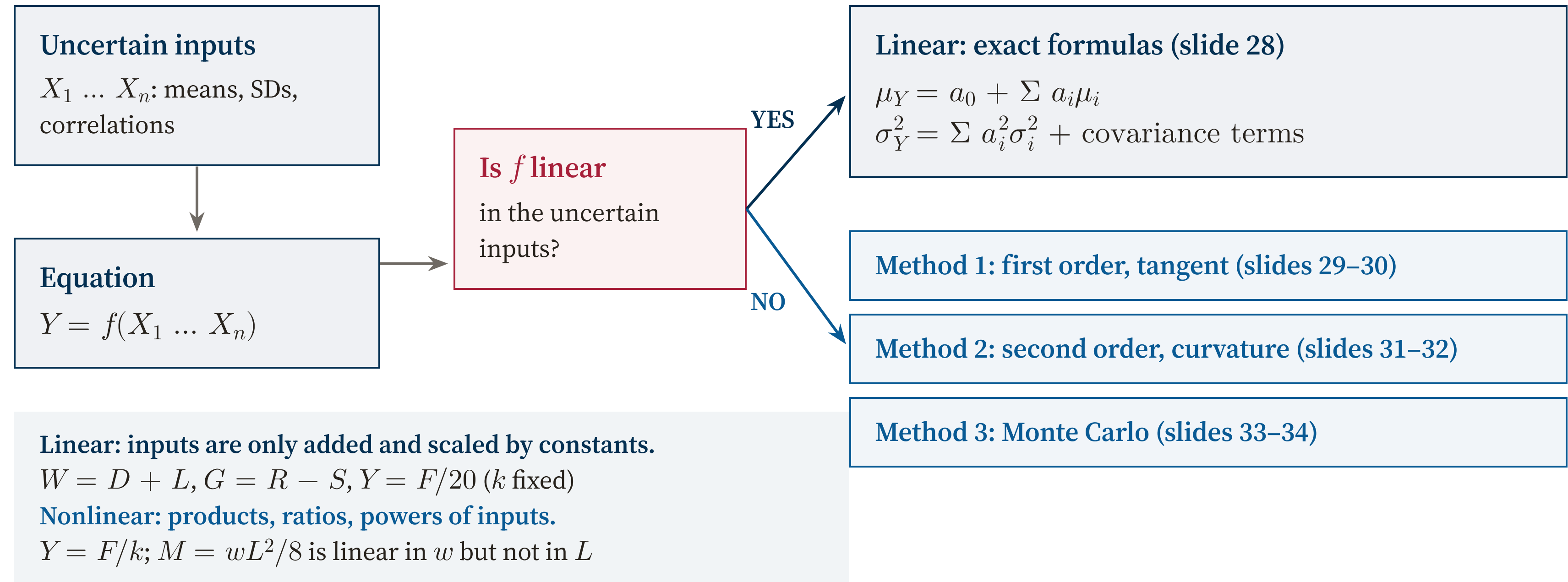
We know how to calculate Y .

Propagation tells us how uncertain Y is; the method depends on whether f is linear (next slide).

Propagation roadmap: is f linear in the uncertain inputs?

This one question decides the method. Linear: exact formulas.

Nonlinear: approximate (first or second order) or simulate (Monte Carlo).



First check linearity, then choose the method. The rest of this section follows this map.

Linear f : mean and variance are exact

If Y is a weighted sum of the inputs, the mean and standard deviation follow exactly. No approximation is needed.

Linear model: exact moments

$$Y = a_0 + a_1 X_1 + a_2 X_2$$

$$\mu_Y = a_0 + a_1 \mu_1 + a_2 \mu_2$$

$$\sigma_Y^2 = a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + 2a_1 a_2 \text{Cov}(X_1, X_2)$$

Load: $W = D + L$; $D 400 \pm 20$, $L 150 \pm 40$ kN

$$\mu_W = 550 \text{ kN}; \quad \sigma_W = \sqrt{(20^2 + 40^2)} = 44.7 \text{ kN}$$

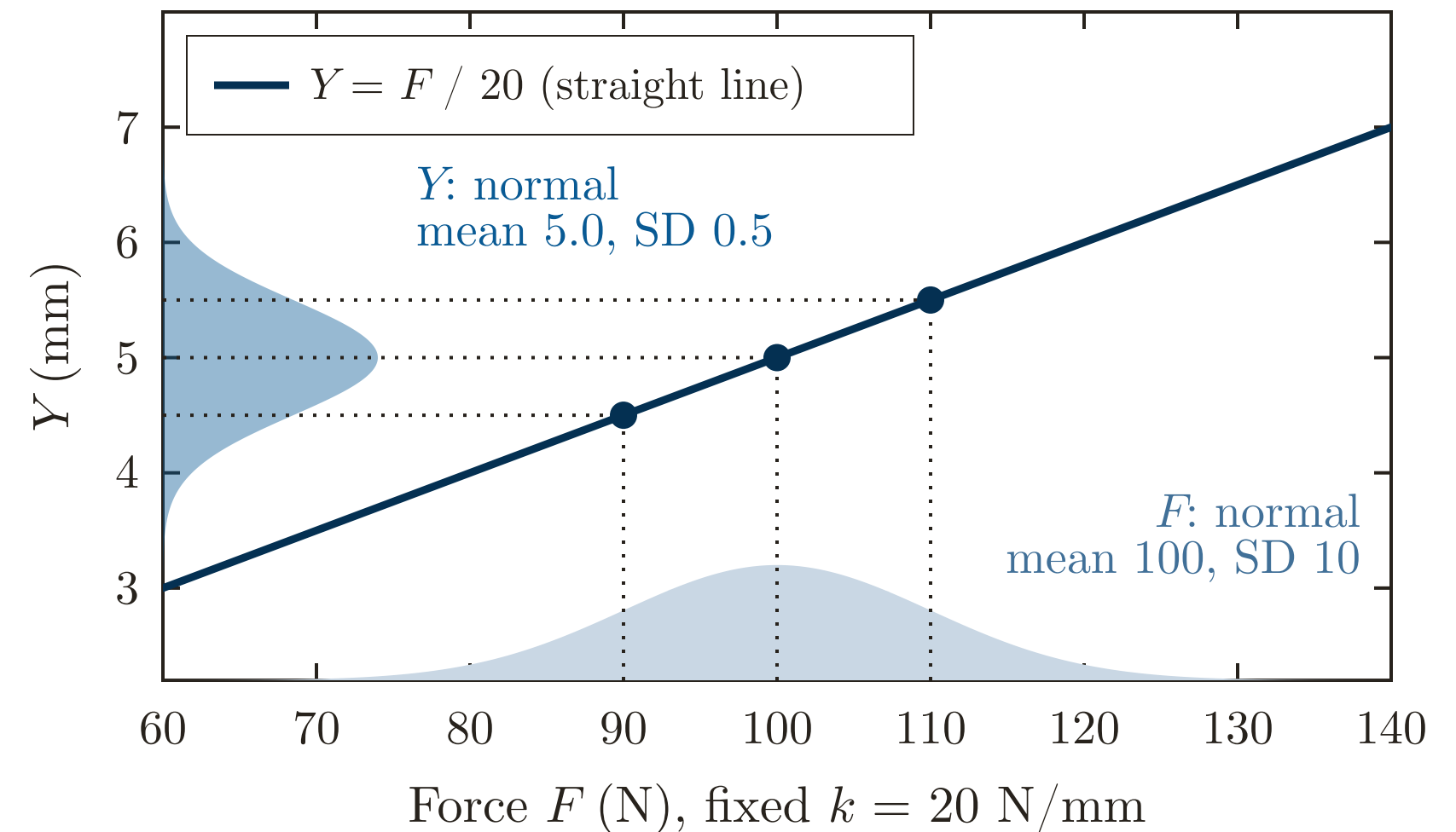
Spring with k fixed at 20 N/mm

$$Y = F/20 = 0.05 F$$

$$\mu_Y = 5.00 \text{ mm}; \quad \sigma_Y = 0.50 \text{ mm}$$

Equal steps in F give equal steps in Y :

$F = 90, 100, 110$ N gives $Y = 4.5, 5.0, 5.5$ mm.



Normal in, normal out:
a linear model keeps the shape of
the input distribution. $\pm$ denotes one SD.

Linear: plug in the means for the mean; scale and add the variances for the spread. Exact.

Nonlinear f , method 1: first order (tangent line)

For a curved f , replace it by its tangent at the input means, then apply the linear formulas.
If f is already linear, this is exact.

1 Approximate the function near the input means

$$\delta_i = X_i - \mu_i \quad c_i = \partial f / \partial X_i \text{ at } X = \mu \quad Y \approx f(\mu) + \sum_i c_i \delta_i$$

δ_i is a deviation; c_i is the local slope. $f(\mu)$ means evaluate f with every input set to its mean.

2 Take the expectation

Each deviation has mean zero: $E[\delta_i] = 0$.

Therefore $\mu_Y = E[Y] \approx f(\mu)$.

For a linear model, the tangent is the model.
For a curved model, this mean can be biased.

3 Square the deviations and average

$$\begin{aligned} \text{Var}(Y) &\approx E[(\sum_i c_i \delta_i)^2] \\ &= \sum_i c_i^2 \sigma_i^2 + 2 \sum_{i < j} c_i c_j \text{Cov}(X_i, X_j) \end{aligned}$$

Squared terms give the individual variances.
Cross terms describe joint movement.
For independent inputs, cross terms vanish.

Sensitivity coefficients appear because they are the slopes in a Taylor expansion.

Nonlinear f : why the tangent misses the mean

First order plugs in the mean stiffness and gets $Y = 5.00$ mm.

A three-point check, with no calculus, shows why the true mean is a little higher.

$F = 100$ N, three stiffness values

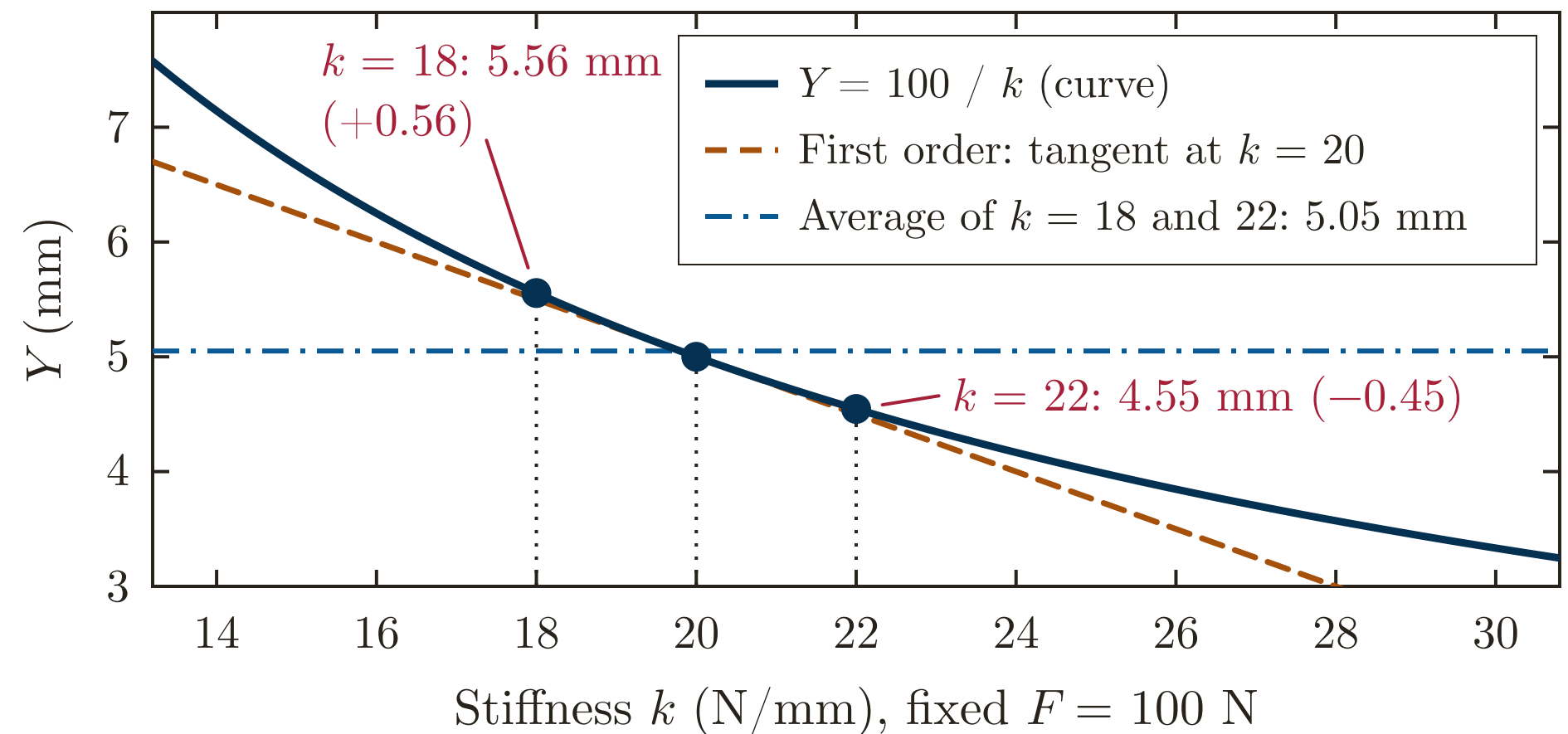
$k = 18$ N/mm (softer): $Y = 5.56$ mm, $+0.56$

$k = 20$ N/mm (mean): $Y = 5.00$ mm

$k = 22$ N/mm (stiffer): $Y = 4.55$ mm, -0.45

Average of the 18 and 22 cases: 5.05 mm

Softer springs stretch extra.



Compare: Y is a straight line in F

$F = 90$ or 110 N gives $Y = 4.5$ or 5.5 mm

Average 5.00 mm: no shift

The tangent (orange) cannot see the bend,
so first order misses this 0.05 mm.

Next: second order adds the curvature

$$\begin{aligned} \frac{1}{2} \times (\partial^2 Y / \partial k^2) \times \sigma_k^2 &= \frac{1}{2} \times 0.025 \times 2^2 \\ &= 0.05 \text{ mm, so } \mu_Y \approx 5.05 \text{ mm} \end{aligned}$$

Curved model: the average output is not the output at the average input.

Nonlinear f , method 2: second order (add curvature)

“Order” is the highest power of the input deviation kept. First- and second-order (Taylor) approximations are the standard tools in textbooks and research papers.

One input: $\delta = X - \mu$, local slope $a = f'(\mu)$, curvature $b = f''(\mu)$

First order: $Y \approx f(\mu) + a\delta$ Second order: $Y \approx f(\mu) + a\delta + \frac{1}{2}b\delta^2$

Why curvature matters

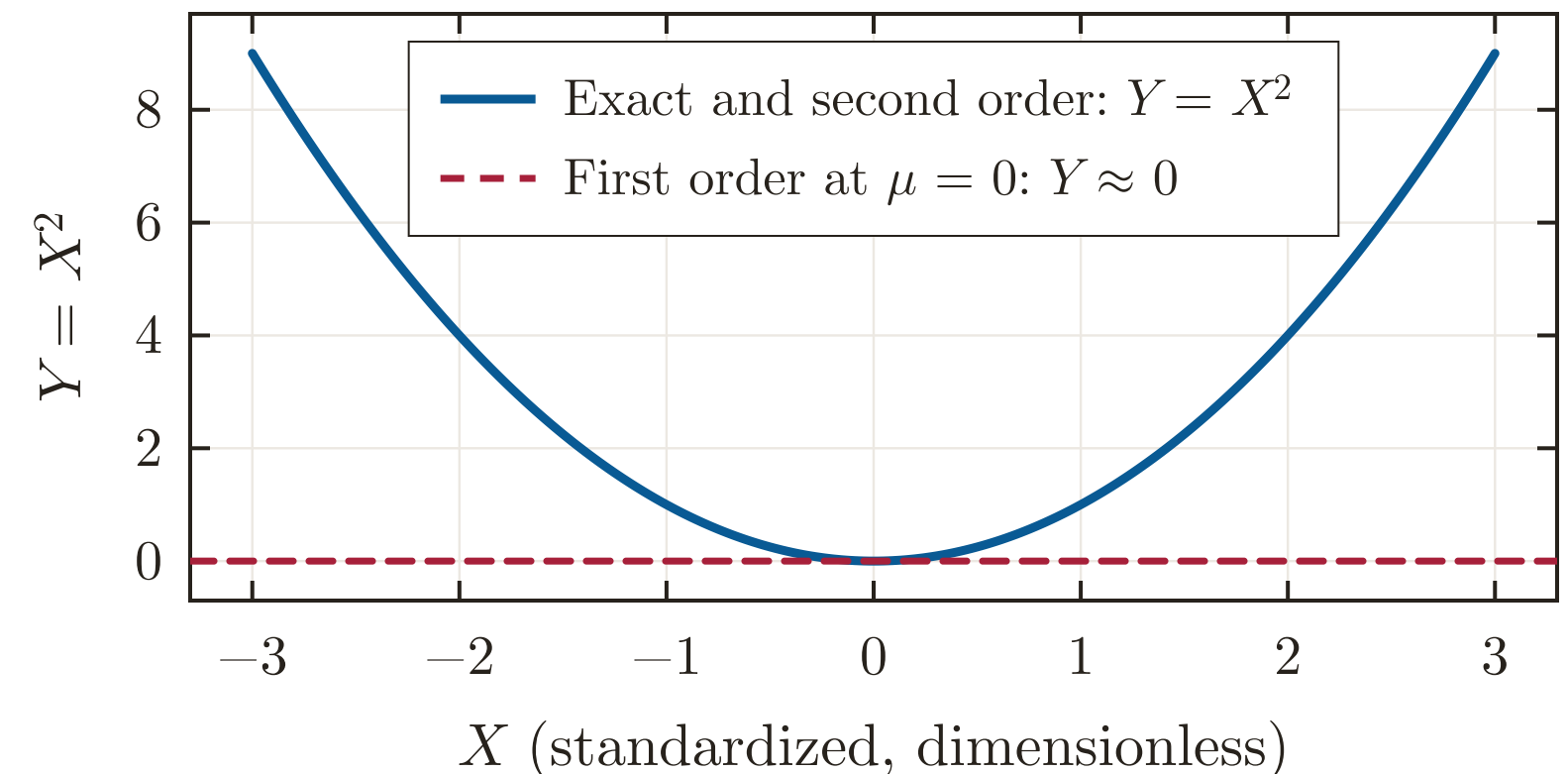
$E[\delta] = 0$, but $E[\delta^2] = \sigma^2$.

So $E[Y] \approx f(\mu) + \frac{1}{2}b\sigma^2$.

For a normal input, the quadratic model gives:

$\text{Var}(Y) \approx a^2\sigma^2 + \frac{1}{2}b^2\sigma^4$.

For nonnormal inputs, skewness and the fourth central moment also enter.



$X \sim N(0, 1)$, $Y = X^2$: first order predicts zero.

Second order is exact: mean 1, variance 2.

A zero slope does not imply zero uncertainty when the response is curved.

Nonlinear f : spring example, first vs. second order

Y is a straight line in F but a curve in k . The curve is what shifts the mean.

First order (tangent line at the means)

$$c_F = \partial Y / \partial F = 1/20 = 0.05 \text{ mm/N}$$

$$c_k = \partial Y / \partial k = -100/20^2 = -0.25 \text{ mm}^2/\text{N}$$

$$\mu_Y \approx 5.00 \text{ mm}$$

$$\sigma_Y^2 \approx (0.05 \times 10)^2 + (-0.25 \times 2)^2$$

$$\sigma_Y \approx 0.707 \text{ mm}$$

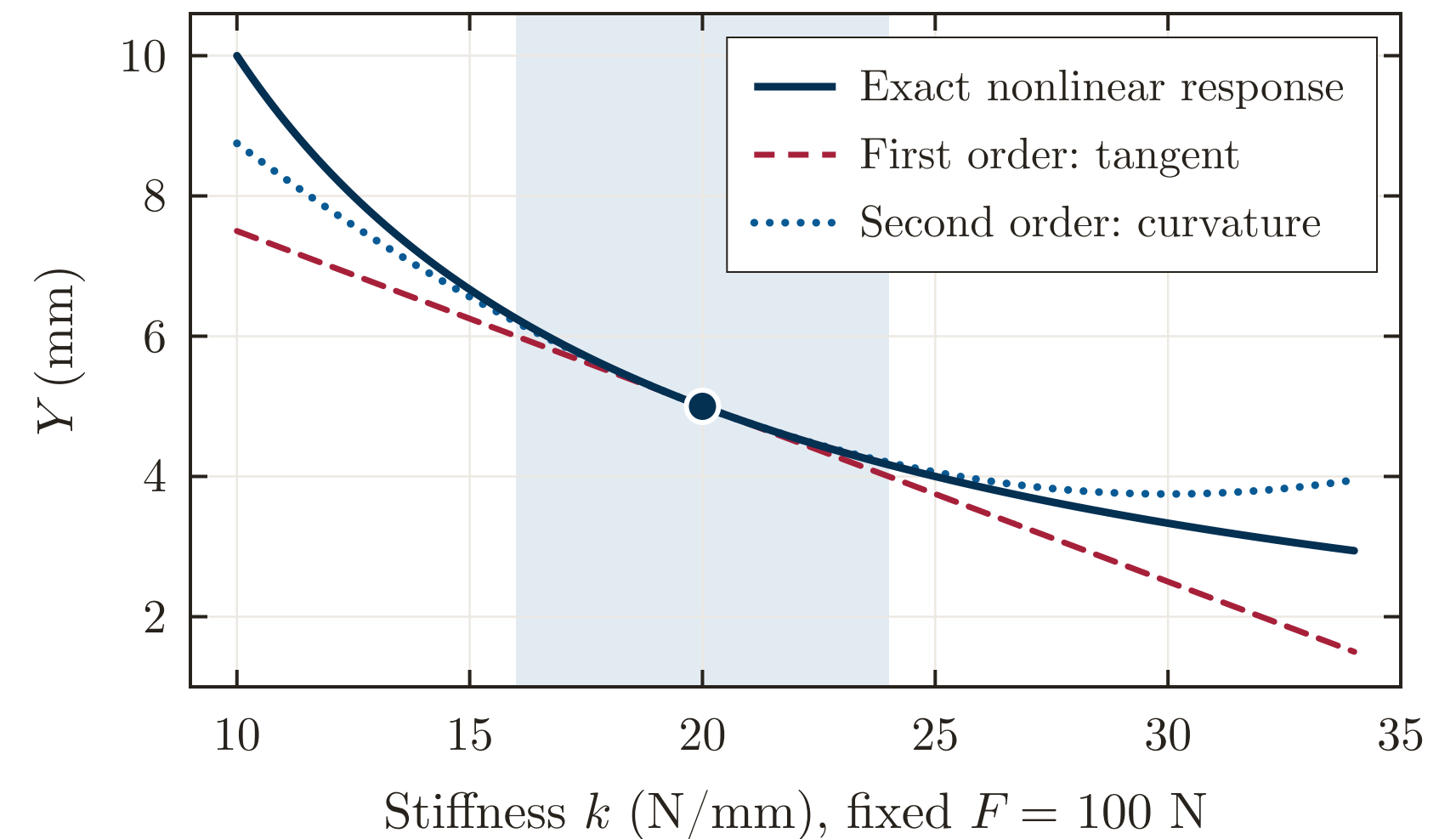
Second order (tangent plus curvature)

$$\partial^2 f / \partial k^2 = 2F/k^3 = 0.025 \text{ at the means}$$

$$\mu_Y \approx 5 + \frac{1}{2}(0.025)(2^2) = 5.05 \text{ mm}$$

Only the stiffness-curvature term survives here:

$$f_{FF} = 0 \text{ and } \text{Cov}(F, k) = 0.$$



Exact model moments for these inputs:

Mean = 5.050 mm; SD = 0.716 mm

Monte Carlo (slides 33–34) agrees.

Curvature shifts the mean displacement from 5.00 mm to 5.05 mm.

Nonlinear f , method 3: Monte Carlo simulation

Evaluate the original model many times with random inputs; the outputs approximate the distribution of Y .

- Works for linear and nonlinear models; no derivatives needed.
- Most useful when the model is complex, curvature is strong, or tail probabilities matter.
- Here it confirms the second-order mean of about 5.05 mm.

Four repeatable steps

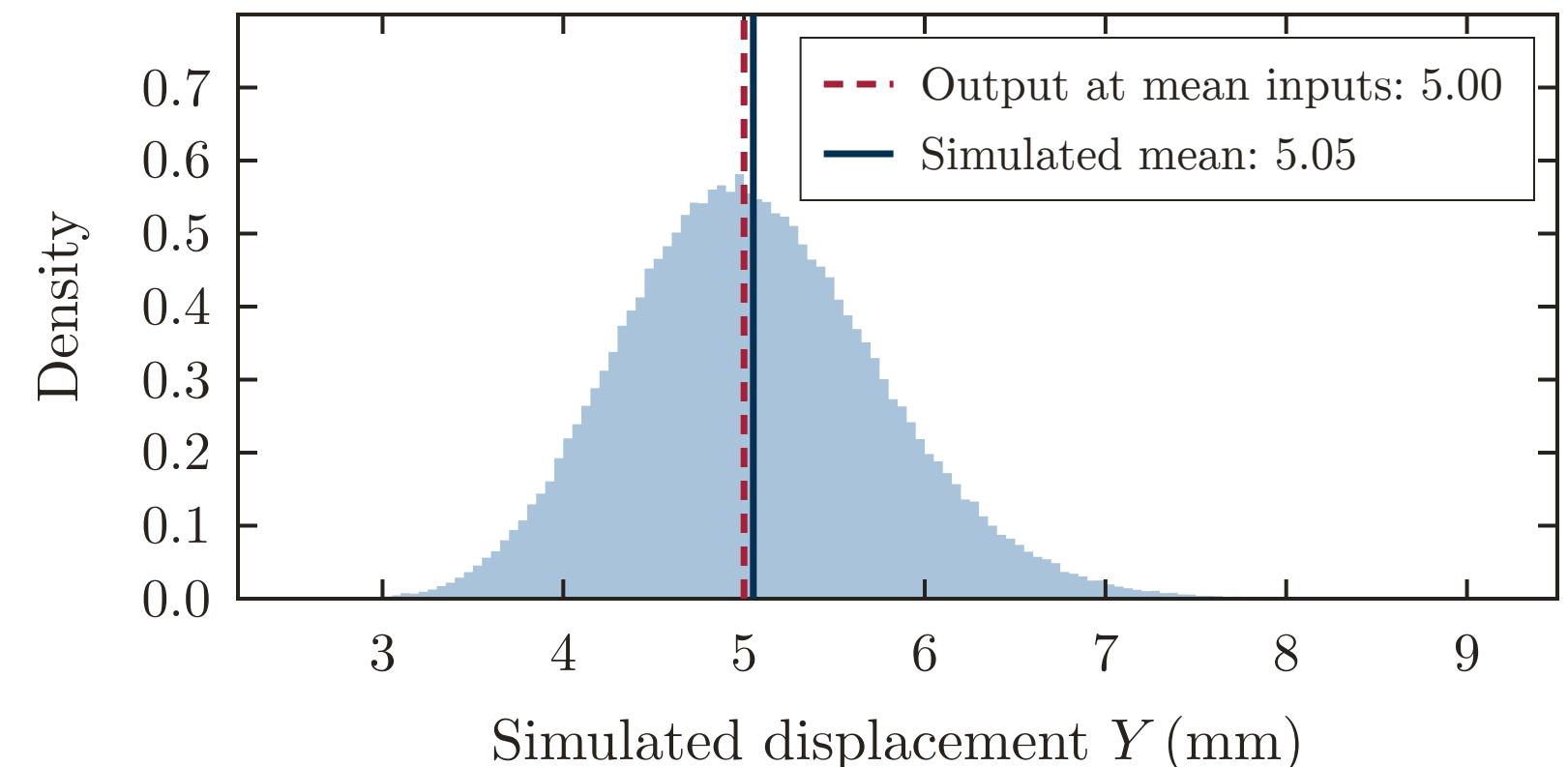
- 1 Draw F and k from their input distributions.
- 2 Compute $y = F/k$ for that pair.
- 3 Repeat N times to obtain $y_1, \dots, y_N$.
- 4 Plot a histogram; calculate mean, SD, quantiles or a threshold probability.

Illustrative draws (F , k) and outputs:

(100 N, 20 N/mm) gives 5.00 mm

(110 N, 18 N/mm) gives 6.11 mm

(90 N, 22 N/mm) gives 4.09 mm



200,000 simulations: mean $\approx$ 5.052 mm

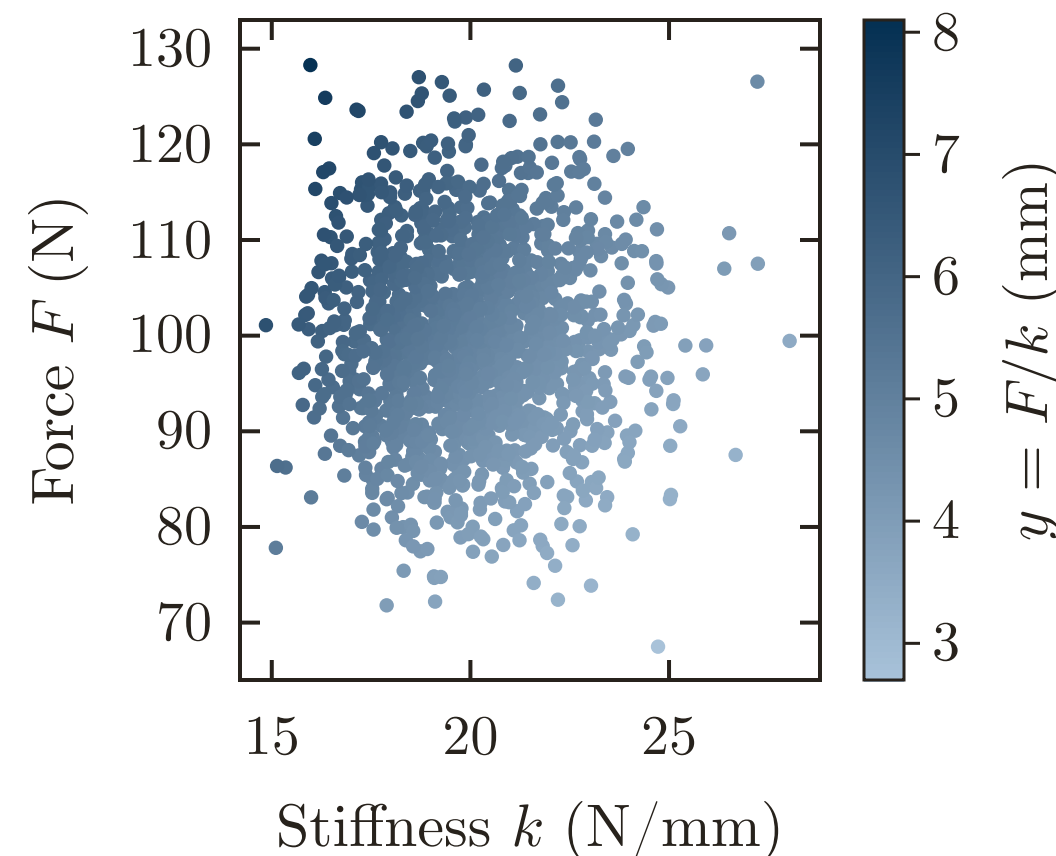
Output SD $\approx$ 0.717 mm

Monte Carlo uses the original model; input assumptions and simulation error still matter.

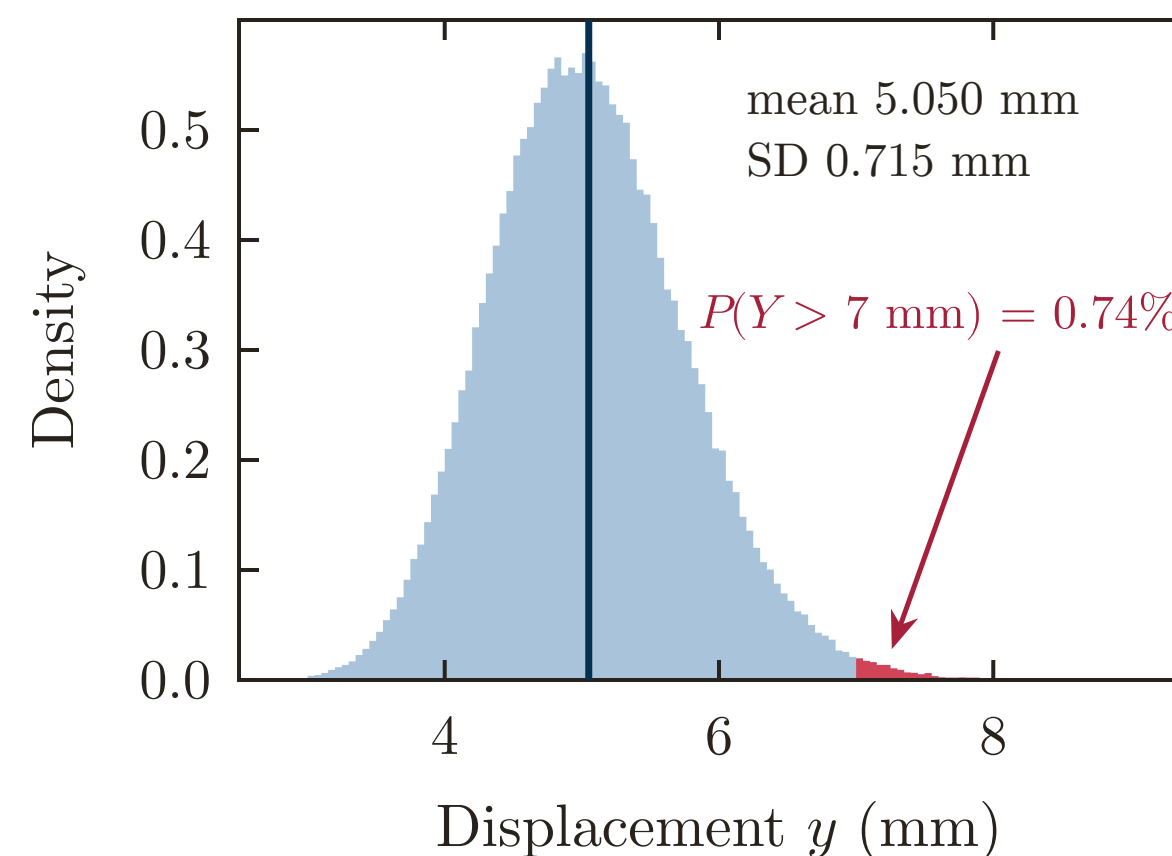
Nonlinear f , method 3: Monte Carlo in pictures

Sample the inputs, run the original model for every sample, then read everything from the collected outputs, including the tail.

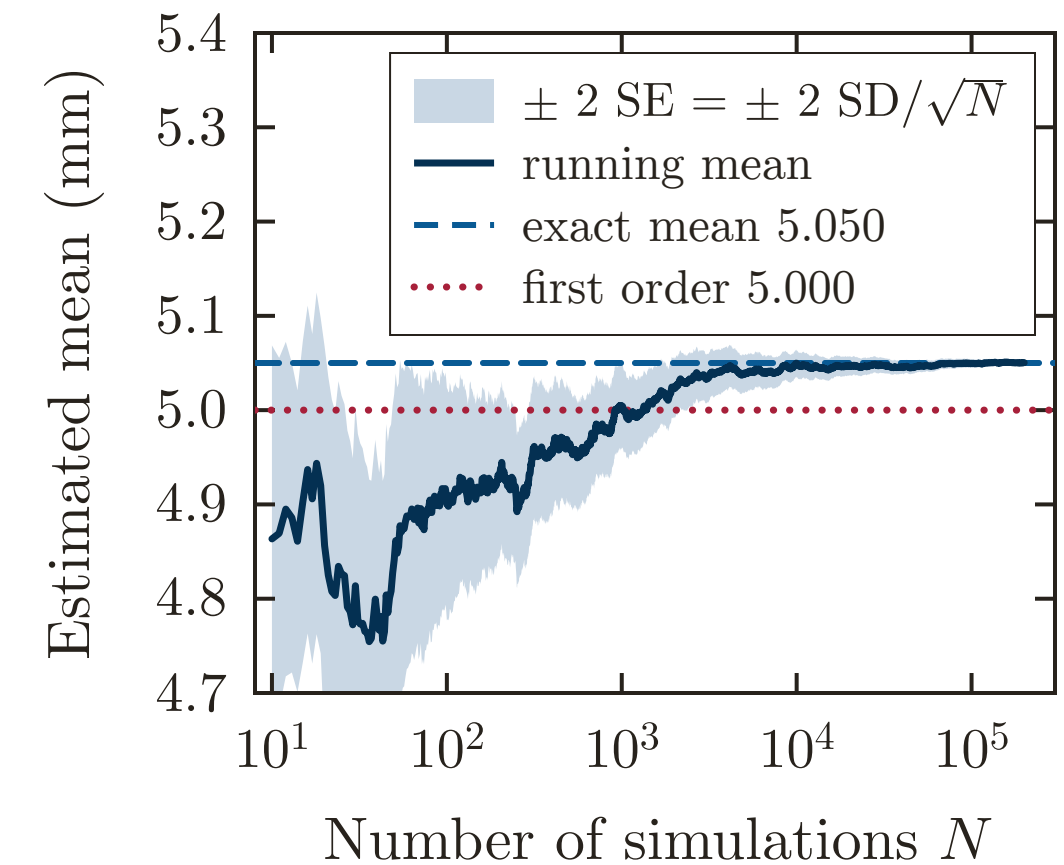
1 Draw input pairs (F, k)



2 Compute y for each pair, collect



3 More samples, smaller error



Tails: first order assumes a normal Y and gives $P(Y > 7 \text{ mm}) \approx 0.23\%$. Simulation gives 0.74%, about 3 times more.

Accuracy: the error of the simulated mean shrinks like $\text{SD}/\sqrt{N}$. With $N = 200,000$ it is about 0.0016 mm.

Monte Carlo needs no derivatives and captures skew and tails; its price is many model runs.

Choosing a propagation method: linear or nonlinear?

First ask: is f linear? Linear models have exact moments. Nonlinear models need a tangent, a curvature correction, or a simulation.

Method	What it does	Useful when	What to check
Exact linear moments	For $Y = b + \sum a_i X_i$, calculate mean and variance exactly.	The model is linear over the input range.	Moments alone do not determine every output distribution.
First-order Taylor	Replace a curved response by its local tangent.	Uncertainty is narrow; curvature is small.	Compare the function with its tangent over plausible inputs.
Second-order Taylor	Keep quadratic terms to represent curvature.	Smooth, moderate curvature affects mean or spread.	It is still local; neglected higher terms may matter.
Monte Carlo	Sample inputs; evaluate the original model many times.	Complex response, skewed outputs or quantiles matter.	Dependence, convergence, computing cost and rare events.

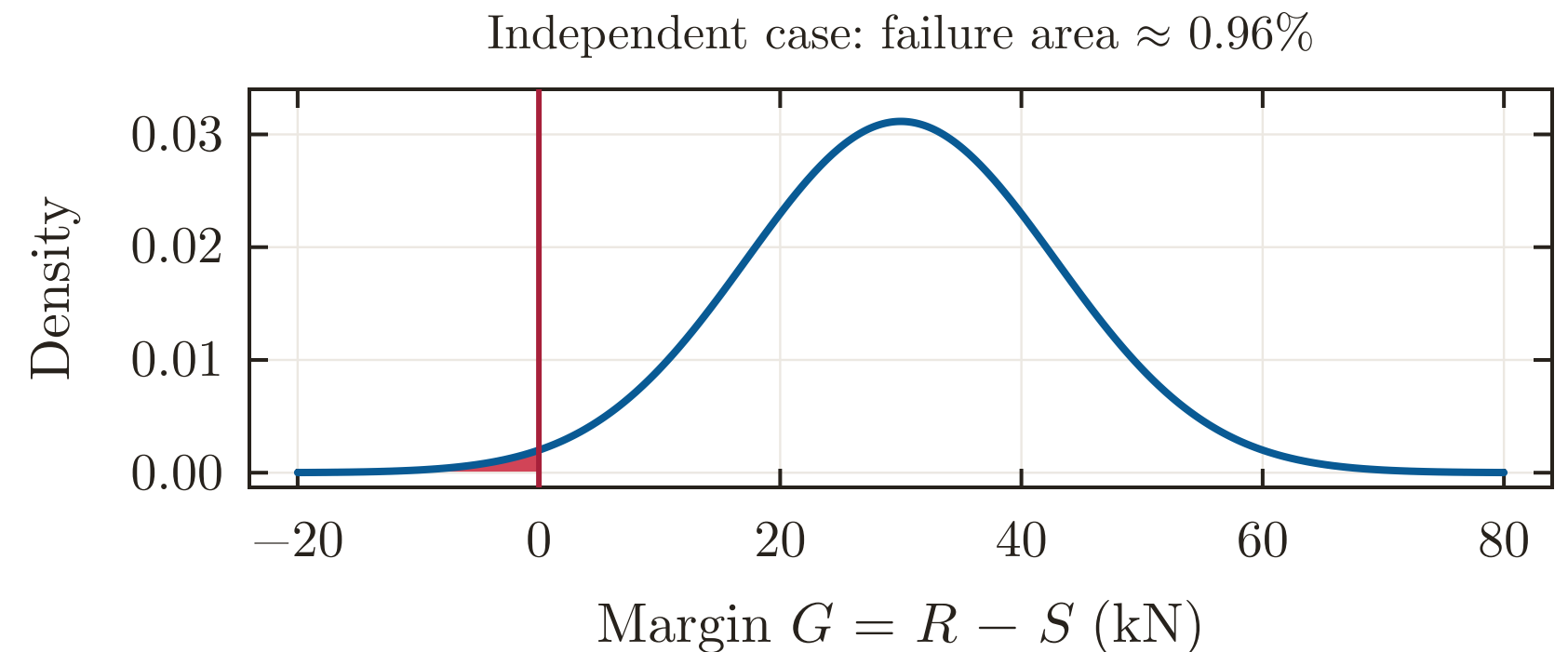
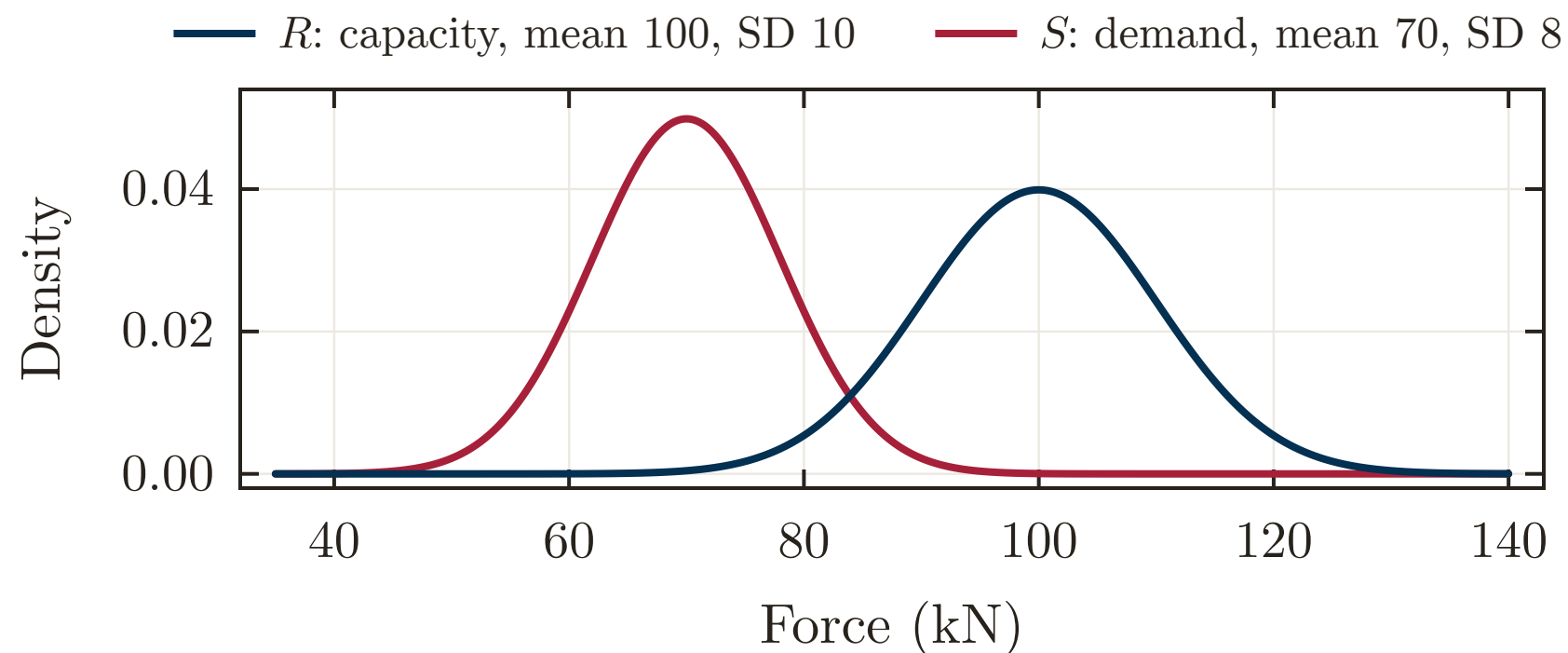
Monte Carlo is useful for nonlinear propagation, but nonlinearity does not automatically require simulation.

Capacity, demand and the safety margin

R = resistance or capacity. S = applied load effect or demand. $G = R - S$ is the remaining safety margin.

Model: $R \sim N(100, 10^2)$ kN and $S \sim N(70, 8^2)$ kN; assume independence for this first case.

Example realization: $R = 95$ kN, $S = 80$ kN gives $G = 15$ kN, a positive margin.



Derivation: $\Delta R = R - \mu_R$ and $\Delta S = S - \mu_S$

Mean: $E[R - S] = \mu_R - \mu_S = 30$ kN

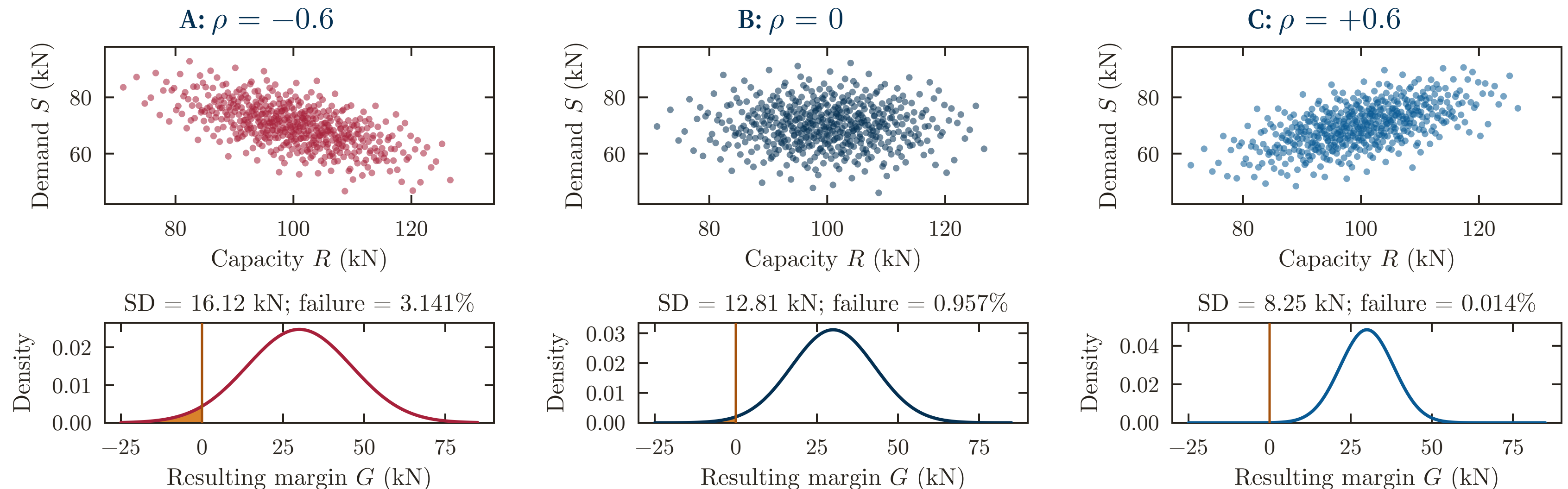
Variance: $E[(\Delta R - \Delta S)^2] = \sigma_R^2 + \sigma_S^2 - 2\text{Cov}(R, S) = 164$ kN²

Failure is the joint event $S > R$, equivalently $G < 0$. Marginal curve overlap is not the failure probability.

Three correlation scenarios with the same marginals

Jointly normal R and S in each case: capacity mean 100, SD 10 kN; demand mean 70, SD 8 kN. Only correlation changes.

$$\rho = \text{Cov}(R, S) / (\sigma_R \sigma_S) \quad \mu_G = 30 \text{ kN} \quad \sigma_G^2 = 10^2 + 8^2 - 2\rho(10)(8) = 164 - 160\rho$$



Top: paired capacity and demand values. Bottom: resulting G distribution; orange tail is $P(G < 0)$.

Positive correlation narrows this difference because R and S tend to rise or fall together.

Reliability: probability of an unwanted event

Return to independent normal R and S . FOSM means first-order, second-moment propagation.

Propagate input uncertainty

Capacity: 100 ± 10 kN

Demand: 70 ± 8 kN

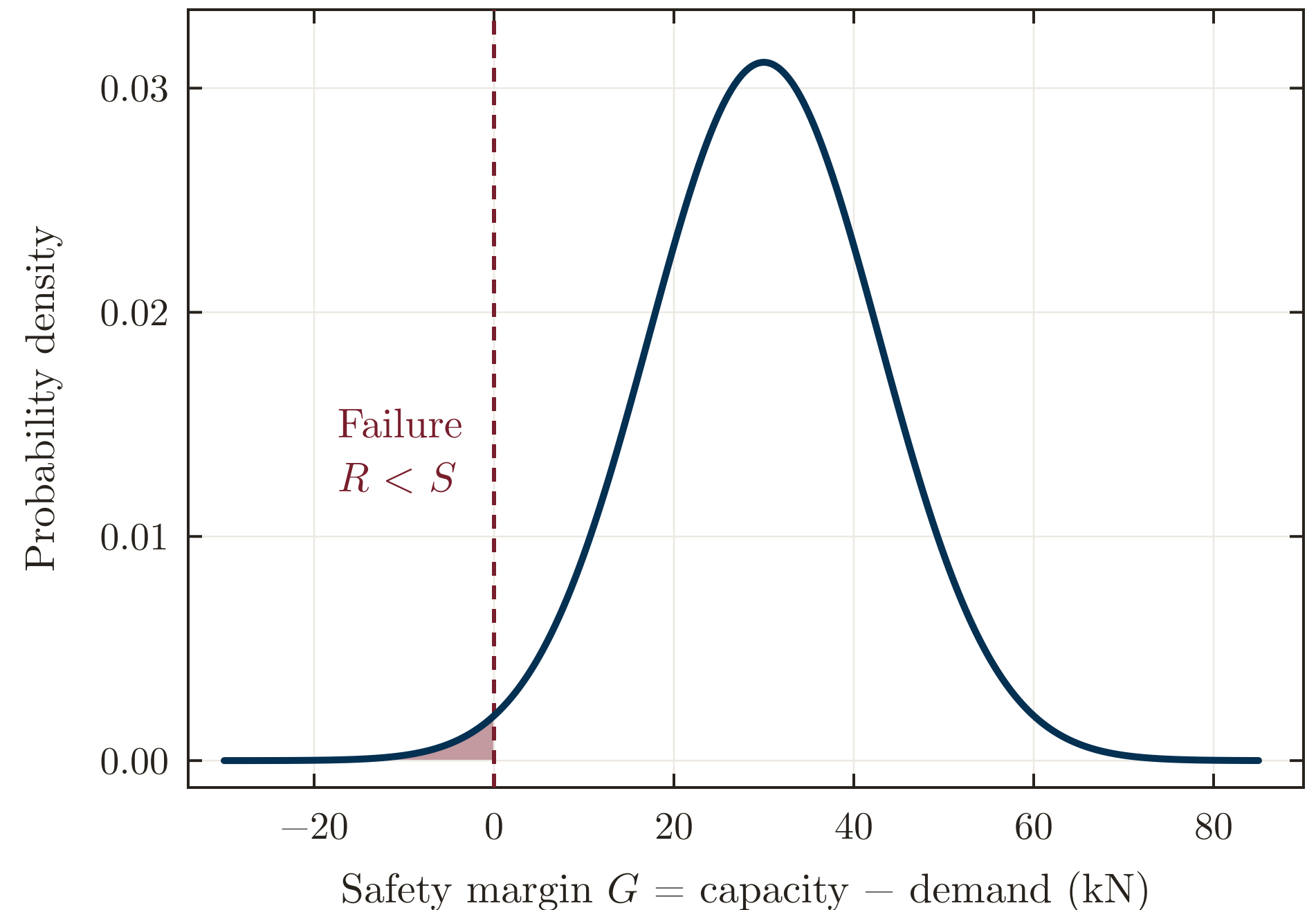
Here $\pm$ denotes one standard deviation.

Calculate an event probability

$$\beta = \mu_G / \sigma_G = 30 / 12.81 = 2.34$$

The mean is 2.34 SDs above failure.

$$P(G < 0) \approx 0.96\% \text{ for this model.}$$



This linear Gaussian case is exact; nonlinear FOSM uses a local approximation.

Model outputs and decision rules are different

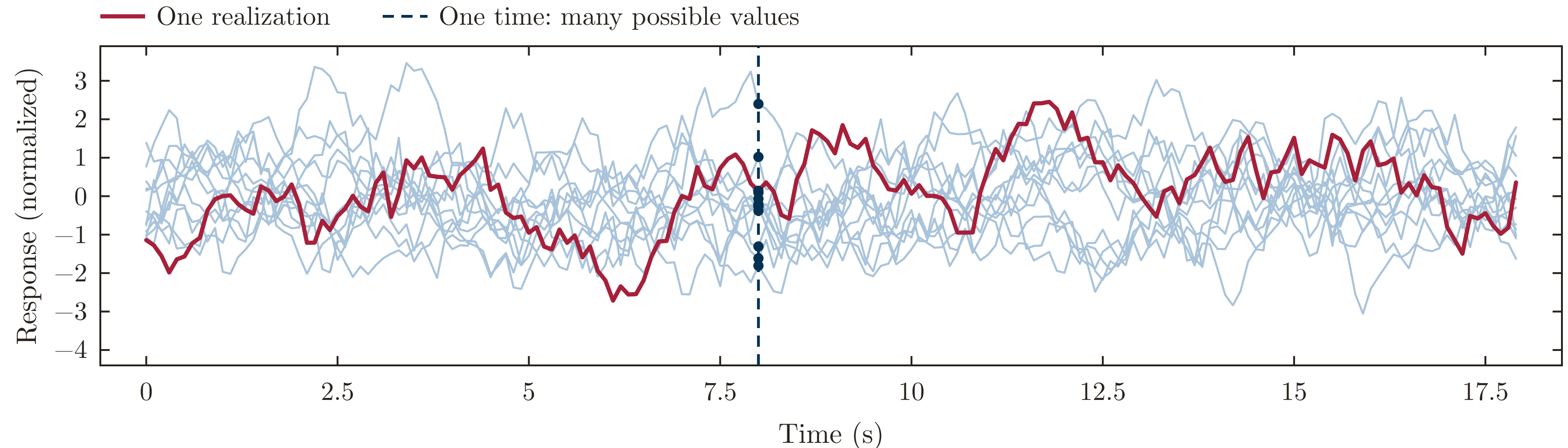
A single number can still contain probabilistic information. A decision needs a rule for using that information.

Model output	What it tells you	Possible decision rule
Point prediction $\hat{y}$	A predicted response, often a mean	Compare with an operating limit
Quantile $q_{0.05}$ or $q_{0.99}$	A specified model tail level	Use a chosen design percentile
Exceedance probability	Chance an outcome crosses a limit	Require probability below a target
Reliability index β	A scalar reliability summary	Compare with a consistent β target

Comparing β with a target is a reliability-based decision, even though β is a scalar.

A stochastic process models a whole history

A random variable describes one quantity. A process describes a family of random variables indexed by time or space.



Random variable X

Strength of one cylinder

Random vector (X, Y)

Strength and density together

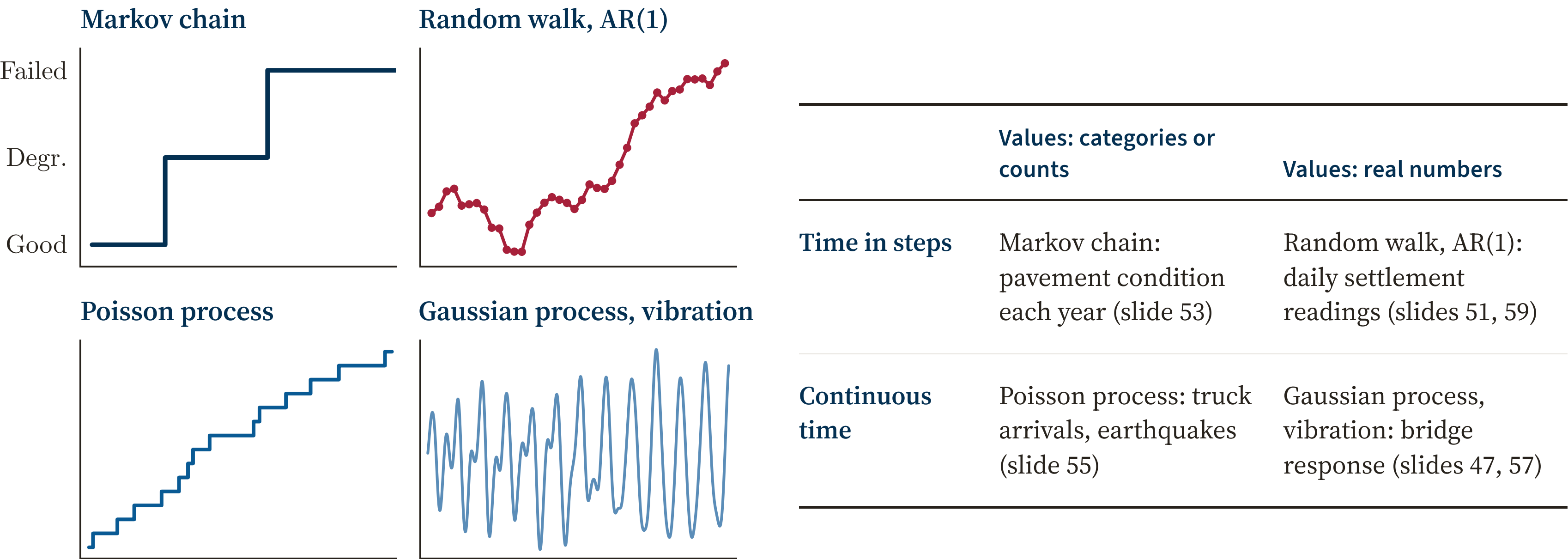
Process $X(t)$

Bridge response over time

One time gives a distribution across realizations. One realization gives a history across time.

Four families of stochastic processes

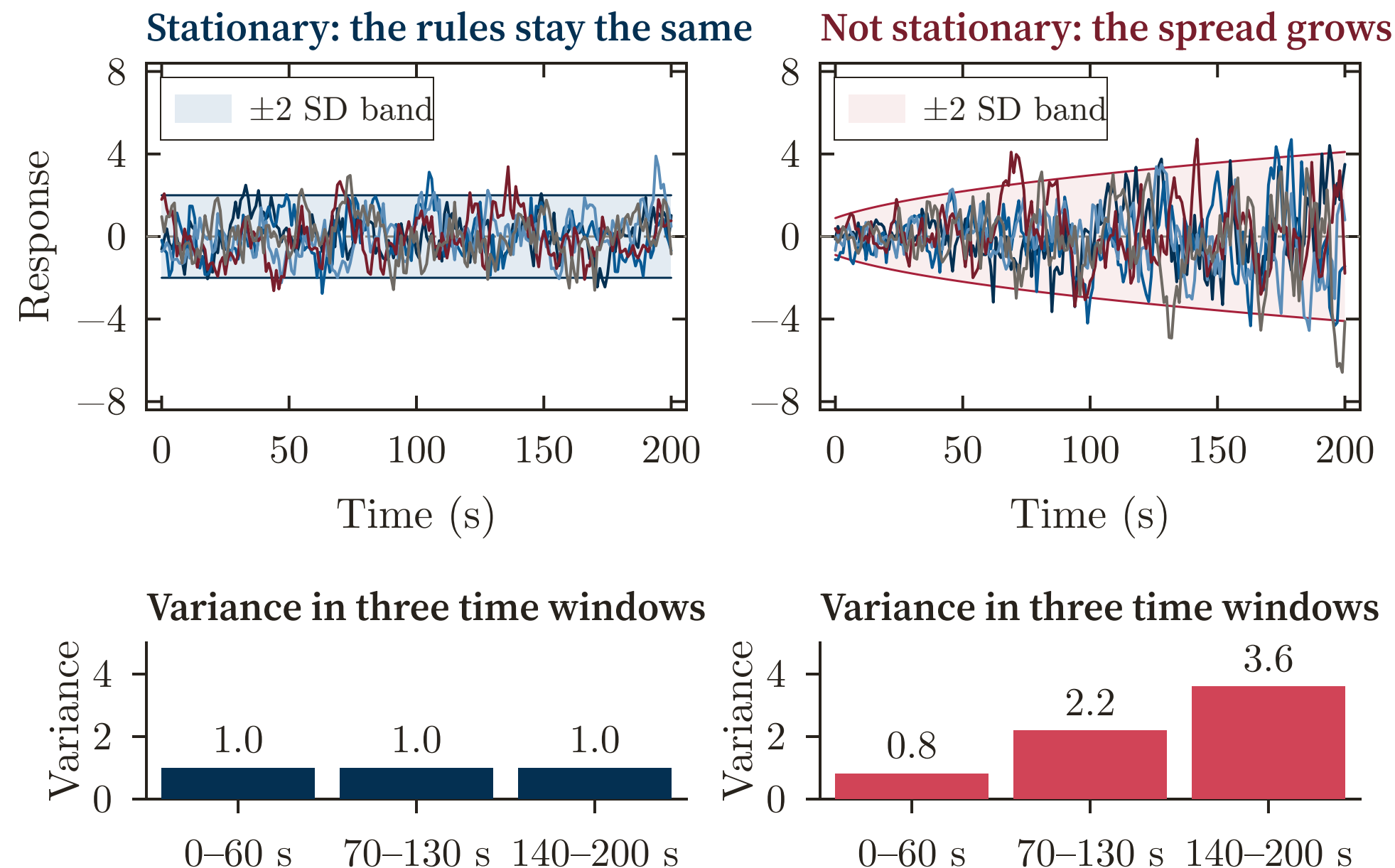
Two questions classify a process: does time move in steps or continuously, and are the values categories/counts or real numbers?



Same tools (mean, variance, correlation) for all four; only the indexing changes.

Stationarity: do the statistical rules change with time?

Stationary means the process follows the same rules at every time: same mean, same spread, same correlation pattern. We check it by comparing statistics measured at different times.



How to read the figures

Each panel: 5 possible records of one process.
Band: ± 2 SD across records at each time.
Left: flat band, variance 1.0 in every window.
Right: band widens, variance $0.8 \rightarrow 3.6$.

Definition (weak stationarity)

$$E[X(t)] = \mu$$

$$\text{Var}[X(t)] = \sigma^2$$

$$\text{Cov}[X(t), X(t+\tau)] = C(\tau)$$

The covariance depends only on the time gap τ , not on when you look.

Statistics measured early and late agree: stationary. If not, model the change or analyze stable windows.

Where stationarity helps, and where it does not

Many methods summarize a stable window. Others explicitly model trends, changing variability or transient events.

Vibration / Welch PSD

Pool segments when their second-order behavior is reasonably stable.

Speech / short-time spectra

Analyze short frames so changing sounds remain visible.

Fatigue / spectral methods

Stationary stress models support spectral approximations. Rainflow also handles variable histories.

Econometrics / ARIMA and ADF

Differencing can remove a unit root. ADF tests for a unit root, not every form of nonstationarity.

EEG / windowed band power

Compare short epochs rather than blur different brain states into one spectrum.

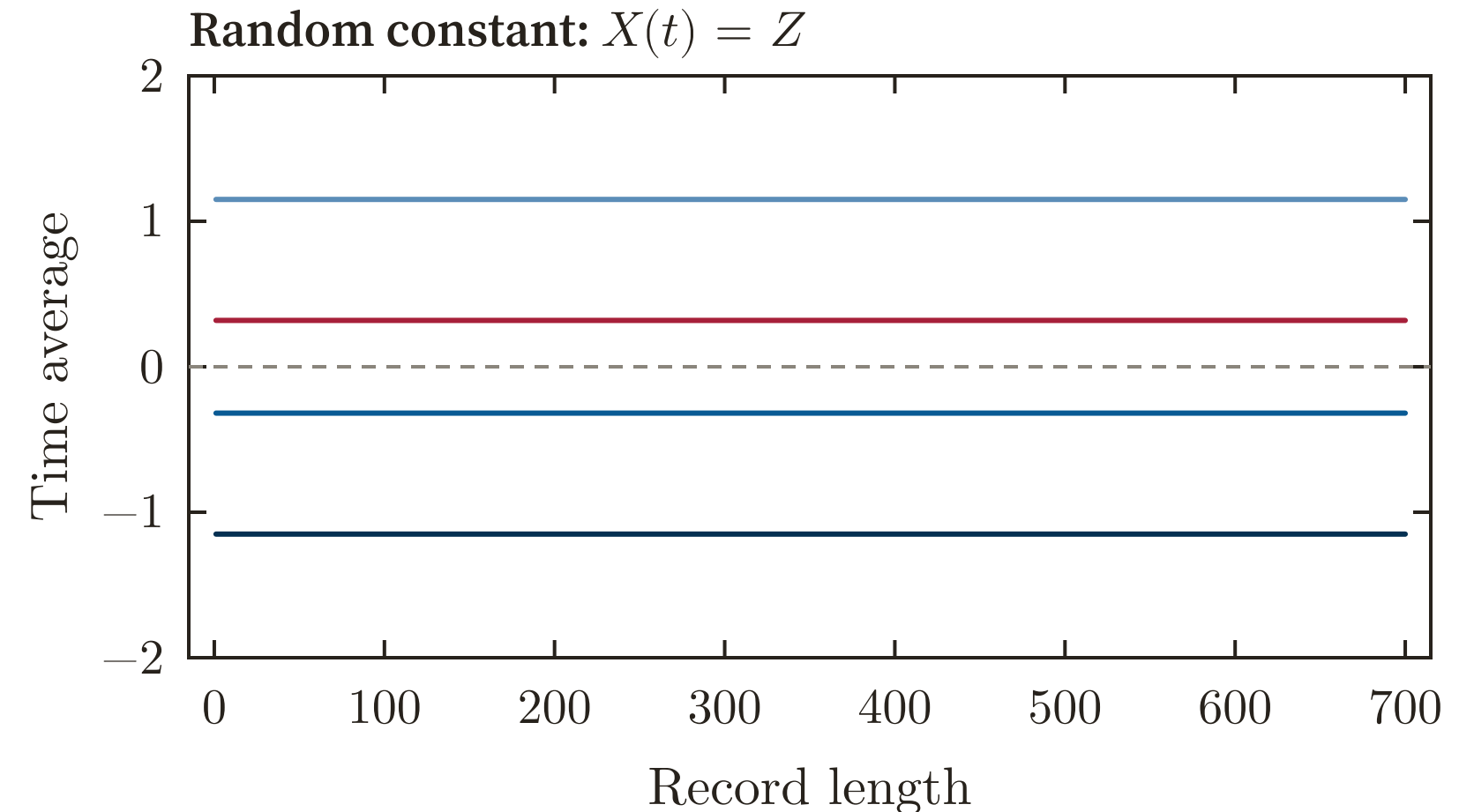
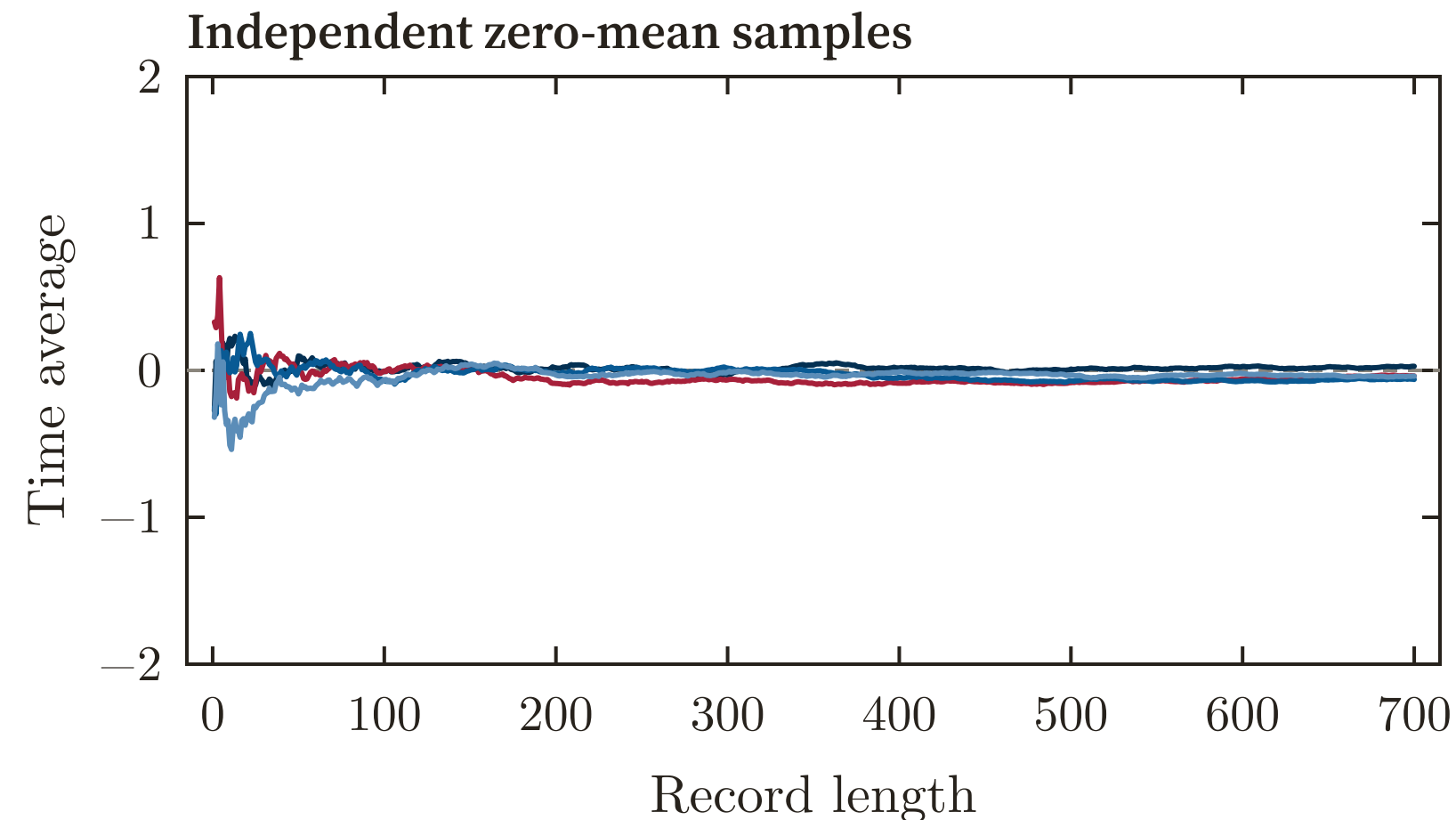
Process control / Shewhart, CUSUM

Compare with a stable baseline. Serial dependence can require adjusted chart limits.

Stationarity is a modeling assumption to assess, not a prerequisite for every time-series method.

Ergodicity: can one long record represent the model?

Can one long record estimate the average across many possible histories (the ensemble)? Mean ergodicity answers this.



Many independent fluctuations

Longer records reveal the common mean.

One random offset per realization

Longer records reveal only that offset.

A non-ergodic time average may settle perfectly, but at the wrong ensemble value.

Why engineers care about representative records

The practical question is whether averaging the available history estimates the population quantity we want.

Operational modal analysis

FDD and SSI use estimated spectra or covariances from ambient vibration records.

Statistical mechanics

Compare long-time averages of system observables with the relevant ensemble average.

Wind and wave spectra

A record should represent the wind regime or sea state being modeled.

Finance / compounding

Average growth along one wealth path can differ from average wealth across paths.

Noise interferometry

Long cross-correlations can reveal propagation information under suitable source conditions.

Simulation checks

Compare many simulated realizations with target statistics. One time average is only one check.

Longer data help only if the recording explores the relevant behavior.

Autocorrelation: what persists after a time shift?

Compare a signal with a delayed version of itself. The lag tells us how far apart the compared values are.

What to look for

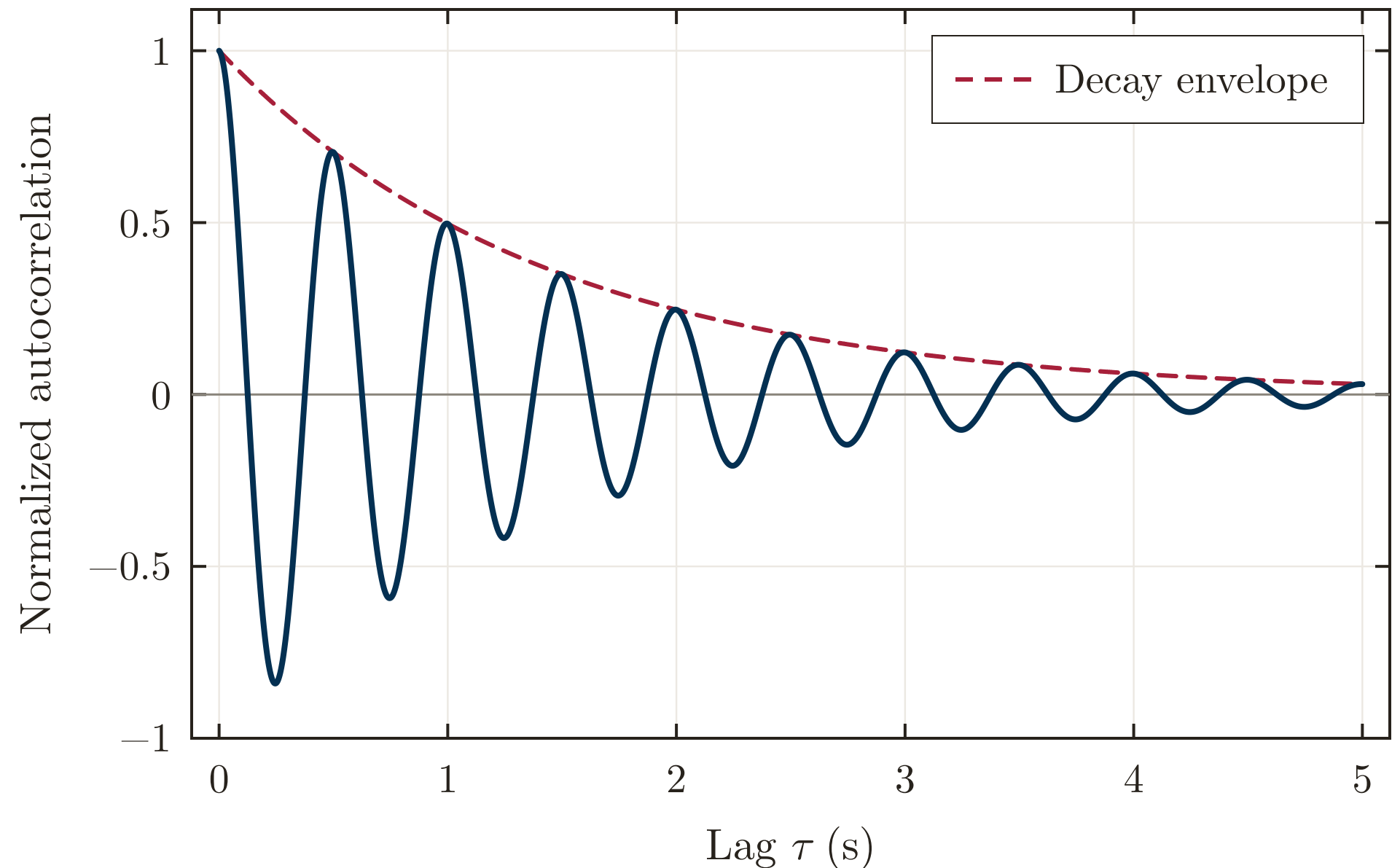
Repeating peaks suggest periodicity.
A slow envelope suggests persistence.
Alternating signs suggest oscillation.

Mean-square versus variance

Autocorrelation $R(0) = \sigma^2 + \mu^2$

Autocovariance $C(0) = \sigma^2$

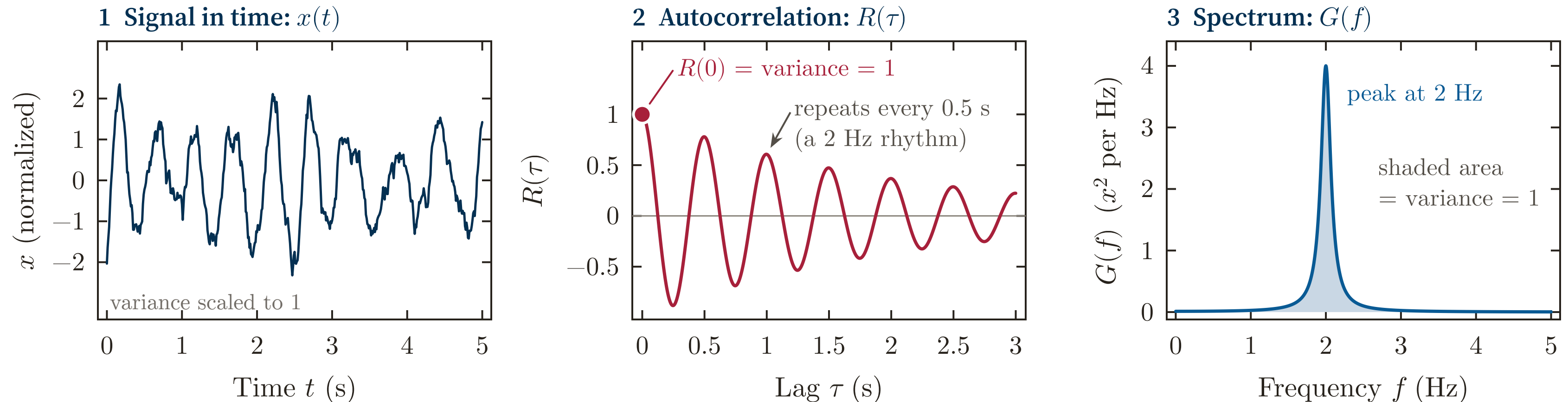
They coincide when the mean is zero.



Correlation describes linear dependence. Zero correlation does not generally mean independence.

From time to frequency: correlation gives the spectrum

Three views of the same random vibration. The spectrum is the Fourier transform of the autocorrelation, so the two carry the same information, arranged differently.



1 Variance

$\sigma^2 = E[x^2]$ for zero mean. The signal was divided by its SD, so $\sigma^2 = 1$.

2 Autocorrelation

$R(\tau) = E[x(t) \cdot x(t+\tau)]$: compare x with a copy shifted by τ . $R(0) = \sigma^2$.

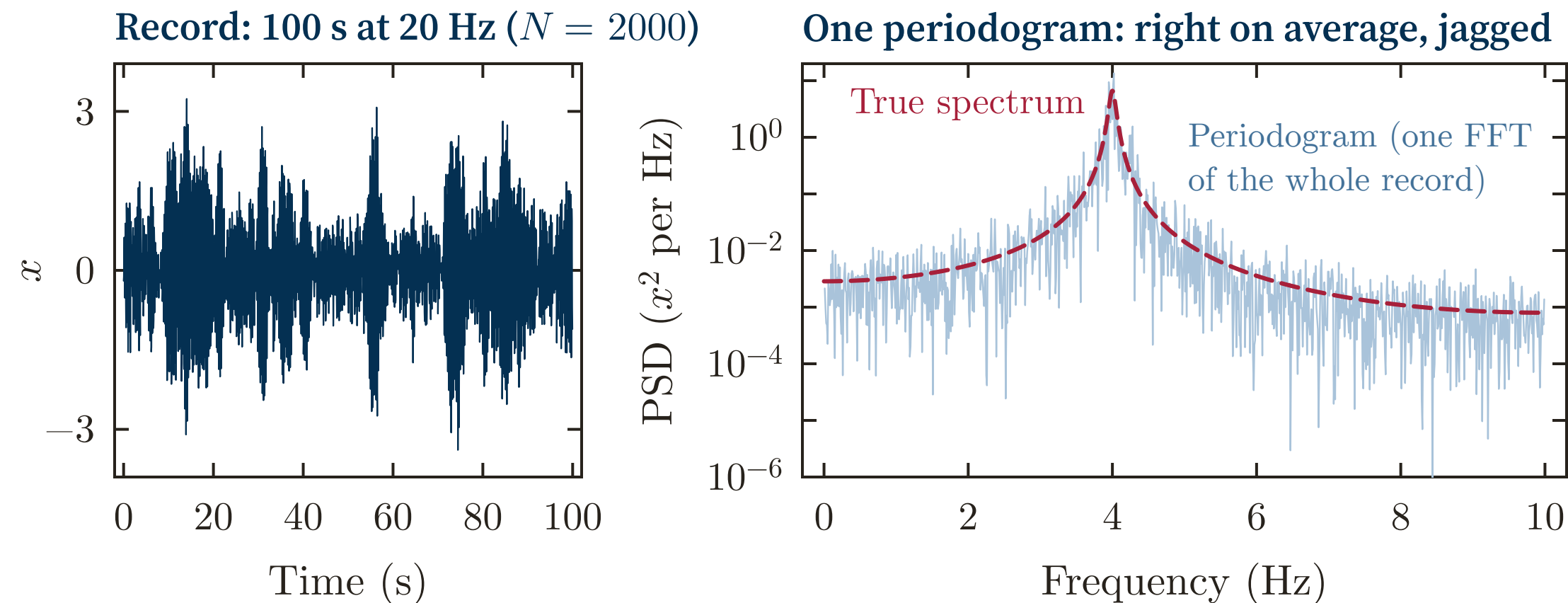
3 Spectrum

$S(f) = \int R(\tau) e^{-i2\pi f\tau} d\tau$.
Area: $\int G(f) df = \sigma^2 = 1$.

The area under the spectrum is the variance: the spectrum shows how the variance is shared among frequencies.

Estimating a spectrum (1): the FFT and the periodogram

Slide 47 used the true $R(\tau)$. A finite record only gives an estimate. The FFT is the calculator; the periodogram is the estimate built from it.



Three steps

1 FFT: $X(f) = \sum x[n] e^{-i2\pi fn\Delta t}$

A fast way to get the Fourier coefficients: complex, with amplitude and phase.

2 Periodogram: $P(f) = (\Delta t/N) \cdot |X(f)|^2$

Square the magnitude, scale to x^2 per Hz, drop the phase.

3 Link to slide 47

$P(f)$ is the Fourier transform of the sample autocorrelation: the same time-to-frequency link, applied to data.

Why so jagged?

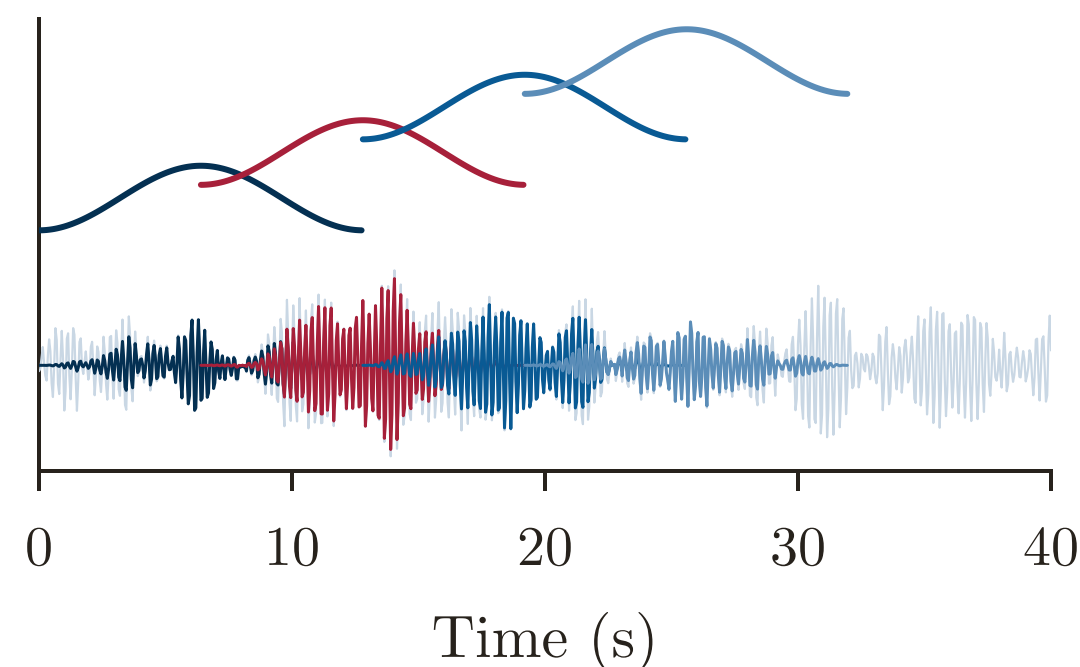
Each value rests on a single Fourier coefficient, so its scatter is about as large as the value. A longer record gives more points, not smoother ones.

The FFT computes; the periodogram estimates. One periodogram is right on average but very jagged.

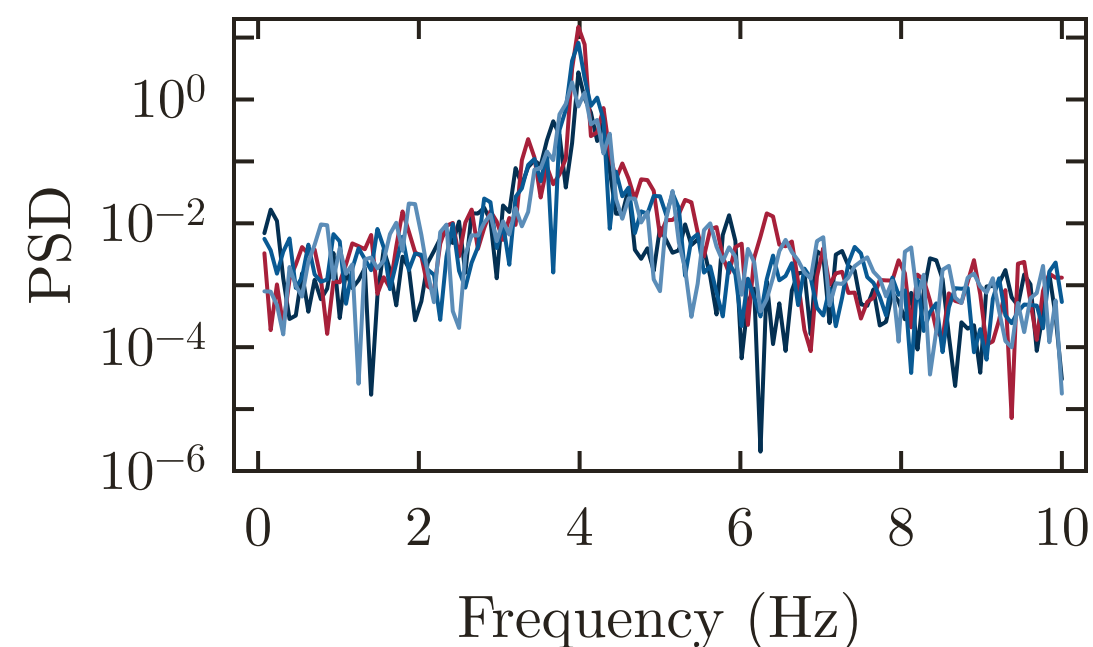
Estimating a spectrum (2): Welch's method

Cut the record into overlapping segments, taper each with a window, compute a periodogram for each, and average. Averaging removes most of the jaggedness.

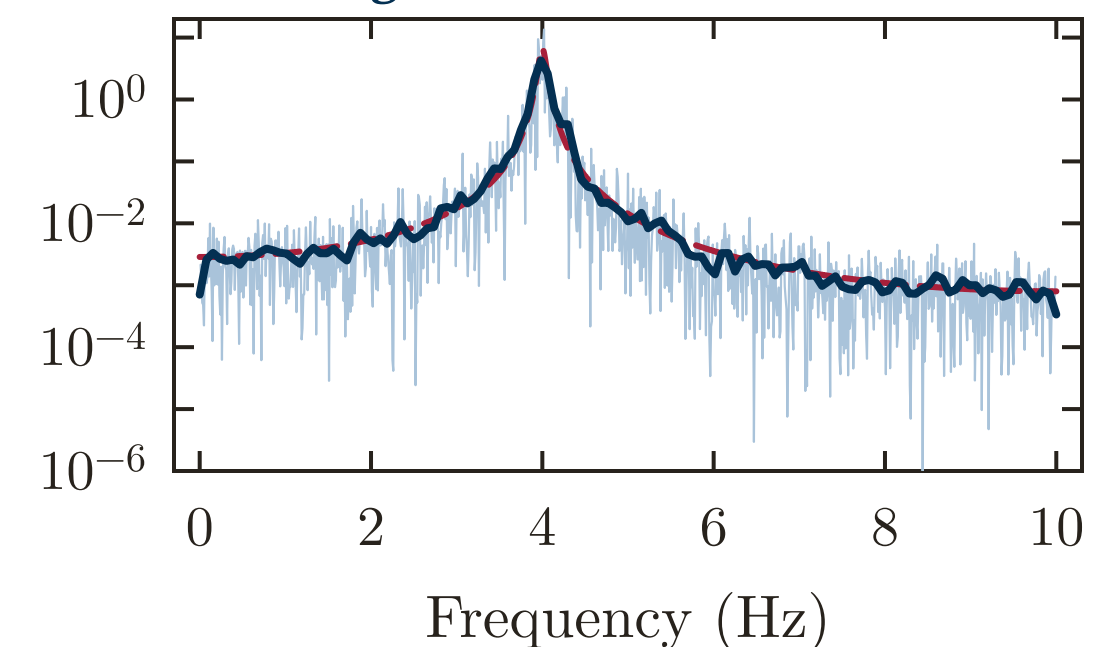
1 Cut into overlapping segments, taper each with a window



2 Periodogram of each segment (still noisy)



3 Average them: smooth estimate



Welch average

$$P_W(f) = (1/K) \cdot \sum P_k(f),$$

K windowed segments

Here

$L = 256$ samples = 12.8 s, 50% overlap,
 $K = 14$. Frequency step
 $1/12.8 \text{ s} \approx 0.08 \text{ Hz}$.

- single periodogram
- Welch average of 14 segments
- - - true spectrum

More segments: smoother spectrum. Longer segments: finer frequency detail. Welch balances the two.

What correlation and spectra help us do

The same calculations answer different physical questions, provided the signal and model assumptions fit.

Structural frequencies

PSD peaks can suggest modal frequencies. Separate structural response from forcing peaks.

Damping

Model the decay of modal correlations under suitable linear-system and excitation assumptions.

Speech and music

Repeated autocorrelation peaks suggest a period. Pitch is approximately its reciprocal.

Radar and sonar

Cross-correlate an echo with the transmitted pulse. Delay gives travel time and range.

EEG rhythms

Band power summarizes rhythmic content. Artifacts can create misleading peaks.

Financial returns

Lag correlations reveal dependence to test further, not a guaranteed trading rule.

A spectral peak is evidence to interpret, not an automatic diagnosis.

Random walks: uncertainty accumulates

Each step adds an independent increment. For zero-mean increments, the expected position stays fixed while spread grows.

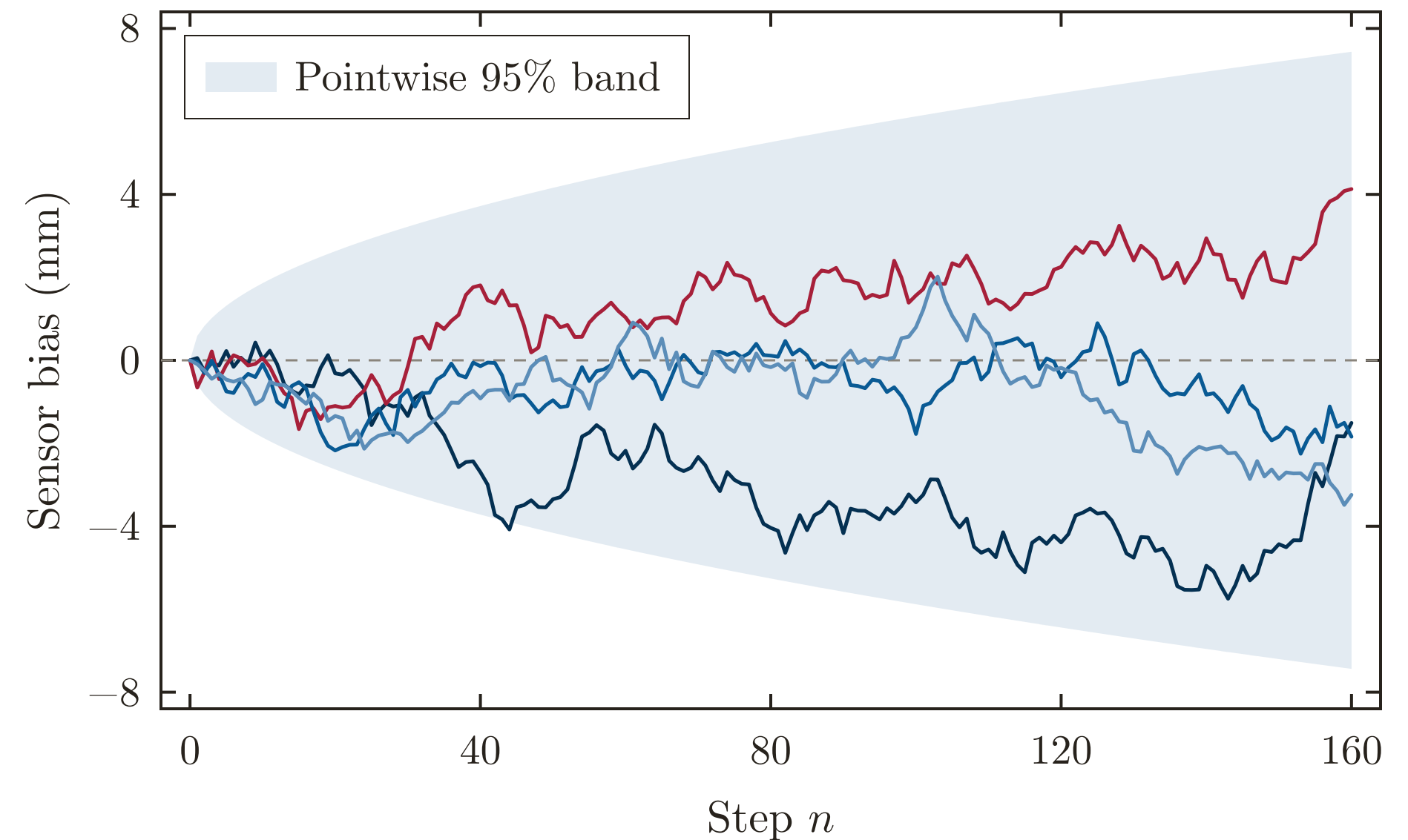
Useful for

Sensor bias drift, diffusion and idealized cumulative disturbances.

Mean and variance

$$E[X_n] = X_0 \quad \text{Var}(X_n) = n\sigma^2$$

Standard deviation grows as $\sqrt{n}$.



A random walk can have stationary increments while its level is nonstationary.

First passage: did the response ever cross the limit?

An instantaneous exceedance and an exceedance at any time during service are different events.

At the final instant

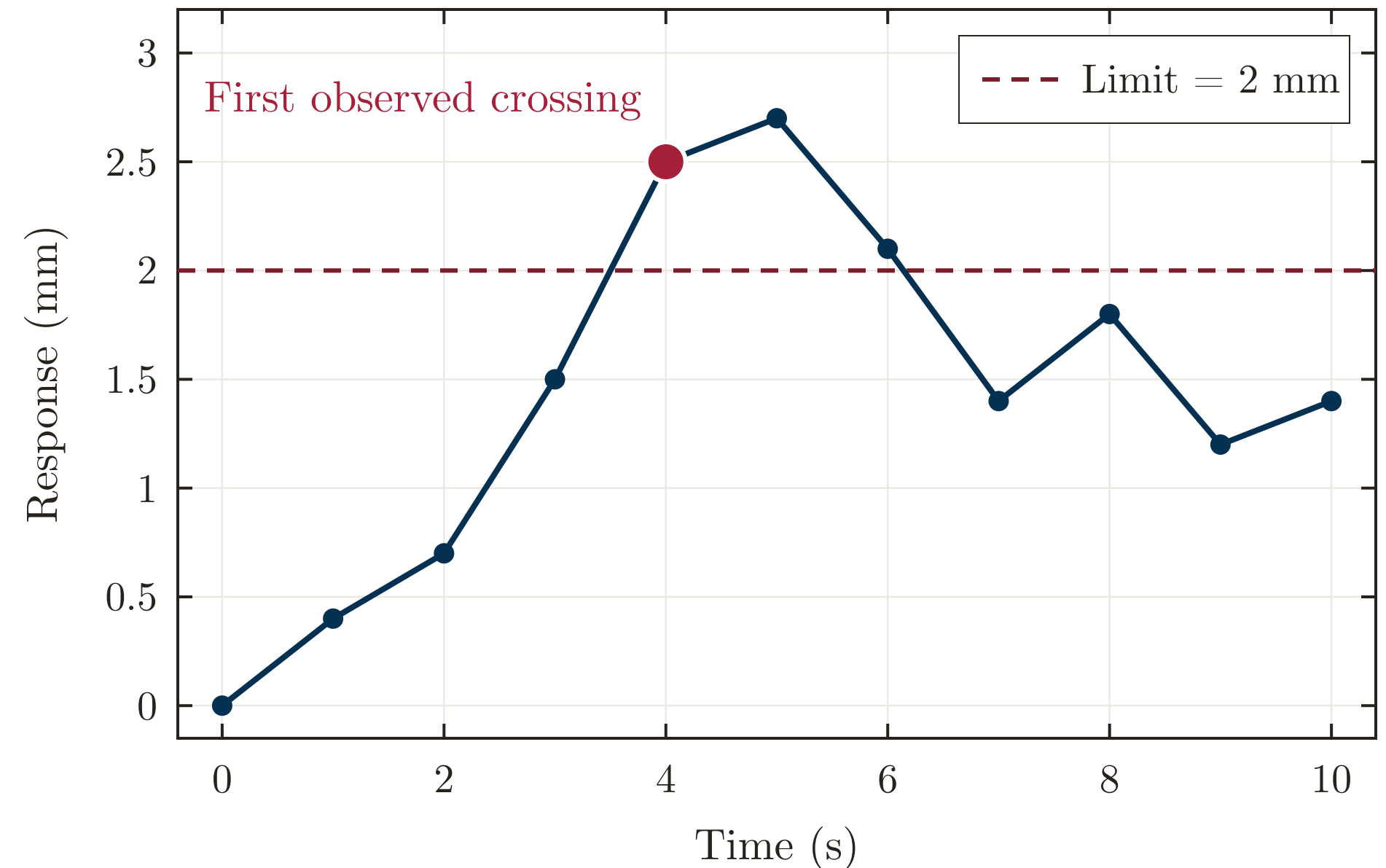
Is $X(T)$ above the limit?

In this path: no.

Over the whole interval

Was $X(t)$ ever above the limit?

In this path: yes.

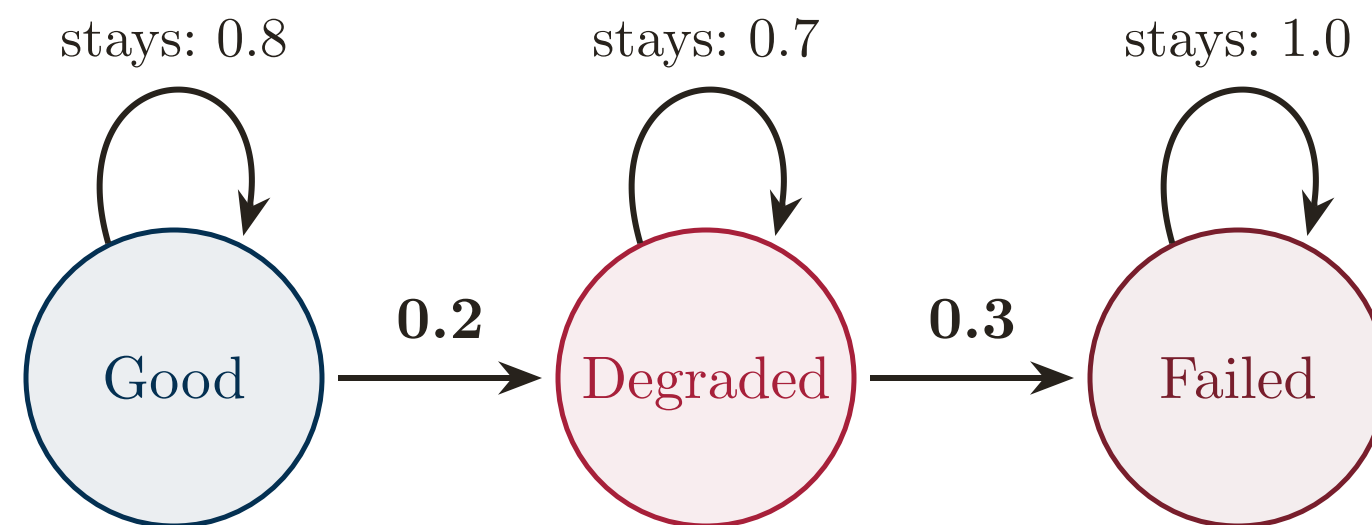


A path can cross a limit and later return below it. The earlier crossing still happened.

Markov chains (1): the rules for moving between states

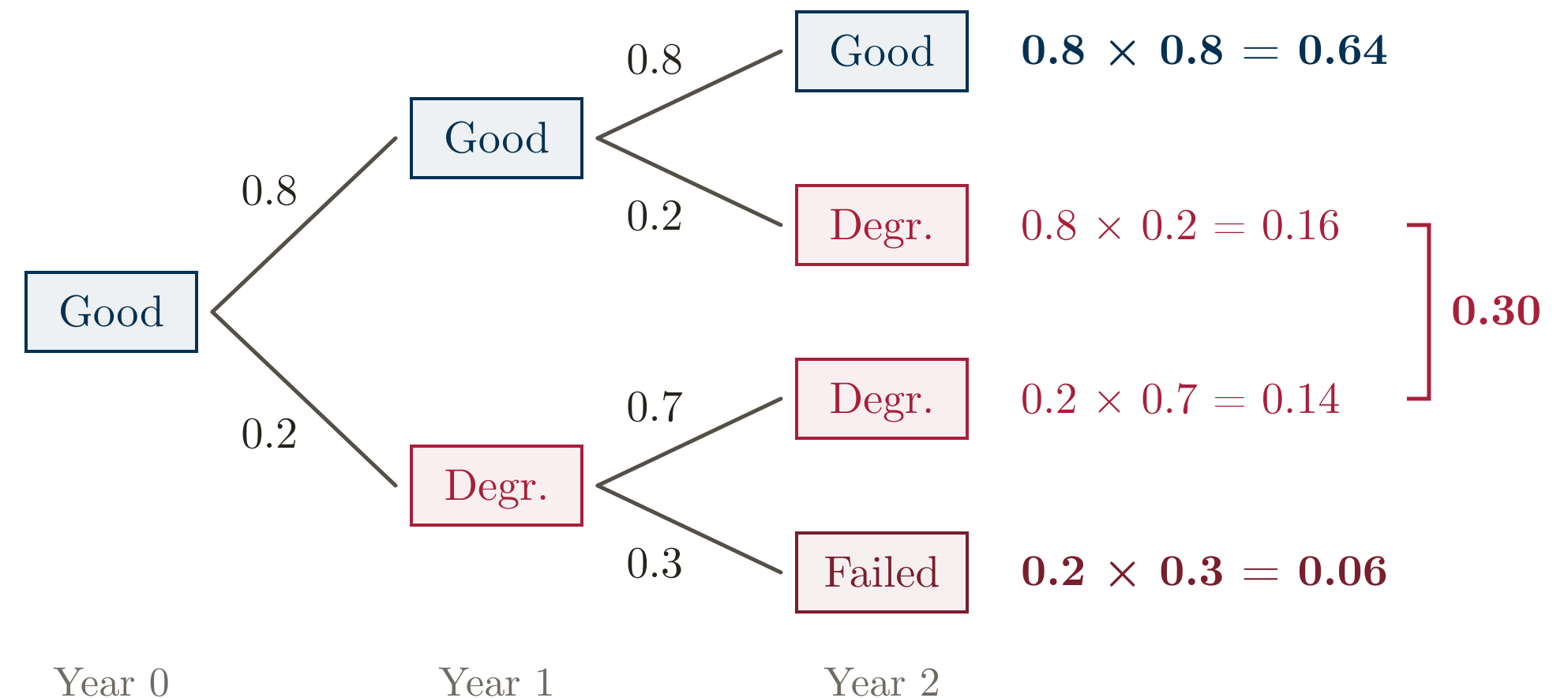
A pavement section is Good, Degraded or Failed. Each year it stays or moves one state worse with fixed probabilities. The next state depends only on the current one.

1 The rules (one inspection interval)



Each year, a section stays or moves one state worse.
The numbers on arrows are the transition probabilities.

2 Follow every path for two intervals, starting Good



After two years: Good $0.8 \times 0.8 = 64\%$, Degraded $0.16 + 0.14 = 30\%$, Failed $0.2 \times 0.3 = 6\%$. The three add up to 100%.

Multiply along each path; add the paths that end in the same state.

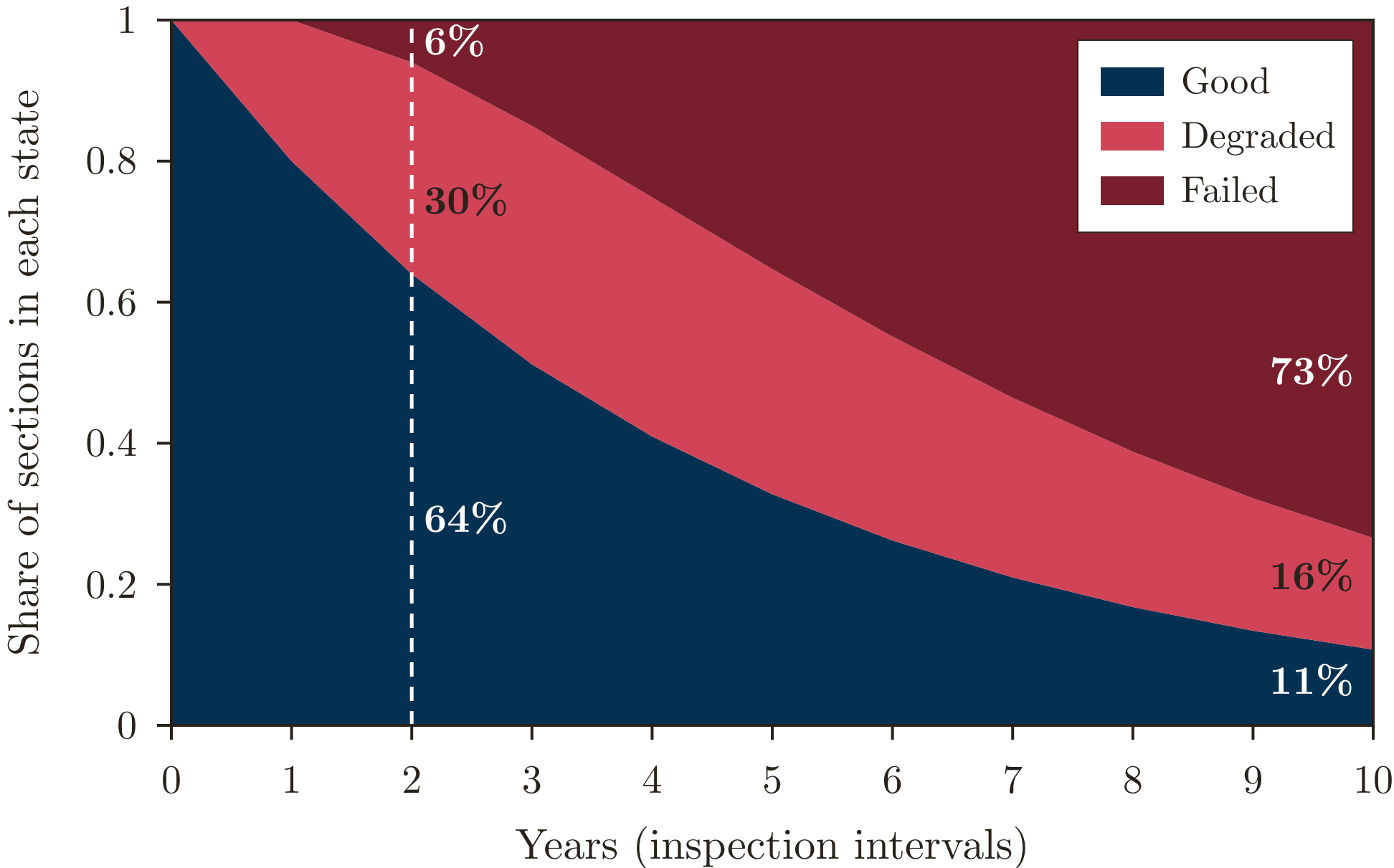
Markov chains (2): the same calculation, year after year

Writing the rules as a table P lets us repeat the tree calculation for any number of years with one line: $\pi(\text{next}) = \pi(\text{now}) \times P$.

From / to	Good	Degraded	Failed
Good	0.8	0.2	0
Degraded	0	0.7	0.3
Failed	0	0	1

Start Good: $\pi_0 = [1, 0, 0]$
 $\pi_1 = \pi_0 P = [0.80, 0.20, 0]$
 $\pi_2 = \pi_1 P = [0.64, 0.30, 0.06]$
 $\pi_{10} = [0.11, 0.16, 0.73]$

Each vertical slice of the chart is one π . At year 2 it reads 64 / 30 / 6%, the tree result.

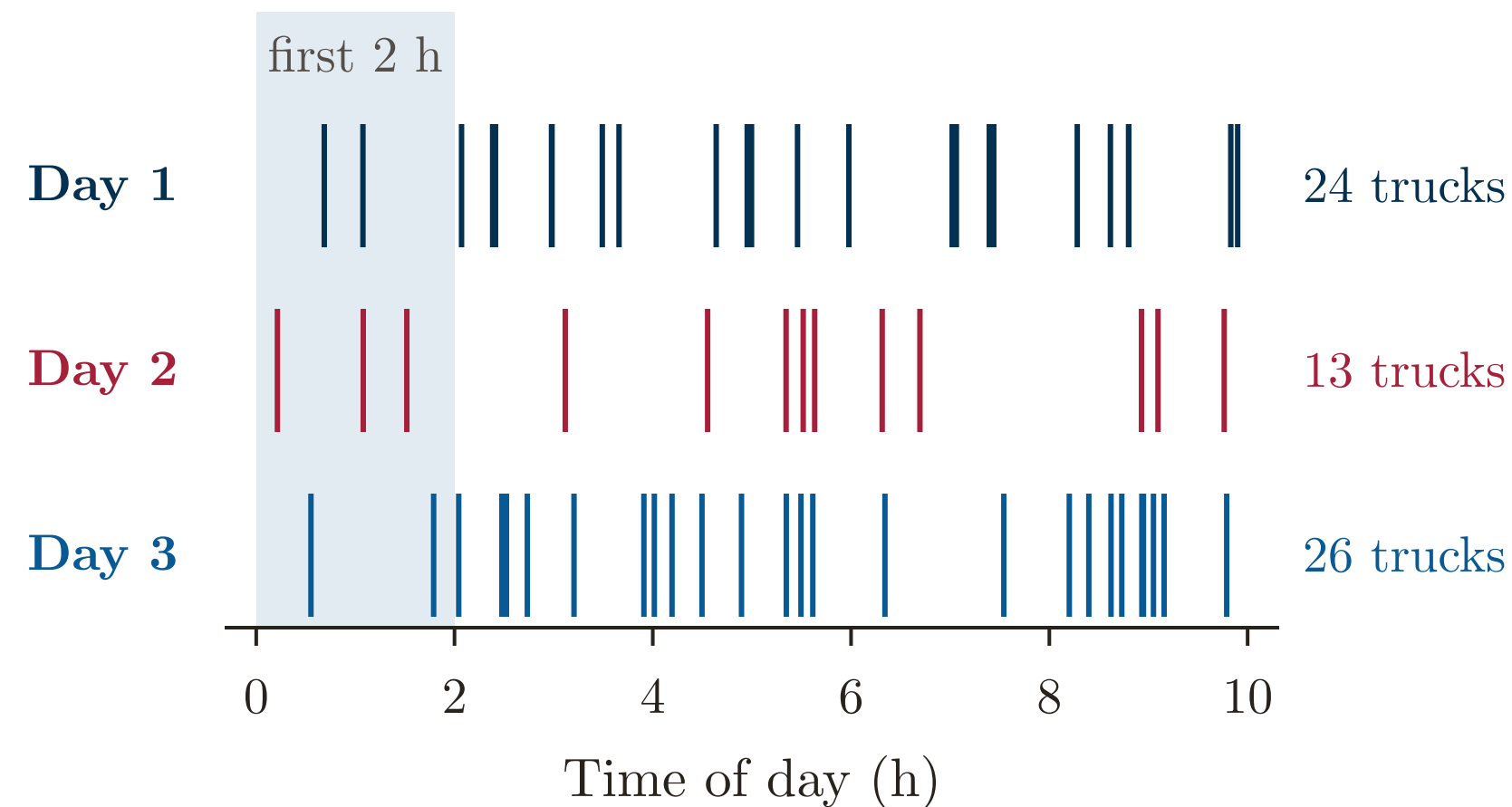


The chart is the tree calculation repeated each year: without repair, 73% of sections fail within 10 years.

Poisson processes (1): counting random events

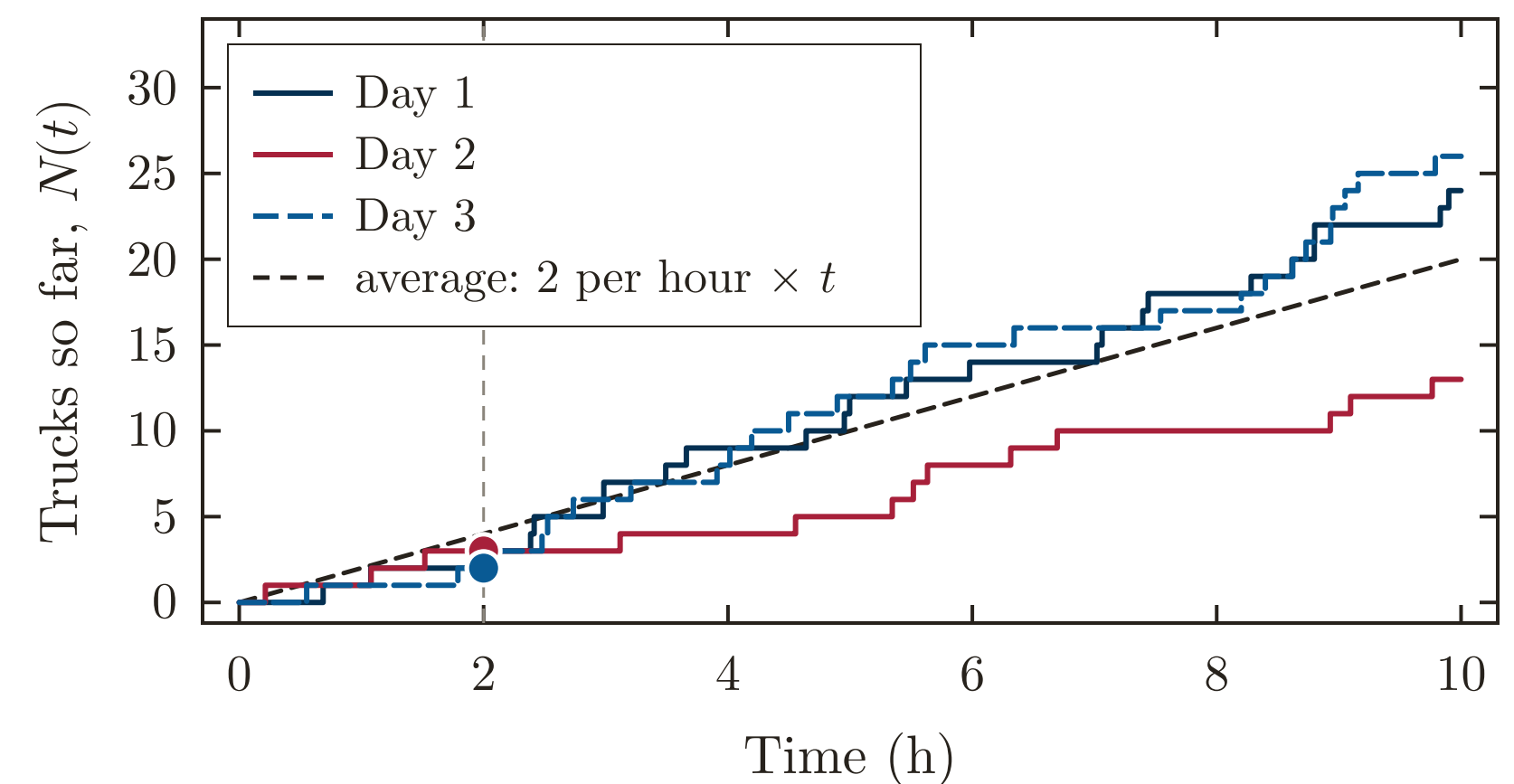
Example: trucks arrive at a weigh station at random, 2 per hour on average (rate $\lambda = 2$ per hour). Arrivals are independent and the rate stays constant.

1 Each tick = one truck arriving (3 different days)



One realization = one possible day. The rate is an average: first 2 h had 2, 3 and 2 trucks; 10 h had 24, 13 and 26 (expected 20).

2 Count so far: each step up = one truck



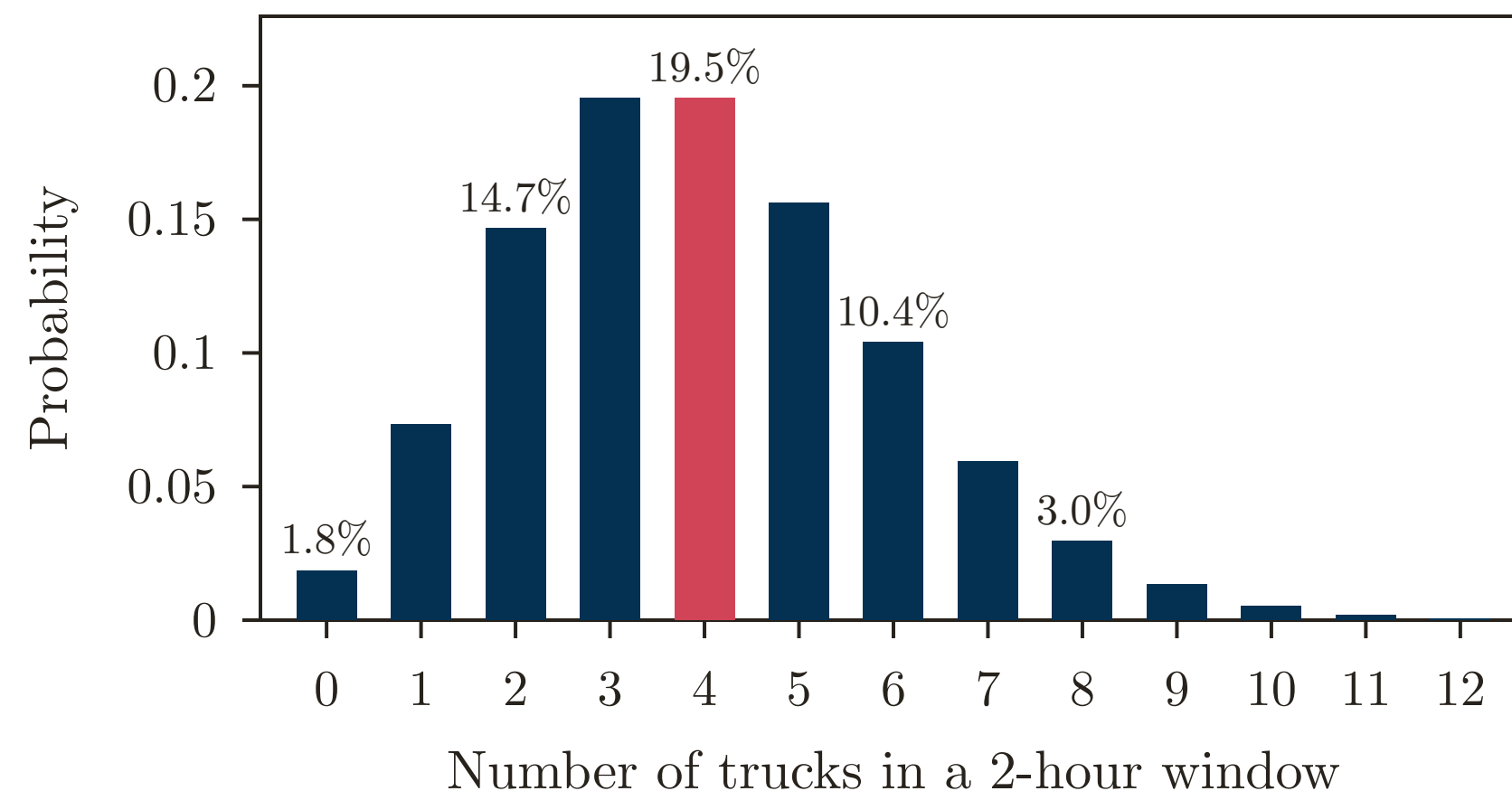
$N(t)$ = trucks so far. Each truck adds a step of 1 at a random time, so every day draws a different staircase around λt .

λ fixes the average; randomness decides the actual arrival times and counts.

Poisson processes (2): two questions the model answers

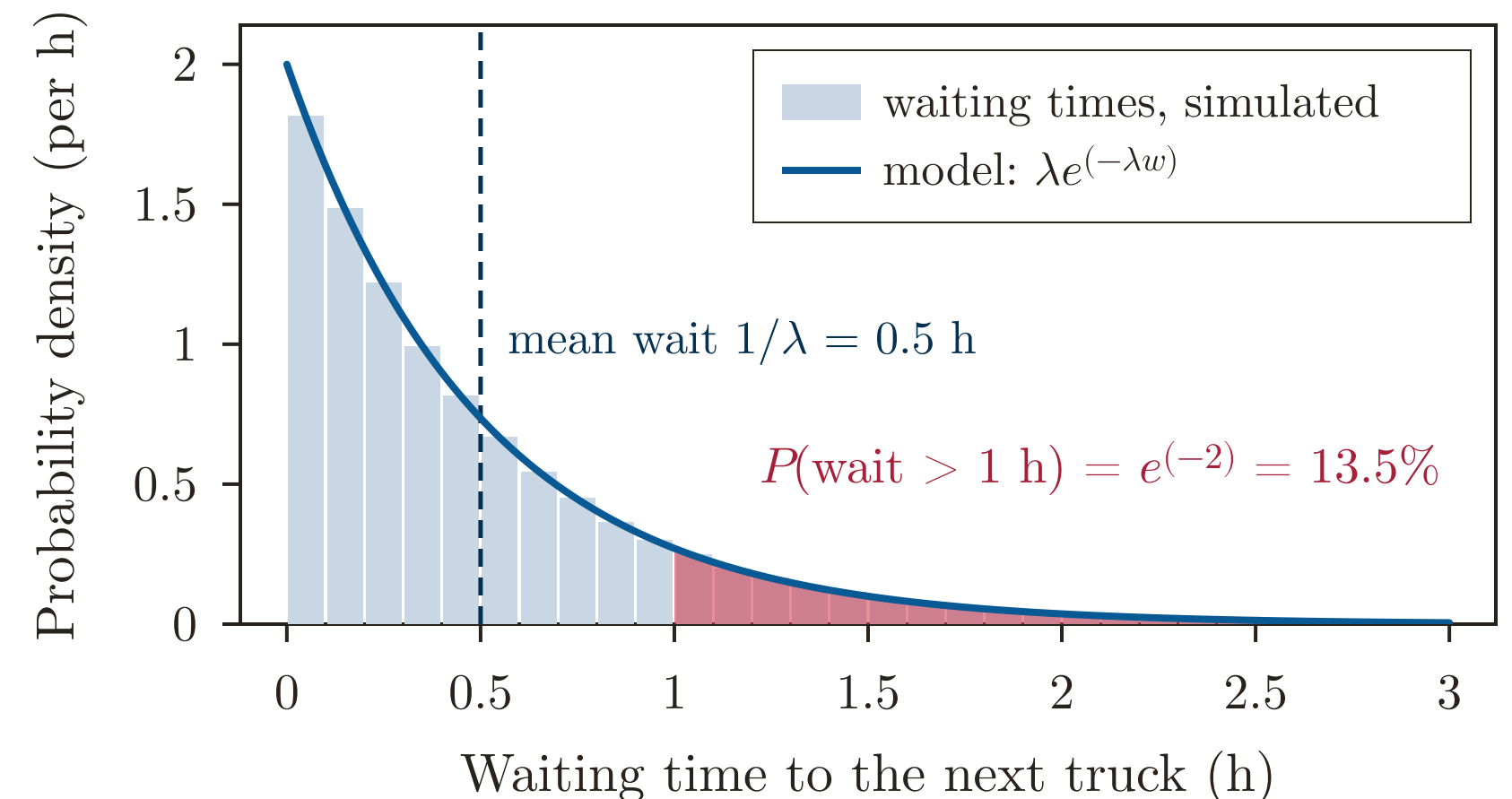
The single number $\lambda = 2$ per hour answers both: how many trucks arrive in a time window, and how long we wait for the next one.

1 How many in 2 hours? Expected $\lambda t = 4$



$P(N = k) = e^{(-\lambda t)} (\lambda t)^k / k!$, $\lambda t = 4$.
 $P(0) = 1.8\%$, $P(4) = 19.5\%$, $P(\geq 8) = 5.1\%$. Bars add up to 1.

2 How long until the next one?



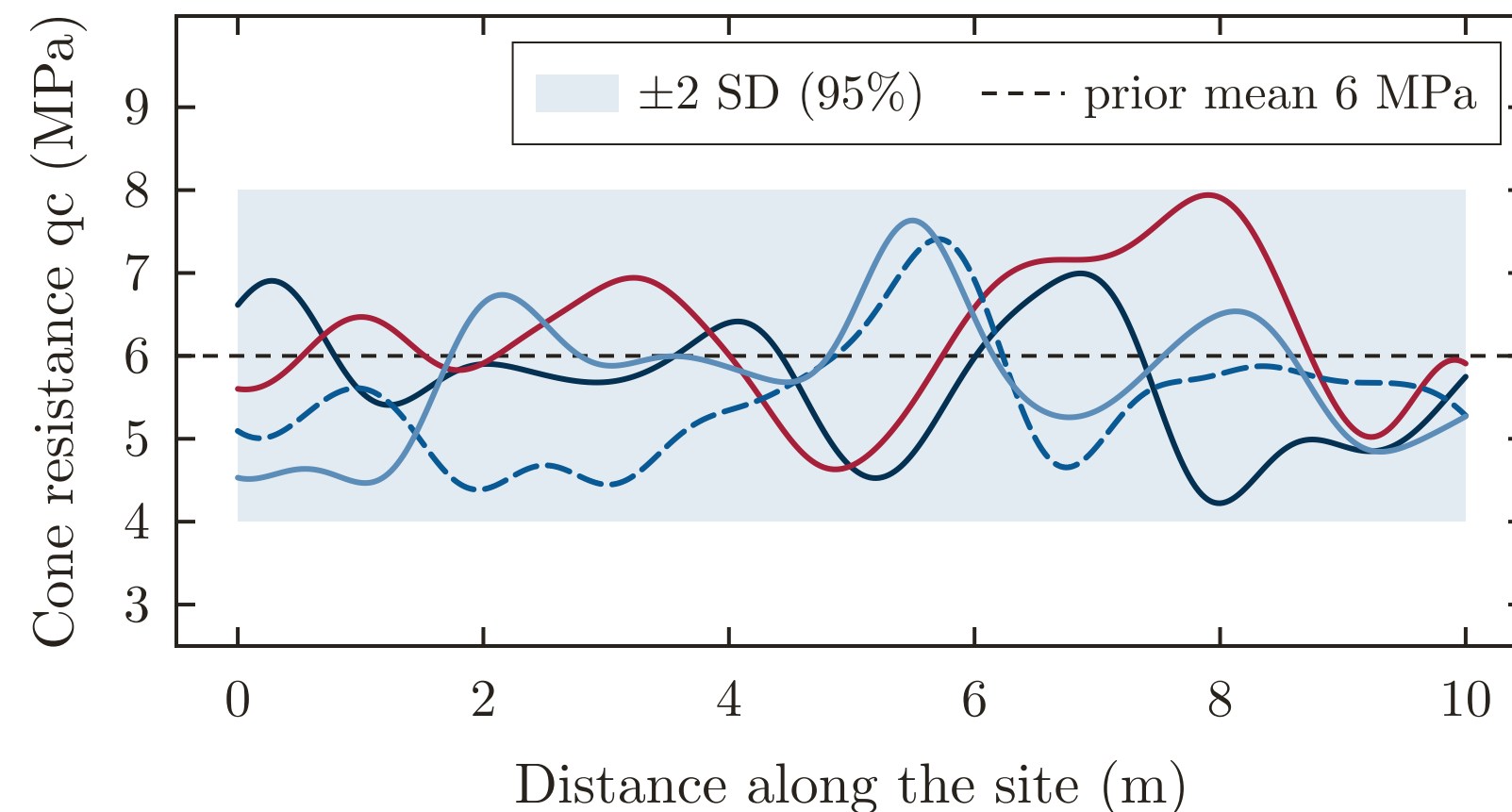
$f(w) = \lambda e^{(-\lambda w)}$, mean $1/\lambda = 0.5$ h.
 $P(\text{wait} > 1 \text{ h}) = e^{(-2)} = 13.5\%$ (shaded area).

Counts are Poisson, gaps are exponential: two faces of the same process.

Gaussian processes (1): a probability model for a curve

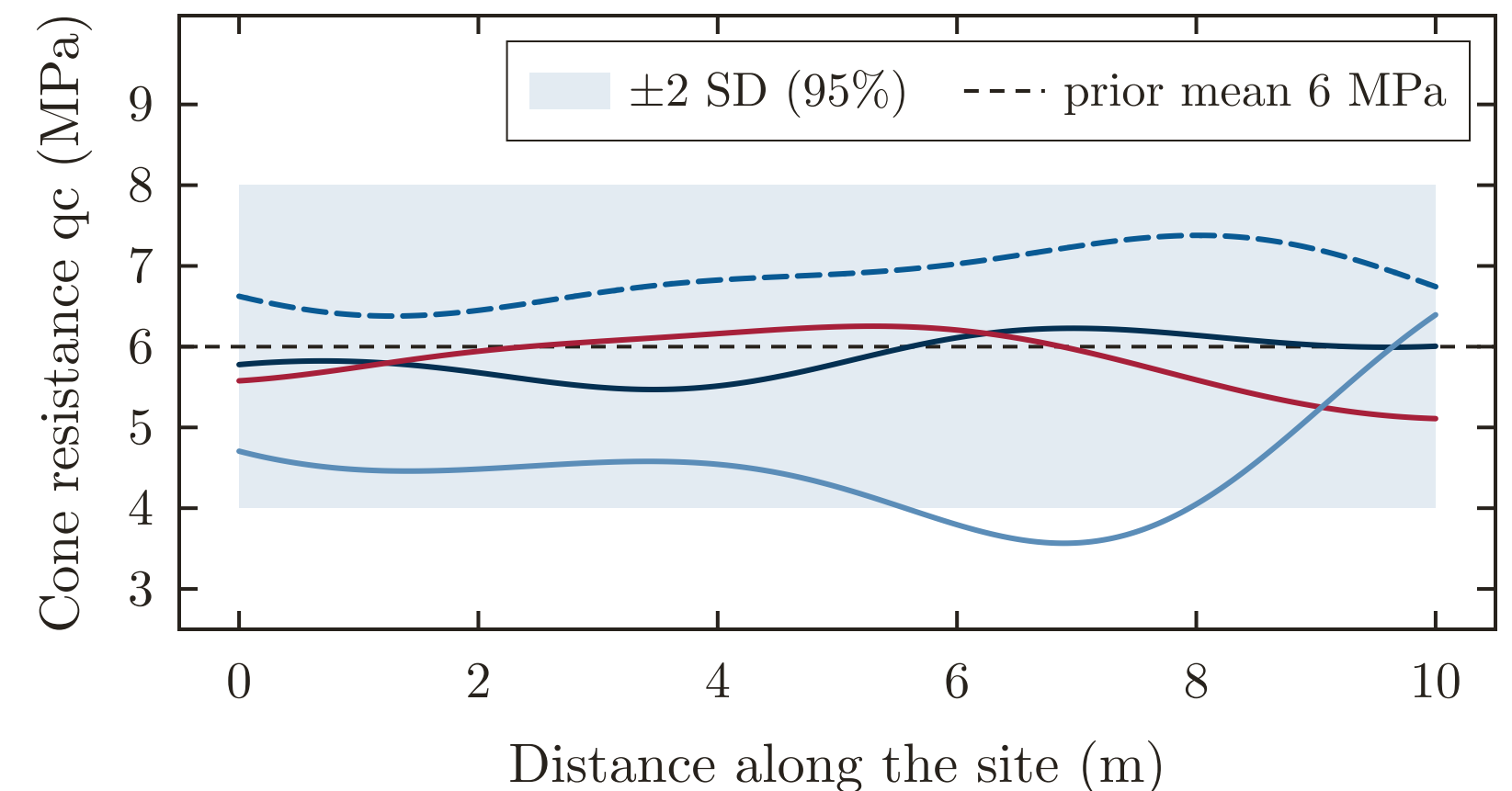
Example: cone resistance q_c along a 10 m site, before any tests. We expect about 6 MPa and a smooth variation. A Gaussian process turns that belief into random curves.

Short correlation length $\ell = 0.6$ m: rough



Mean $m(x) = 6$ MPa: best guess everywhere. Each colored curve is one possible profile (one realization).

Long correlation length $\ell = 2.5$ m: smooth



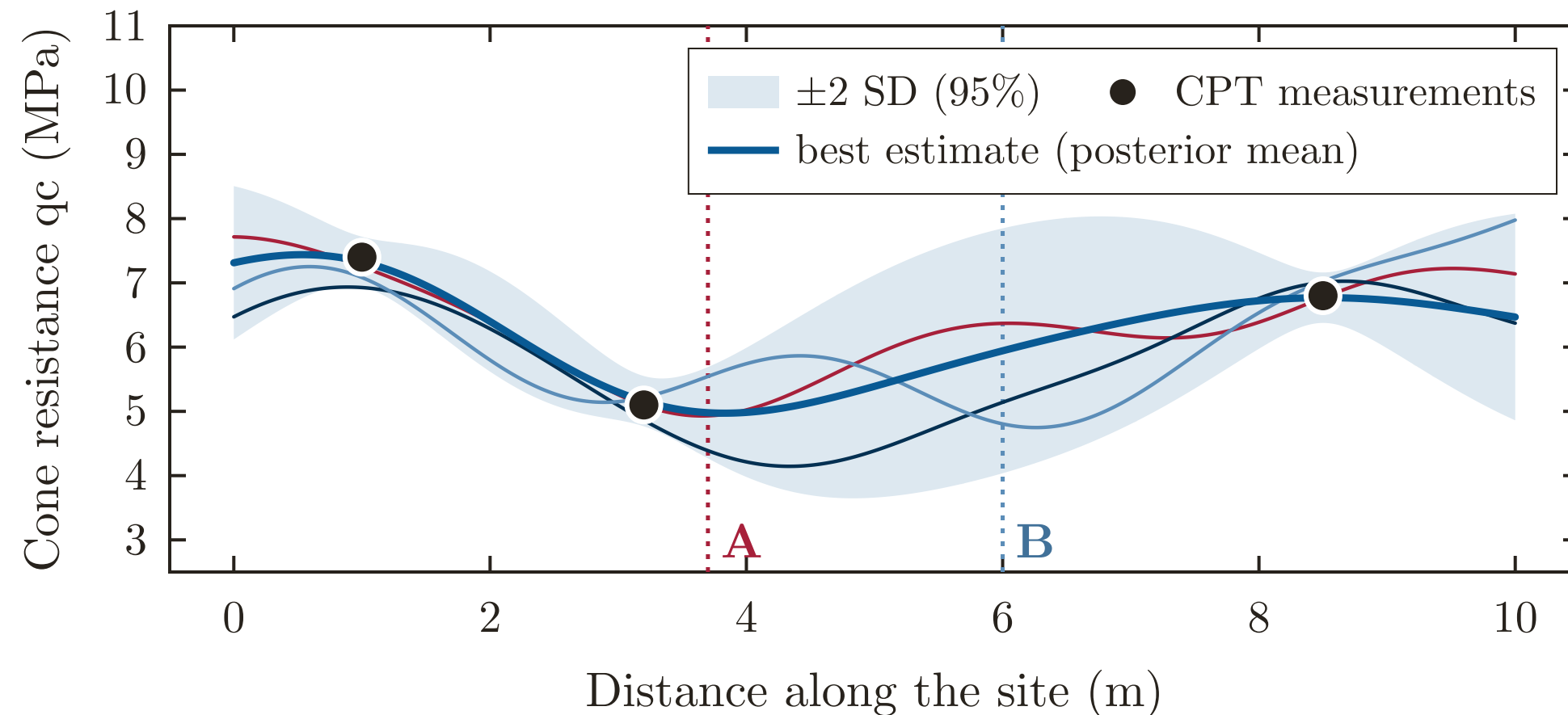
Kernel $k(x, x') = \sigma^2 \exp[-(x - x')^2 / (2\ell^2)]$: points closer than about ℓ have similar values.

Before data, a Gaussian process is a cloud of plausible curves, set by a mean and a kernel.

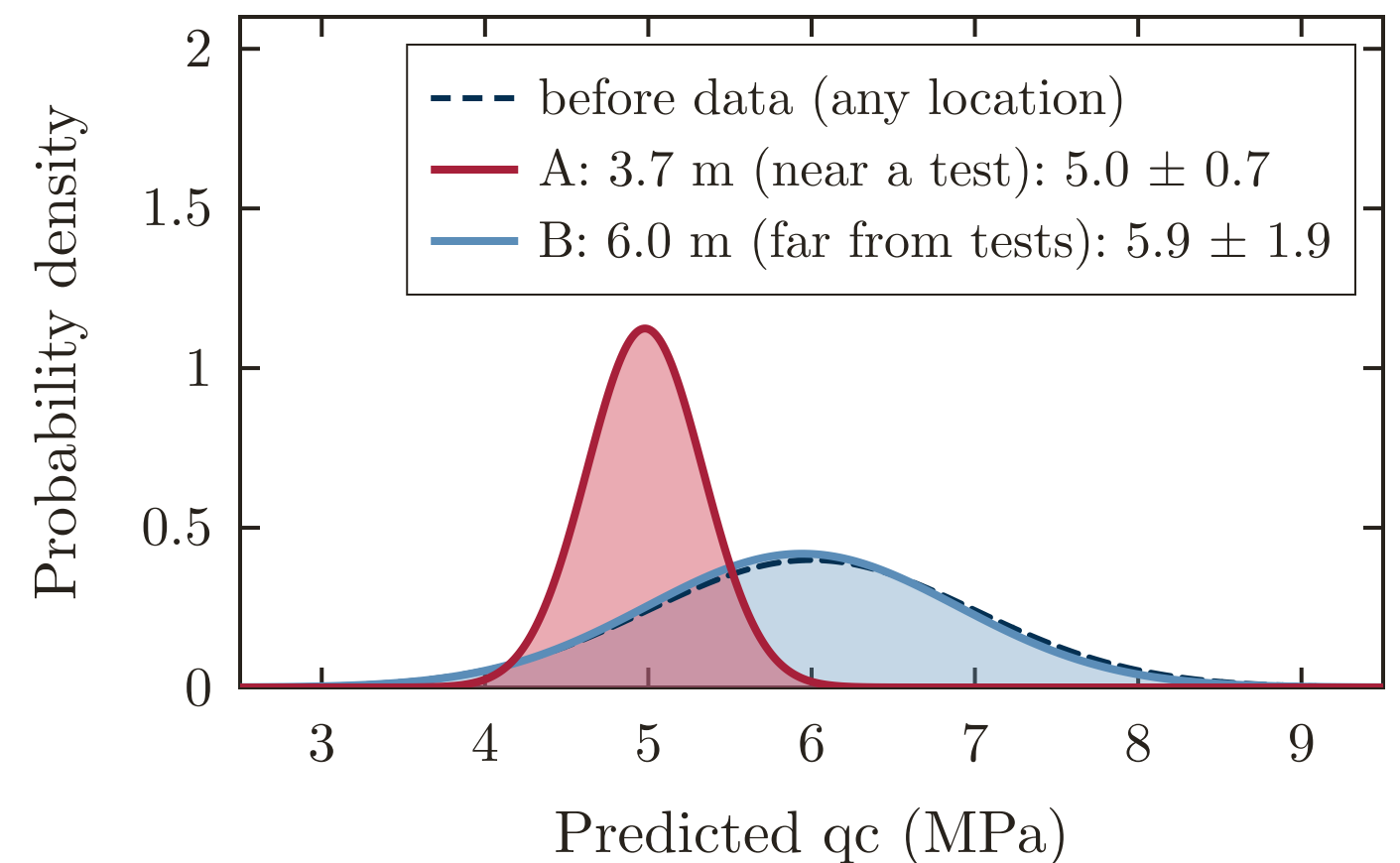
Gaussian processes (2): measurements pin the curves down

Three CPT tests are added ($\ell = 1.5$ m). Only curves that pass near the data stay plausible, and the model predicts q_c with uncertainty at every untested point.

After 3 CPT measurements: curves must pass near the data



Predictions where we did not test



A (3.7 m, near a test): 5.0 ± 0.7 MPa. B (6.0 m, far from tests): 5.9 ± 1.9 MPa, almost the prior 6 ± 2 .
Ranges are 95% (± 2 SD).

Data shrink uncertainty nearby; far from tests, the prior still speaks. Test next where the band is widest.

Autoregression: predict using the recent past

AR(1) uses the previous value and fresh noise. The coefficient controls persistence and possible alternation.

The model

$$X_t - \mu = \varphi(X_{t-1} - \mu) + \varepsilon_t$$

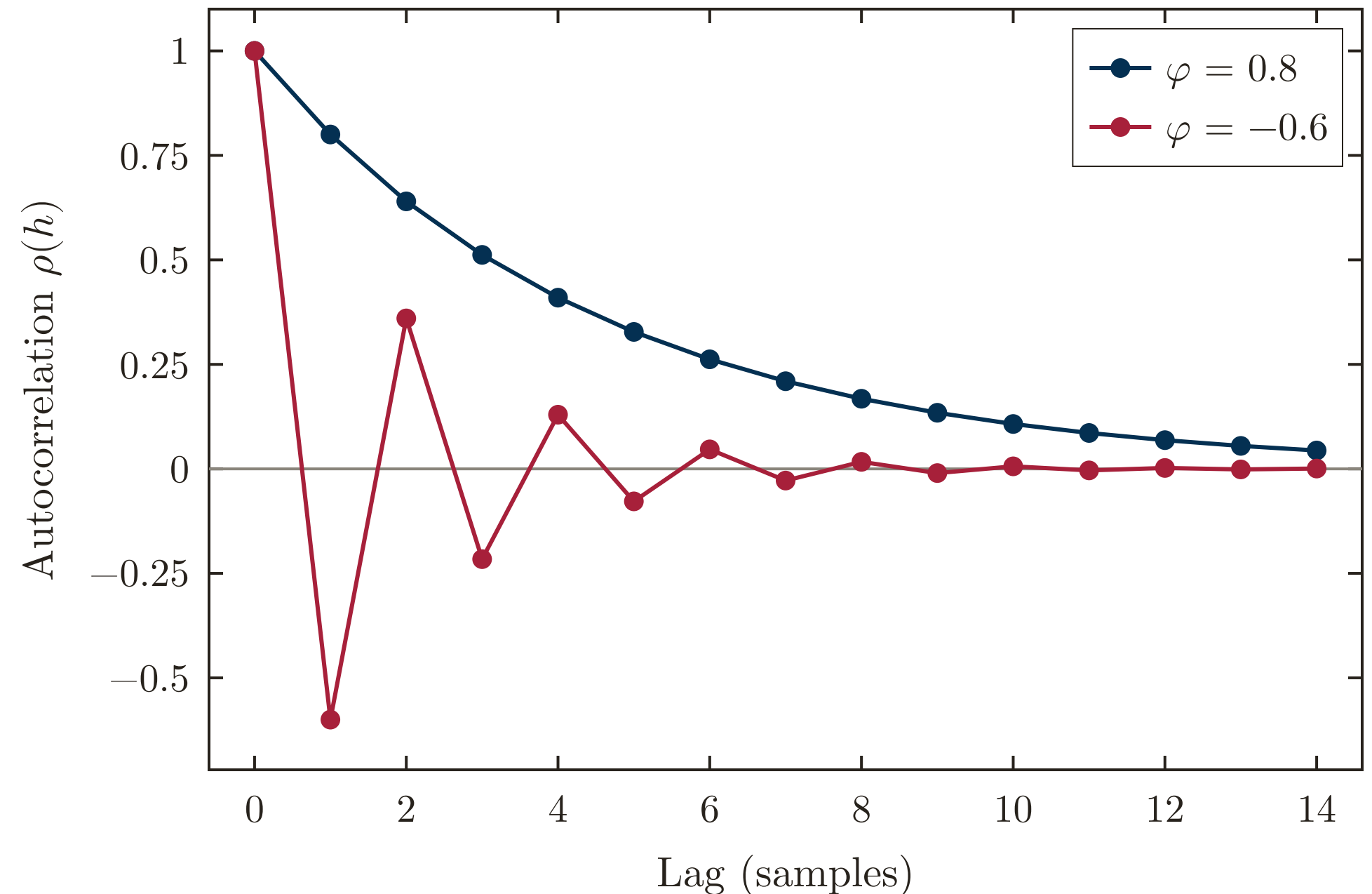
ε_t supplies fresh unpredictable input.

How it behaves

φ near 1: persistent disturbances

φ near 0: little lagged dependence

Negative φ : alternating responses



An AR(1) model is stationary when $|\varphi| < 1$ with stationary initialization.

Process families overlap

These names describe different properties. They are not mutually exclusive bins.

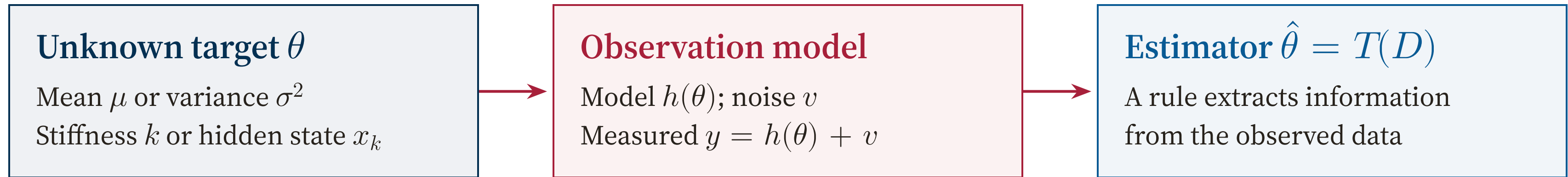
Example	Gaussian?	Markov?	Stationary level?
Gaussian AR(1), $ \varphi < 1$	Yes	Yes	Yes, with stationary start
Gaussian random walk	Yes	Yes	No
Homogeneous Poisson count	No	Yes	No
General Gaussian process	Yes	Depends on kernel	Depends on mean and kernel

A random variable, a regression and a process can all be parts of one engineering model.

Select properties that match the question, rather than choosing a name from a menu.

Estimation theory: designing rules for unknown quantities

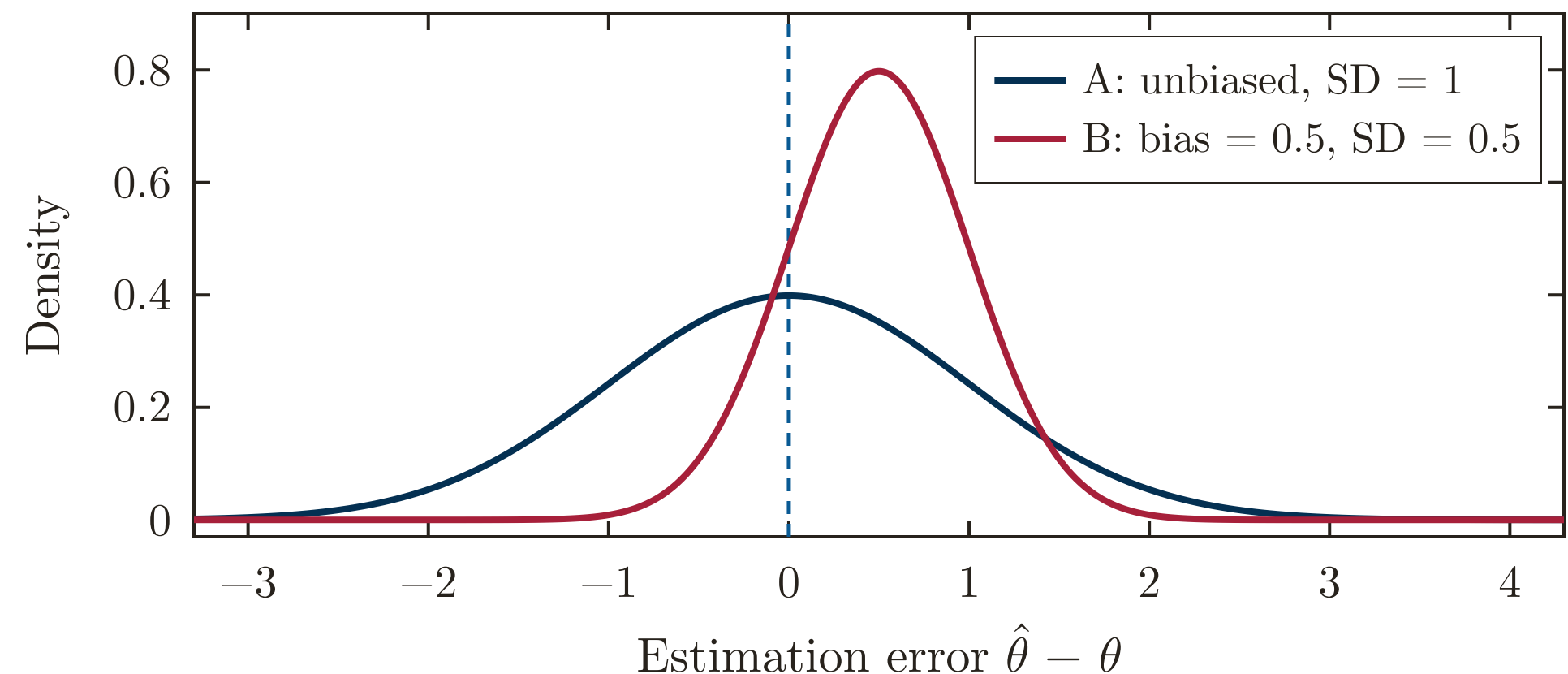
The unknown may be a population parameter, a physical property, or a changing hidden state.



What makes one rule better?

$$\begin{aligned} \text{MSE}(\hat{\theta}) &= E[(\hat{\theta} - \theta)^2] \\ &= \text{variance} + \text{bias}^2 \end{aligned}$$

Bias: systematic offset. Variance: spread
MSE combines both across repeated datasets.



An estimator is a rule applied to data. An estimate is the value that rule returns for one dataset.

Estimation methods: different criteria for choosing an answer

The target, observation model and error criterion determine which estimator makes sense.

Method	What it optimizes / uses	Rule or output	Typical use
Moment estimates	Match sample summaries	$\hat{\mu} = (1/n)\Sigma x_i$ $\hat{\sigma}^2 = (1/n)\Sigma (x_i - \bar{x})^2$	Mean and variance of strength
Least squares	Minimize squared residuals	$\hat{\theta} = \arg \min \Sigma_i [y_i - h_i(\theta)]^2$	Regression and sensor calibration
Maximum likelihood	Maximize likelihood of observed data	$\hat{\theta} = \arg \max p(D \theta)$	Distribution and model parameters
Bayesian MAP / MMSE	Use the posterior distribution	MAP: mode; MMSE: $E[\theta D]$	Prior knowledge plus observations
Kalman filtering	Update a state as measurements arrive	Predict prior $\rightarrow$ update with sensor data	Tracking displacement or sensor bias

Several methods can give the same answer under special assumptions. Their definitions still differ.

Estimation theory across disciplines

The common task is to infer an unknown quantity from imperfect observations. The field changes the target and the data.

Field	Unknown target	Observed data	Shared methods
Statistics	Population mean, variance or regression coefficients	A sample of strength tests	Moments, MLE, Bayesian estimates
Signal processing	Hidden displacement, velocity or a clean signal	Noisy sensor measurements over time	Least squares, MMSE, Kalman filtering
Machine learning	Predictive model weights or a latent representation	Training inputs and observed outcomes	Likelihood, regularized loss, MAP
Data science	Parameters and predictions used in an analysis	Collected, cleaned and combined datasets	Regression, MLE, Bayesian estimation
Optimization	The parameter value that minimizes an objective	A loss function, constraints and possibly data	Numerical solver for LS, MLE or MAP

Same calculation, different language

Gaussian noise: least squares = MLE. Add a Gaussian prior: quadratic regularization = MAP.
Assumptions: independent Gaussian errors of equal, fixed variance.

Optimization computes many estimates; the statistical model and loss explain what those estimates mean.

Estimation: what exactly is unknown?

The target and the available data determine whether we fit a model, filter a state, smooth a history or predict a future value.

Target	Example	Typical output
Parameter	Material mean, regression slope, stiffness	Point estimate and uncertainty
Current hidden state	Bridge displacement from noisy sensors	Filtered state estimate
Past hidden state	Reconstruct yesterday’s response using later data	Smoothed history
Future response	Traffic demand or displacement at the next step	Predictive distribution

“Batch versus recursive” describes computation. “Parameter versus state” describes the target.

Kalman filtering: prediction meets a new measurement

A bridge displacement is estimated every hour. The sensor is noisy (SD 3 mm). The filter combines a simple model of how the displacement evolves with each new reading.

1 Predict

The model moves the estimate one step ahead:
 $\hat{x}^- = F \hat{x}$. Uncertainty grows.

2 Observe

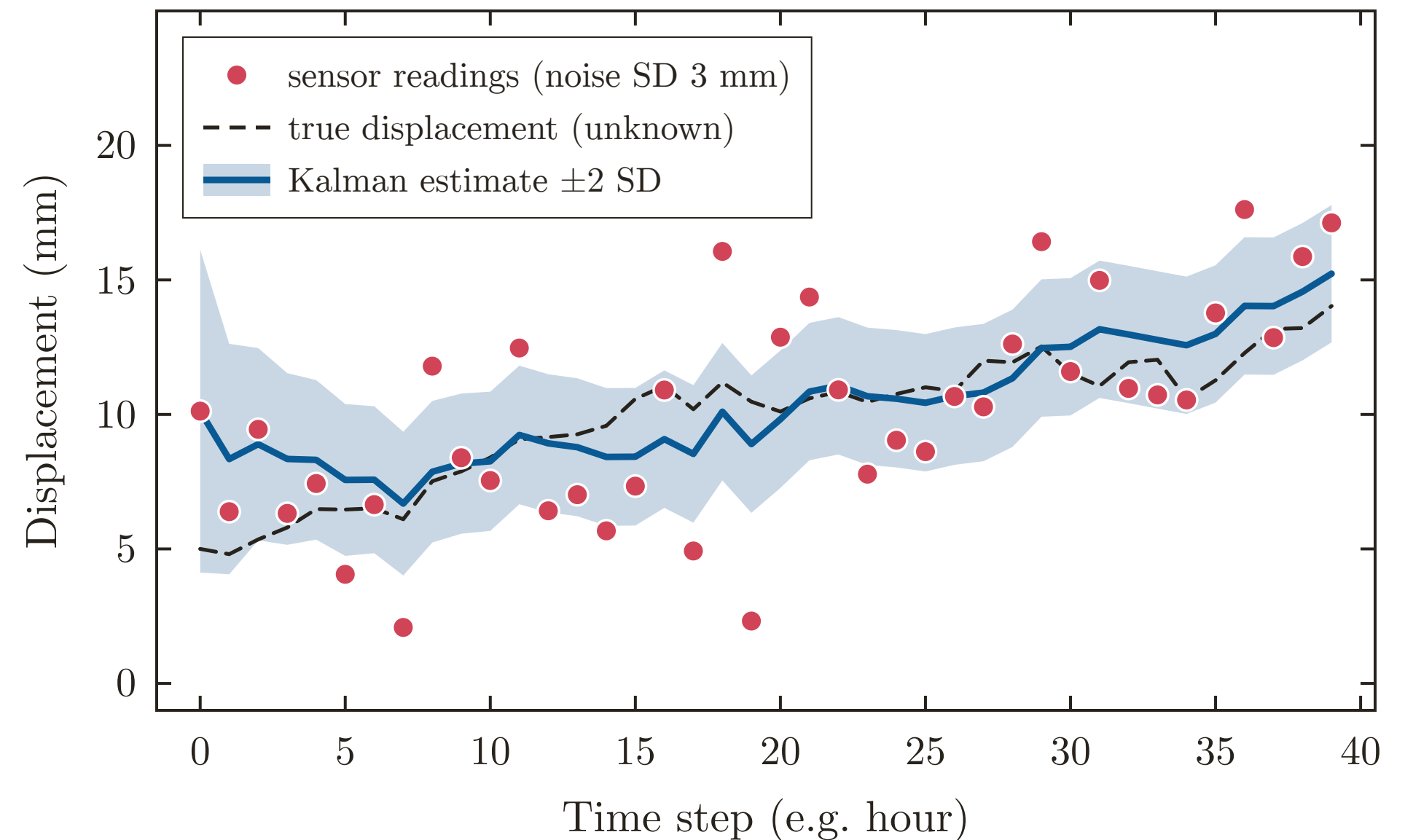
A noisy reading arrives: $z = H x + v$.

3 Update

Move toward the reading by the gain K ; uncertainty shrinks. Repeat every hour.

Result over 40 hours

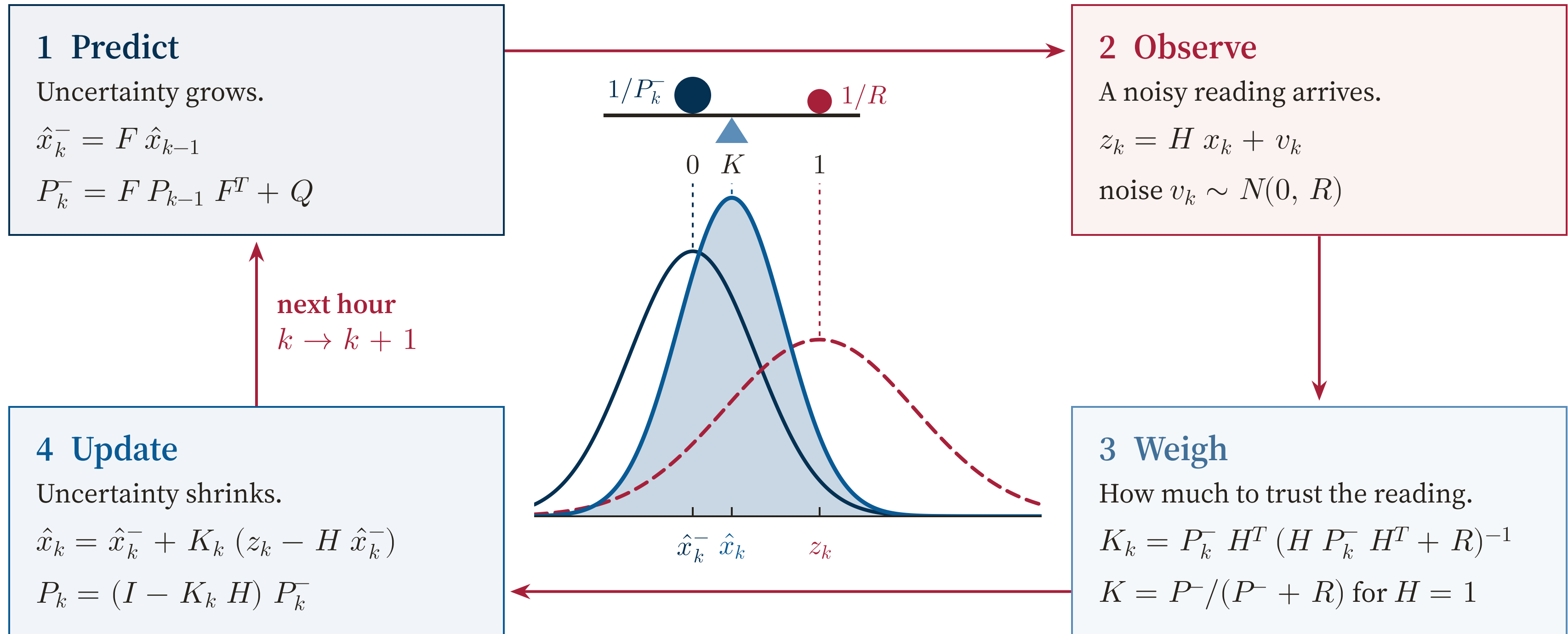
Typical error: sensor alone 3.1 mm, Kalman estimate 1.6 mm.



Predict, observe, update, repeat: the estimate tracks the truth better than the raw readings.

The Kalman loop: predict, observe, weigh, update

Every hour the filter goes once around this loop: the model proposes, the sensor reports, the gain decides.



The gain weighs the two sources: K near 0 keeps the prediction, K near 1 follows the reading.

One Kalman update, step by step

Zoom into one hour of the previous slide. The model predicts 10 mm (SD 2 mm); the sensor reads 14 mm (noise SD 3 mm). Where is the displacement?

The two sources

Prior = model prediction: 10 mm, SD 2 mm

Sensor reading: 14 mm, SD 3 mm

Combine them

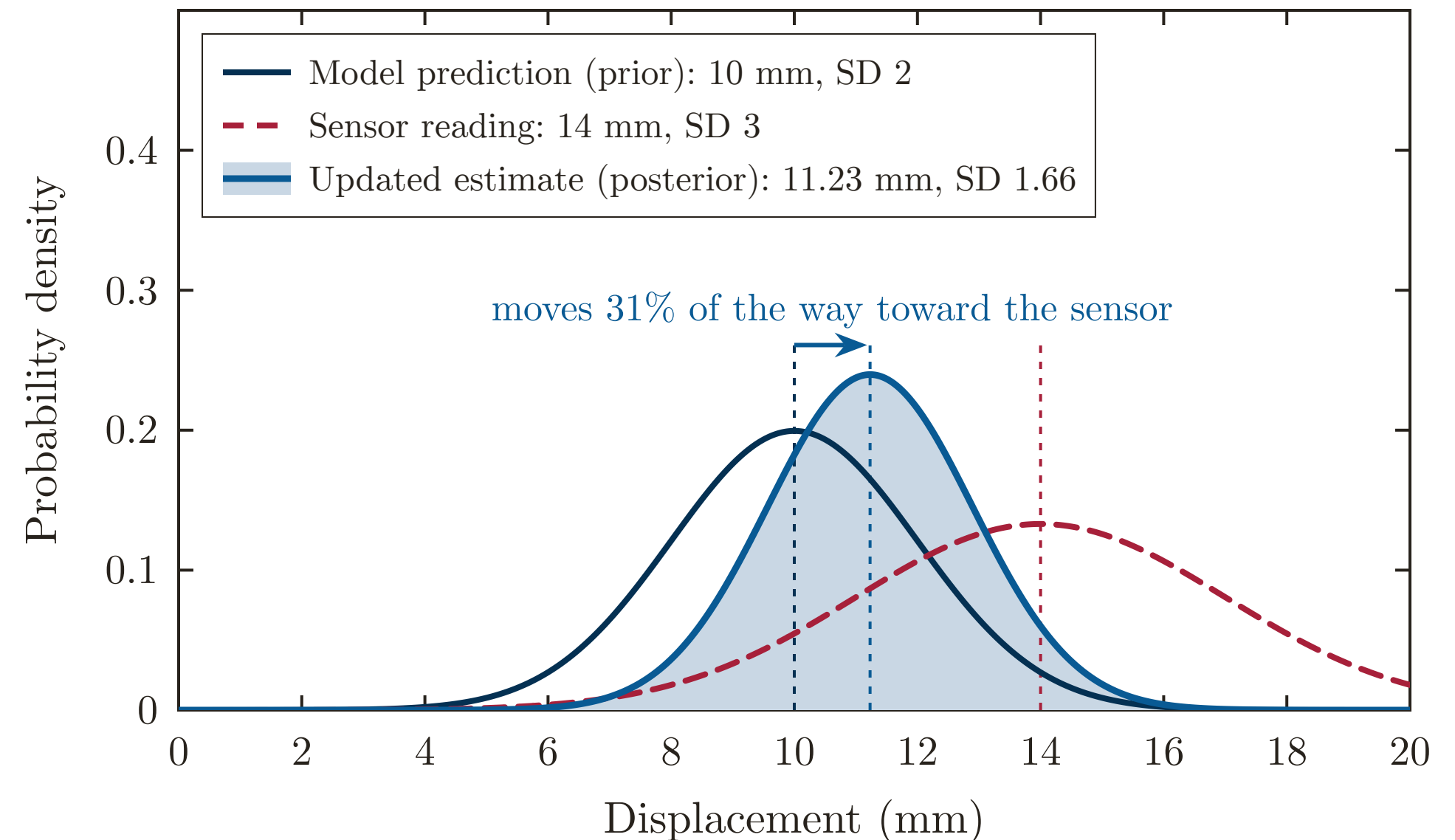
Gain $K = 2^2 / (2^2 + 3^2) = 4/13 = 0.31$

Estimate = $10 + 0.31 \times 4 = 11.23$ mm

New SD = $\sqrt{((1 - K) \times 4)} = 1.66$ mm

Why not just trust the sensor?

Its error (SD 3) is larger than the model's (SD 2), so the model gets more weight.

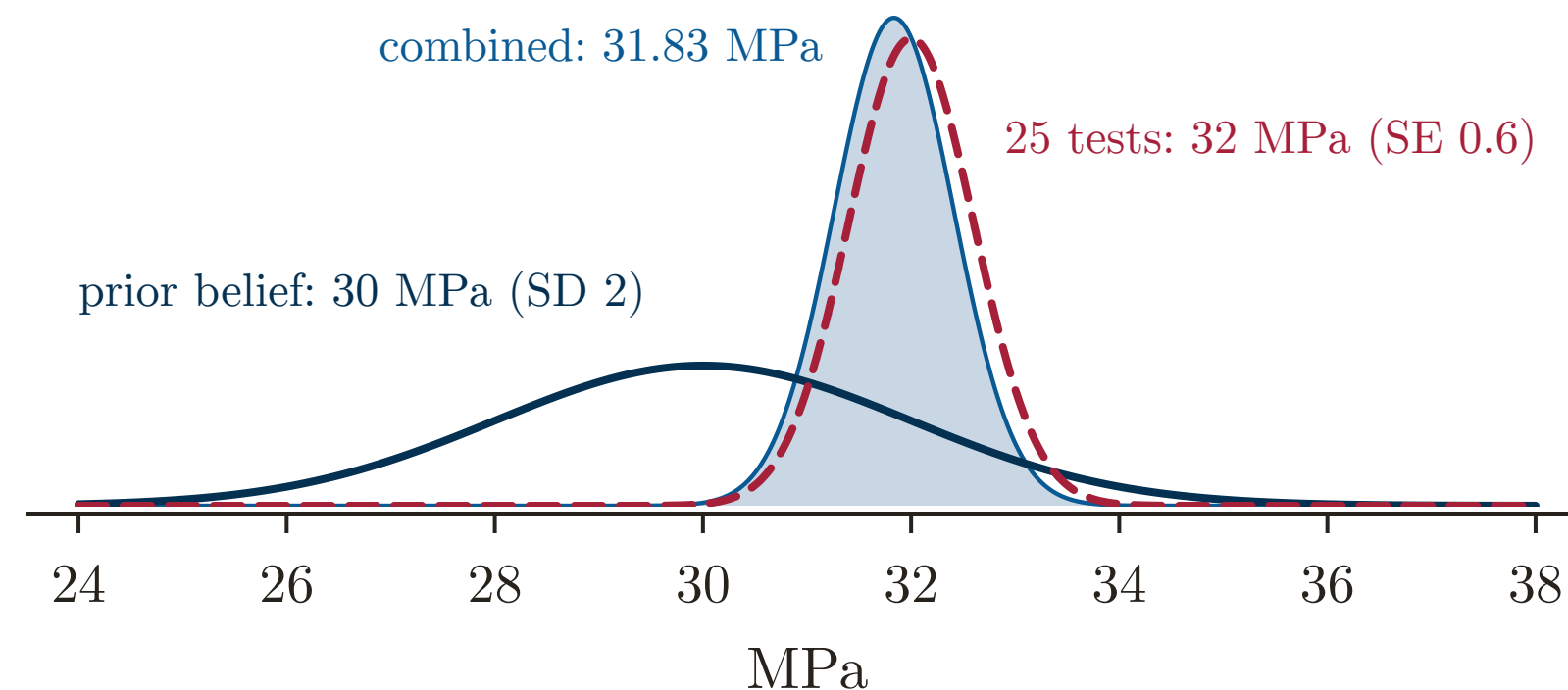


Prior = model prediction. The update moves toward the sensor by K and ends more certain than either source.

One idea behind estimation: weight by precision

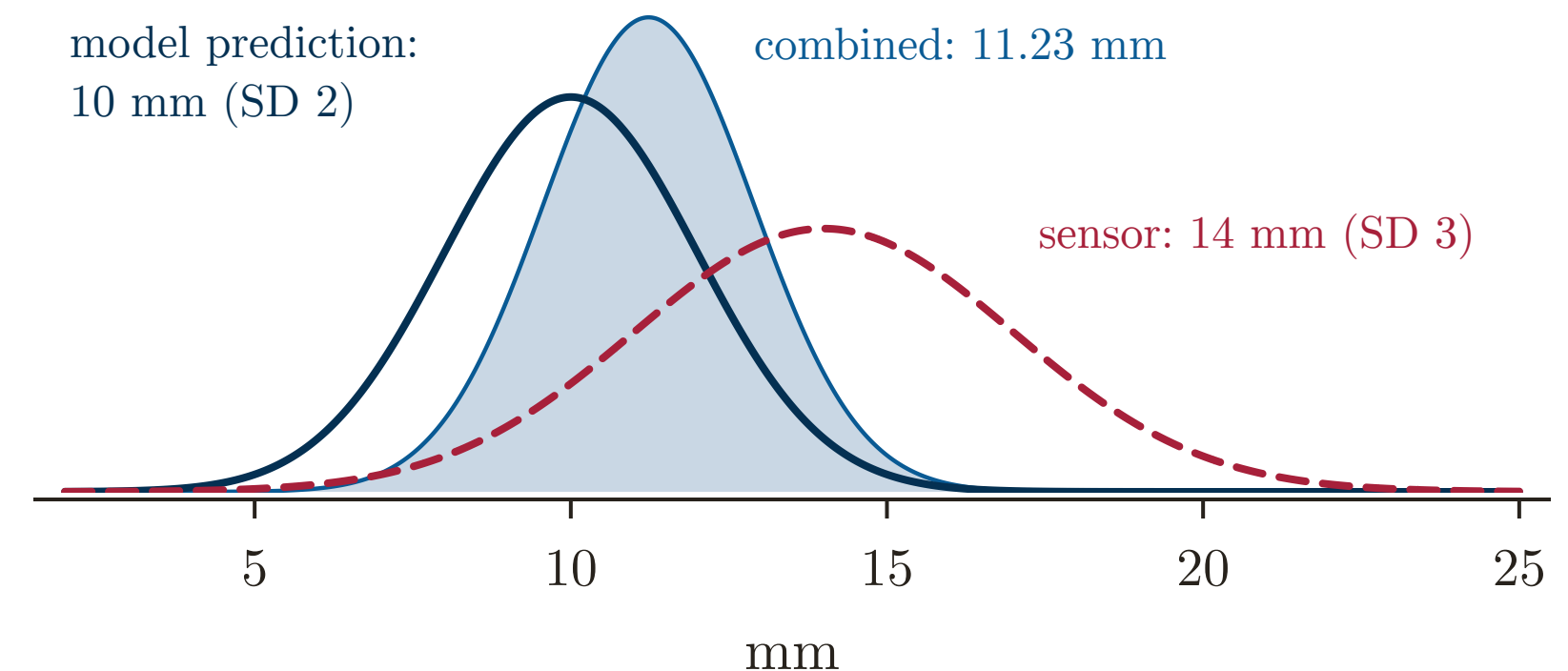
Two different examples from earlier in this guide. They look unrelated, but both combine an old estimate with new data using exactly the same rule.

Case A: concrete strength (slides 18–19)



Old 30 MPa (variance 4), data 32 MPa (variance 0.36).
 $w = 4 / 4.36 = 0.92 \rightarrow 31.83 \text{ MPa}$

Case B: bridge displacement (slides 65–66)



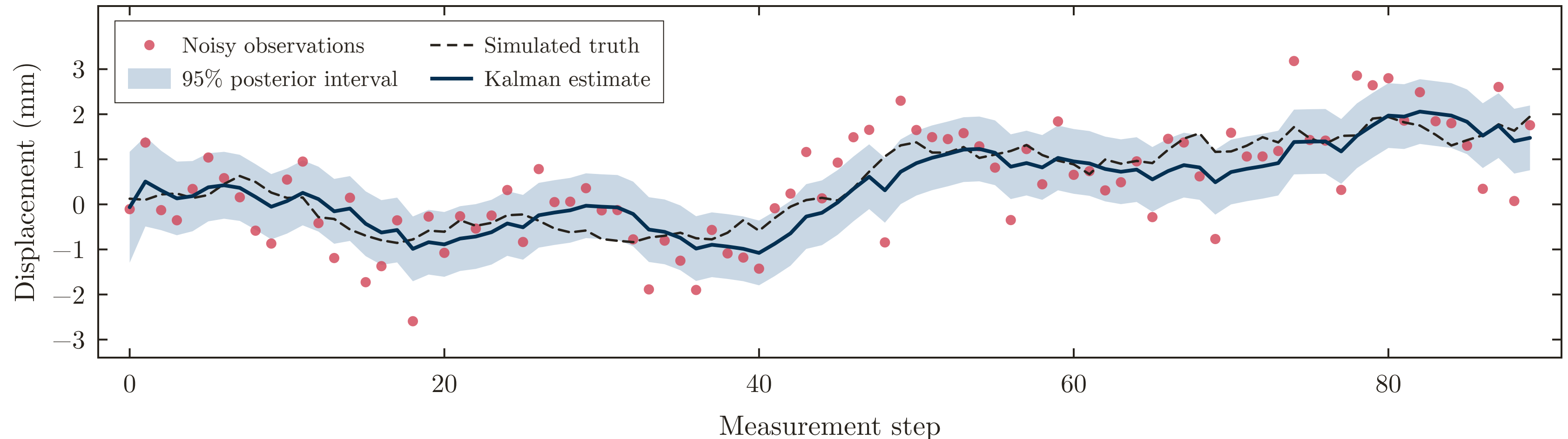
Old 10 mm (variance 4), data 14 mm (variance 9).
 $w = 4 / 13 = 0.31 \rightarrow 11.23 \text{ mm}$

Same rule: $\text{new} = \text{old} + w \times (\text{data} - \text{old})$, $w = \text{var}(\text{old}) / [\text{var}(\text{old}) + \text{var}(\text{data})]$

Different problems, same rule: each source counts in proportion to its precision.

Kalman applications: states, signals and parameters

A model-based estimate can combine noisy measurements with information about how the system evolves.



Track a state

Combine sensor readings with a response model.

Denoise a signal

Estimate the underlying signal while measurements stream in.

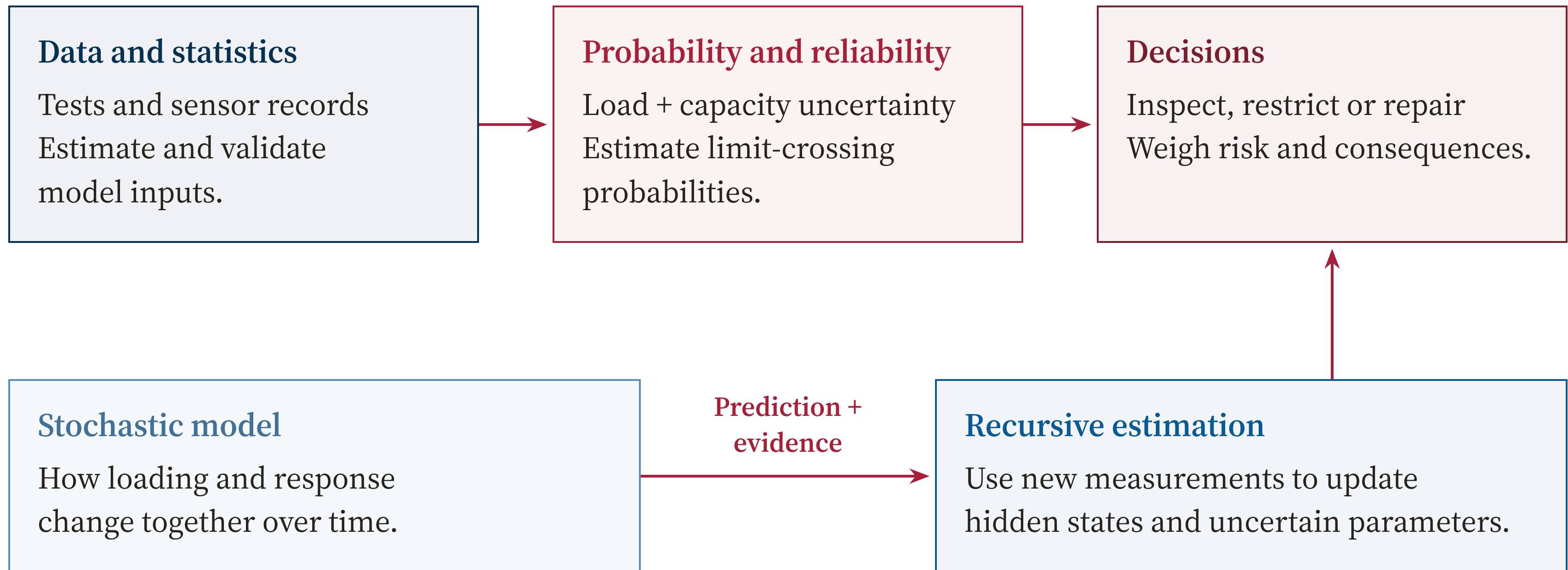
Identify parameters

Estimate stiffness or bias when identifiable.

Useful estimates require observable states, credible noise models and checks against measurement residuals.

One bridge can use the whole toolkit

The methods connect through a physical question: how should we inspect, operate or maintain this structure?



The workflow is iterative: new observations can change the model and the decision.

A guide to the methods and further reading

Use the question to choose a tool, then check the assumptions that make its output meaningful.

Your question	Useful starting point
How likely is an event?	A probability model and its assumptions
What can a finite sample tell me?	Estimation, intervals, tests and diagnostics
How do values depend over time or space?	Stochastic processes, covariance and kernels
Which frequencies matter?	Autocorrelation and spectral estimation
What is the hidden state now?	State-space modeling and filtering

Further reading

NIST Engineering Statistics Handbook • MIT OCW probability and signals notes

Rasmussen & Williams: Gaussian processes • Pei et al.: Kalman filtering

USGS flood guidance • SciPy Welch documentation • Full links in slide notes

The goal is to understand the role of each method before learning its derivation.

Sources: [1], [3], [10], [22], [37], [40]

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Numbers in square brackets on each slide point to this list. Worked-example numbers were constructed for teaching (see the note on the last reference slide).

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Standard results cite textbooks; named methods cite their original papers or standards.

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Standard results cite textbooks; named methods cite their original papers or standards.

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About the worked examples

The numbers in the examples (the 25 concrete tests, the spring $Y = F/k$, the load and capacity values, the flood, traffic and sensor cases) were constructed by the author for teaching. Formulas, definitions and methods follow the cited sources. The $Y = X^2$ example on slide 31 follows the same idea as JCGM 101, 9.4.2 [44].

Standard results cite textbooks; named methods cite their original papers or standards.